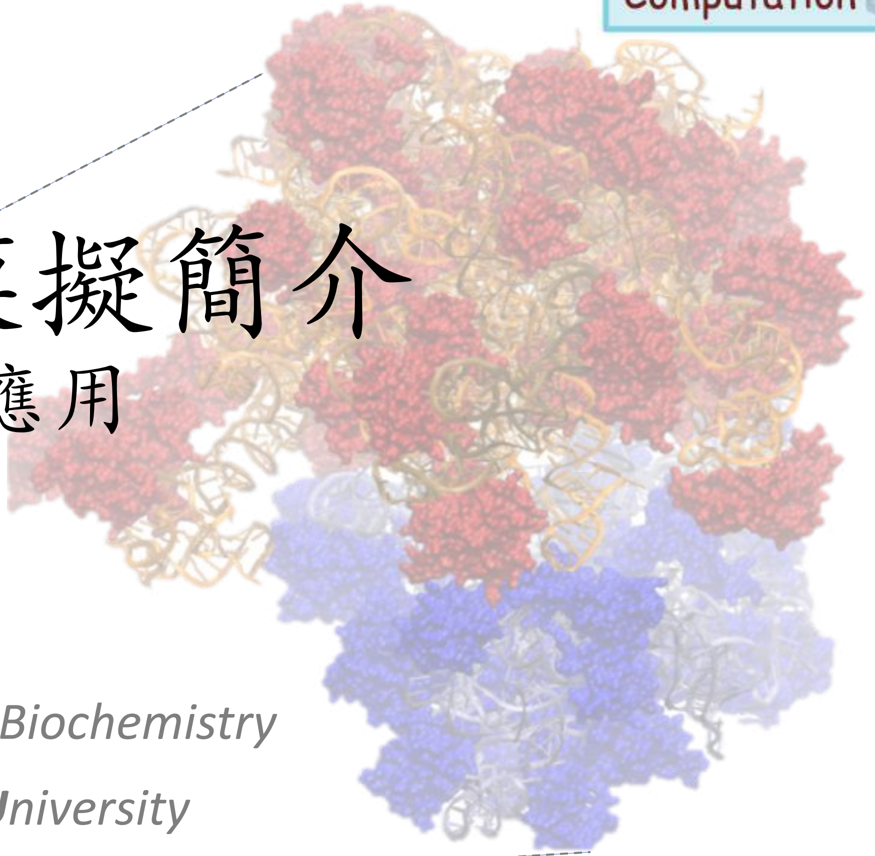
# 分子動力學模擬簡介

## 理論與實務應用

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國立中正大學-化學暨生物化學系


Aug. 25-26, 2025

@臺大計中

**請注意！**



（投影片內容採中[Chinese]/英文[English]夾雜）

MYTLab Computation Bio

# 蔡旻燁
# Min-Yeh Tsai

- Theoretical Phys. Chem.
- Computational Biophysics
- Structural Biology

Chemistry

**BS**

NTNU (01-05)

師大

Physical Chemistry (Protein folding)

**PhD**

NTU (06-11)

台大

S. H. Lin

Applied Chemistry (Protein aggregation)

**Military Duty**

NCTU (11-14)

交大

Protein-protein & protein-DNA interaction

**Postdoc**

Rice (14-18)

萊斯大學

P. G. Wolynes

**AP**

TKU (18-)

淡江大學

**AP**

CCU (23-)

國立中正大學

化生系

助教 1

張育誠

Heery

助教 2

陳宏毅

Sam

# 這兩天(十二個小時)會學些什麼?

## You will learn

- 分子動力學模擬的 ABC
- 從頭快速建立一簡單粒子模型
- 使用軟體VMD視覺化複雜分子系統
- 使用 VMD 建立模型以及 NAMD 進行動力學模擬
- 模擬設定的輸入/輸出 (I/O)
- 快速建立用於生物分子、奈米材料的模擬系統
- 對模擬軌跡進行簡單結構與能量分析
- 了解MD模擬的整個流程,快速找到參考資料

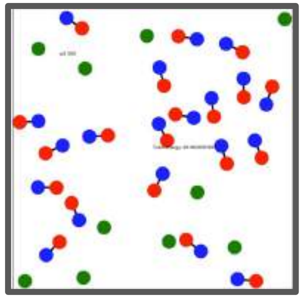
## You will **NOT** learn

- 進階的MD理論
- 分子力場開發
- 建立完全符合您研究上的模擬系統(一天是不夠的…)
- 成為MD的專家

(Rome wasn't built in a day…)

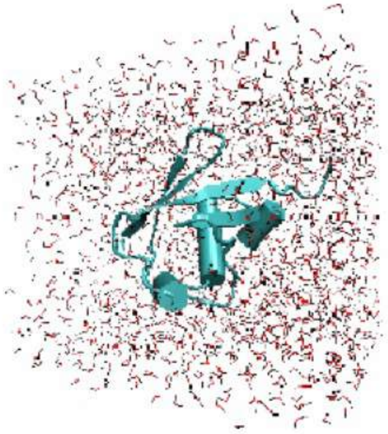
讓各位對於建立模型與設定動態模擬的工具有初步的認識
→激勵學習分子模擬的種子

# Outline



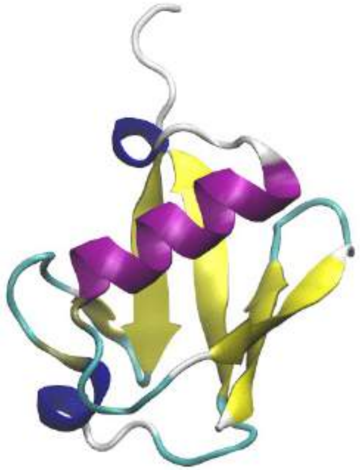
**I. Simple MD**
- Python
- JupyterLab

JupyterLab



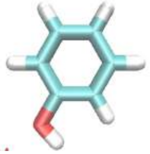
**II. Visualization**
- VMD (Water)
- MolView
- Protein/DNA

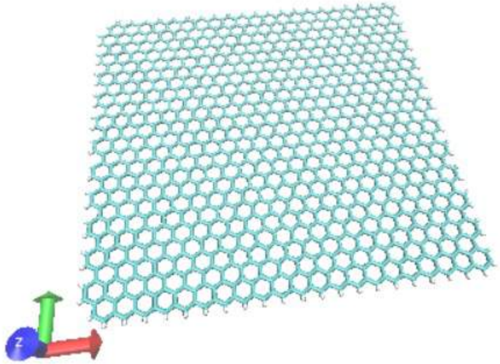
VMD
Visual Molecular Dynamics



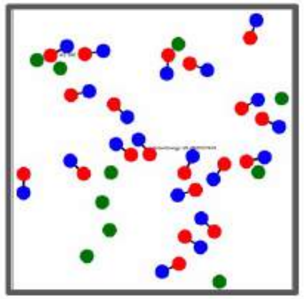
**III. MD (Bio)**
- NAMD/VMD
  - Protein
  - Solvation
  - Add ions

NAMD
Scalable Molecular Dynamics





**IV. MD (Nano)**
- NAMD/VMD
  - Graphene
  - Small molecules



**I. Simple MD**
- Python
- JupyterLab

JupyterLab

# Part I

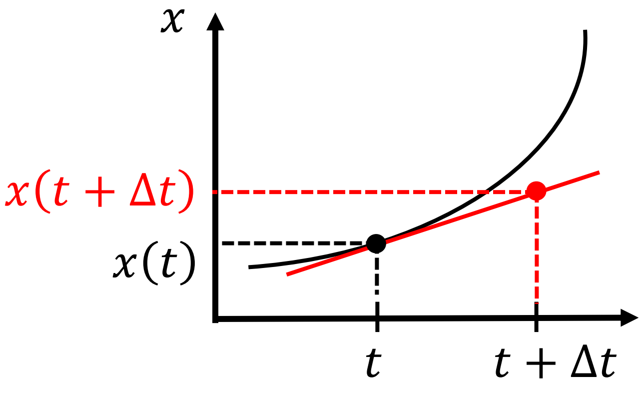
(分子動力學模擬ABC)

# 預測運動軌跡好比預測未來發生的事

- 位置與時間關係 $x(t)$ 的一階線性關係

$$x(t+\Delta t) = x(t) + \dot{x}\Delta t \qquad \dot{x} = \frac{dx}{dt} = v \ \ \text{(velocity)}$$

粒子現在位於 $x(t)$，經過 $\Delta t$ 之後到達 $x(t+\Delta t)$




- 線性思維


$t$ 今日$(t)$不讀書

$t+\Delta t$ 明日$(t+\Delta t)$考試就GG囉

<u>考試表現</u> = <u>目前能力</u> + <u>用功程度</u> x <u>時間</u>

$x(t+\Delta t) \quad x(t) \quad \dot{x} \quad \Delta t$

# 讓粒子動起來的三部曲

初始化 (Initialization)

- 粒子的起始位置與初速度

更新 (Update)

- 粒子運動的演算法

執行 (Run)

- 產生軌跡

視覺化 (Visualization) & 分析 (Analysis)

1. 從Dropbox下載 113.2 MD workshop.zip

2. 解壓縮檔案至 113.2 MD workshop\

3. 進入113.2 MD workshop\ 資料夾，將 **module1** 移至桌面 (Desktop)

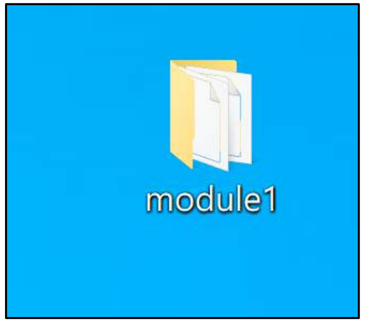
JupyterLab

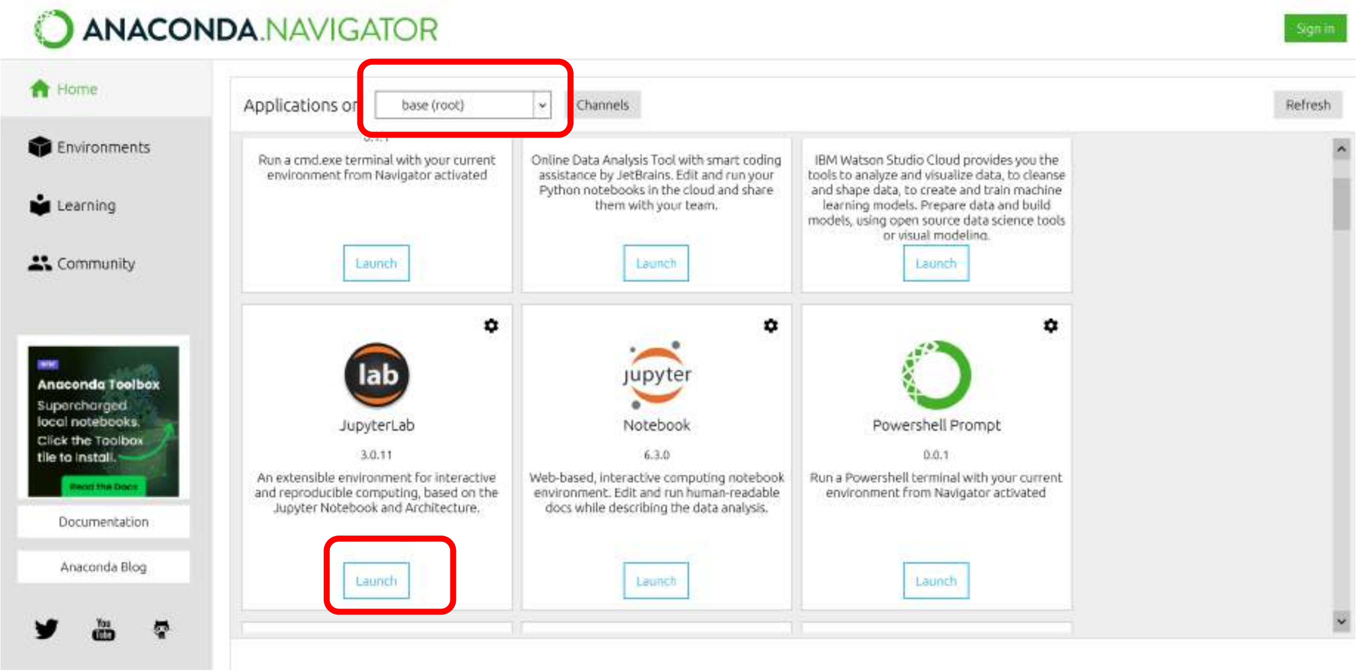
4. 開啟 Anaconda → (wait…) → 在 **base (root)** 環境下開啟 JupyterLab → 點 **Launch**

5. JupyterLab將開啟於瀏覽器新分頁

將左側之目錄切換至『桌面』

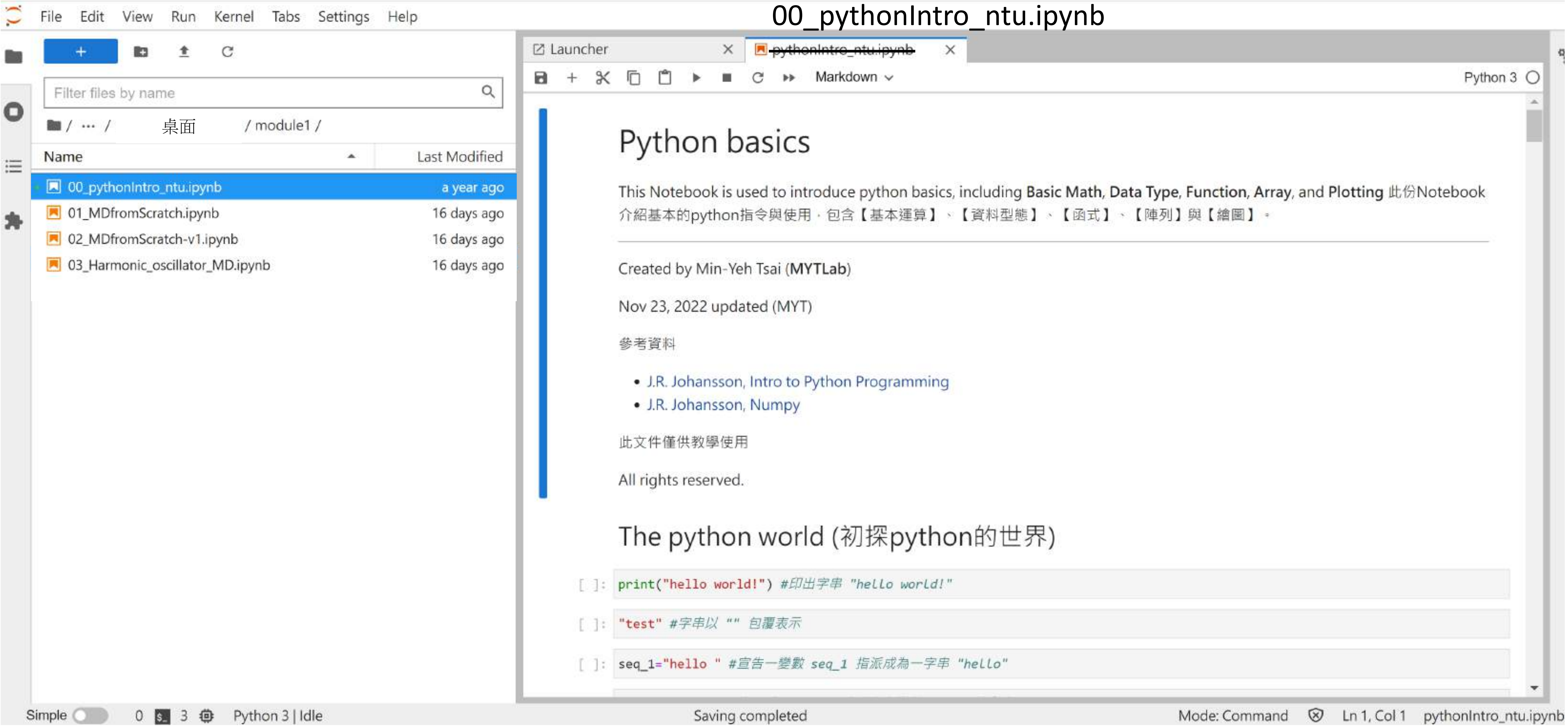
→ 進入 **module1**。



將左側之目錄切換至『桌面』→ 進入 **module1**

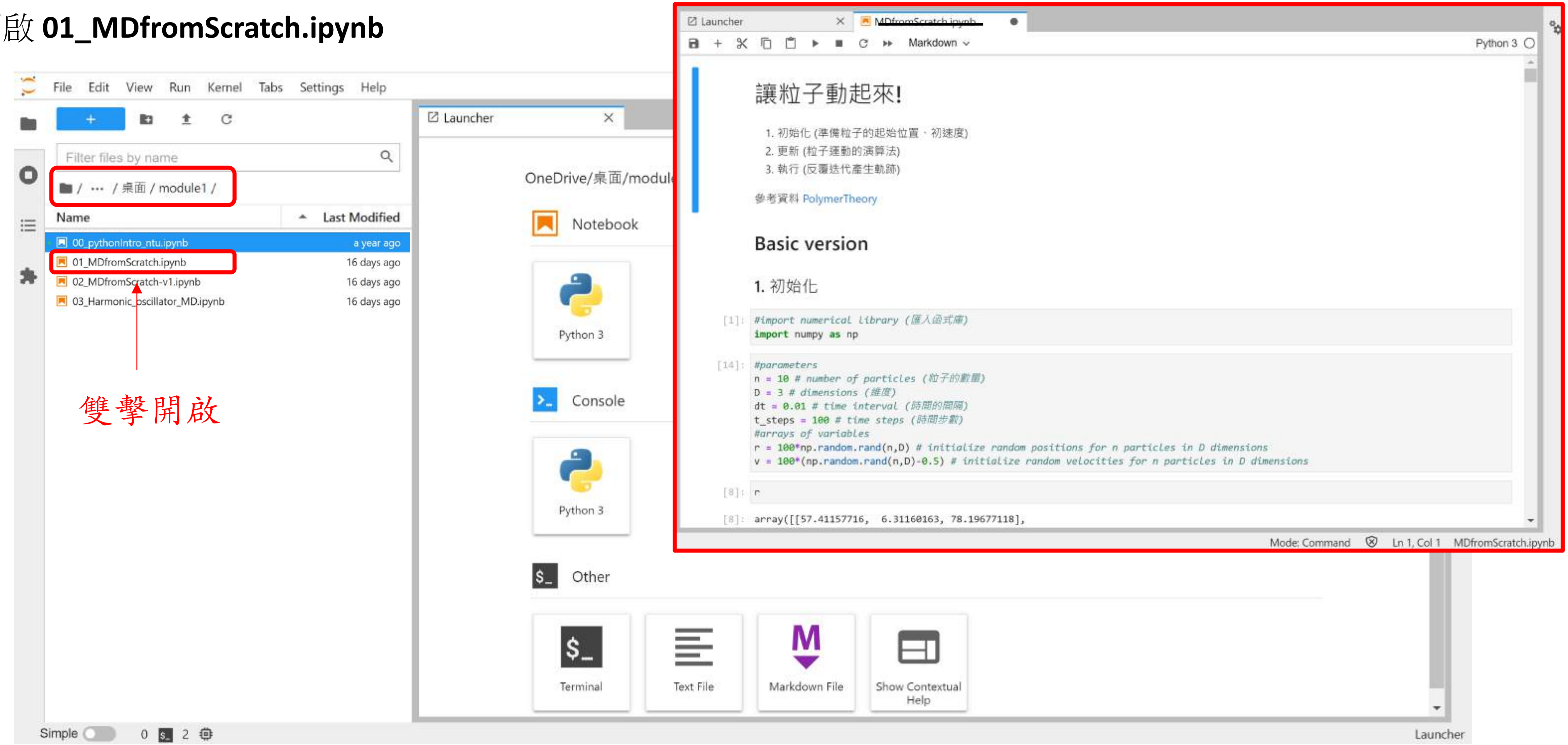
→ 開啟 **00_pythonIntro_ntu.ipynb**



將左側之目錄切換至『桌面』→ 進入 **module1**

→ 開啟 **01_MDfromScratch.ipynb**

01_MDfromScratch.ipynb



將左側之目錄切換至『桌面』→ 進入 **module1**

→ 開啟 **02_MDfromScratch-v1.ipynb**

02_MDfromScratch.ipynb

# 預測的品質與採取的策略有關

- **Verlet algorithm**
  (韋爾萊算法)

$$x(t+\Delta t) = 2x(t) - x(t-\Delta t) + \ddot{x}\Delta t^2$$

表現　目前實力　過去實力　有效努力 x 時間$^2$

實際表現是根據過去與目前的差異，透過一定時間內有效的努力所達成

了解歷史，掌握當下，加上持續的努力 → 成功

# [補充] 運用數值方法解運動方程式

**Position and dynamic properties can be approximated as Taylor series expansions**

Position → $\mathbf{r}(t+\delta t) = \mathbf{r}(t) + \delta t\mathbf{v}(t) + \frac{1}{2}\delta t^2\mathbf{a}(t) + \frac{1}{6}\delta t^3\mathbf{b}(t) + \frac{1}{24}\delta t^4\mathbf{c}(t) + \cdots$

Velocity → $\mathbf{v}(t+\delta t) = \mathbf{v}(t) + \delta t\mathbf{a}(t) + \frac{1}{2}\delta t^2\mathbf{b}(t) + \frac{1}{6}\delta t^3\mathbf{c}(t) + \cdots$

Acceleration → $\mathbf{a}(t+\delta t) = \mathbf{a}(t) + \delta t\mathbf{b}(t) + \frac{1}{2}\delta t^2\mathbf{c}(t) \cdots$

$\mathbf{b}(t+\delta t) = \mathbf{b}(t) + \delta t\mathbf{c}(t) + \cdots$

**Truncated**

## Verlet algorithm

$$\mathbf{r}(t+\delta t) = \mathbf{r}(t) + \delta t\mathbf{v}(t) + \frac{1}{2}\delta t^2\mathbf{a}(t) + \cdot$$

$$\mathbf{r}(t-\delta t) = \mathbf{r}(t) - \delta t\mathbf{v}(t) + \frac{1}{2}\delta t^2\mathbf{a}(t) - \cdots$$

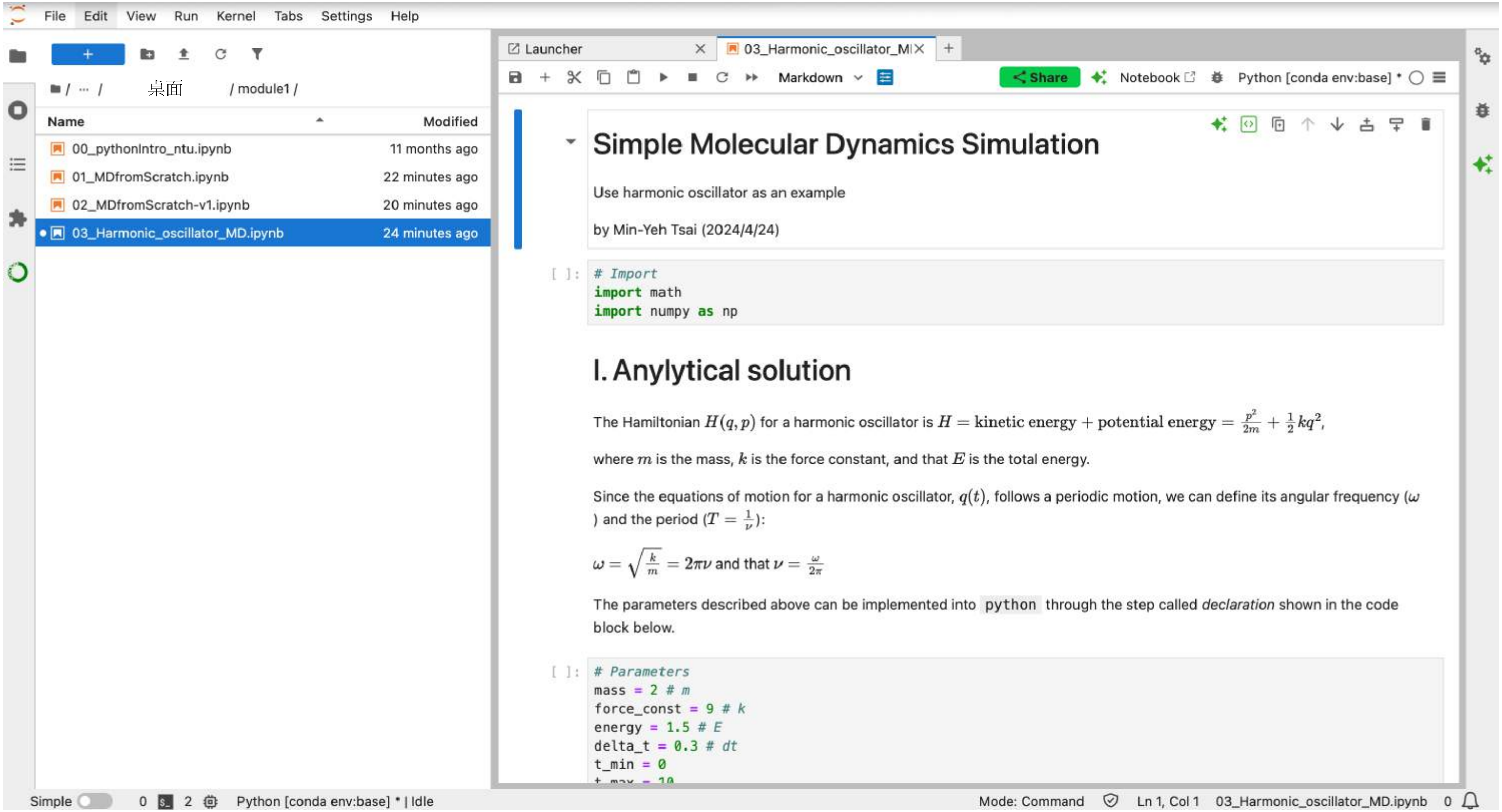
Summation ↓

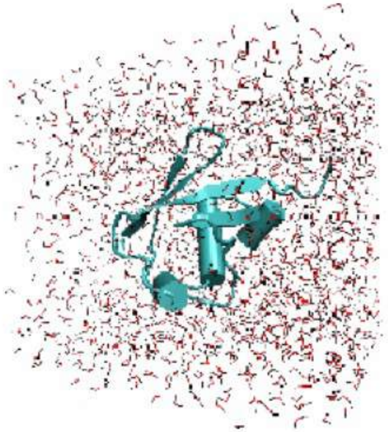
Truncated at O($\delta t^4$)
→Fourth-order method

$$\mathbf{r}(t+\delta t) = 2\mathbf{r}(t) - \mathbf{r}(t-\delta t) + \delta t^2\mathbf{a}(t)$$

Leach, Molecular Modelling: Principles and Applications 2nd 2001 Chapter 7

# Tutorial: 簡諧運動模擬 (Harmonic Oscillator)





**II. Visualization**
- VMD (Water)
- MolView
- Protein/DNA

VMD
Visual Molecular Dynamics

# Part II

(分子結構 & 視覺化)

# Tutorial: 前置作業

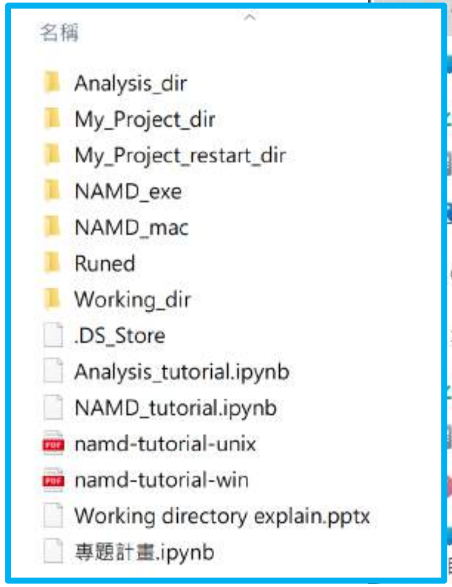
## 方法一：

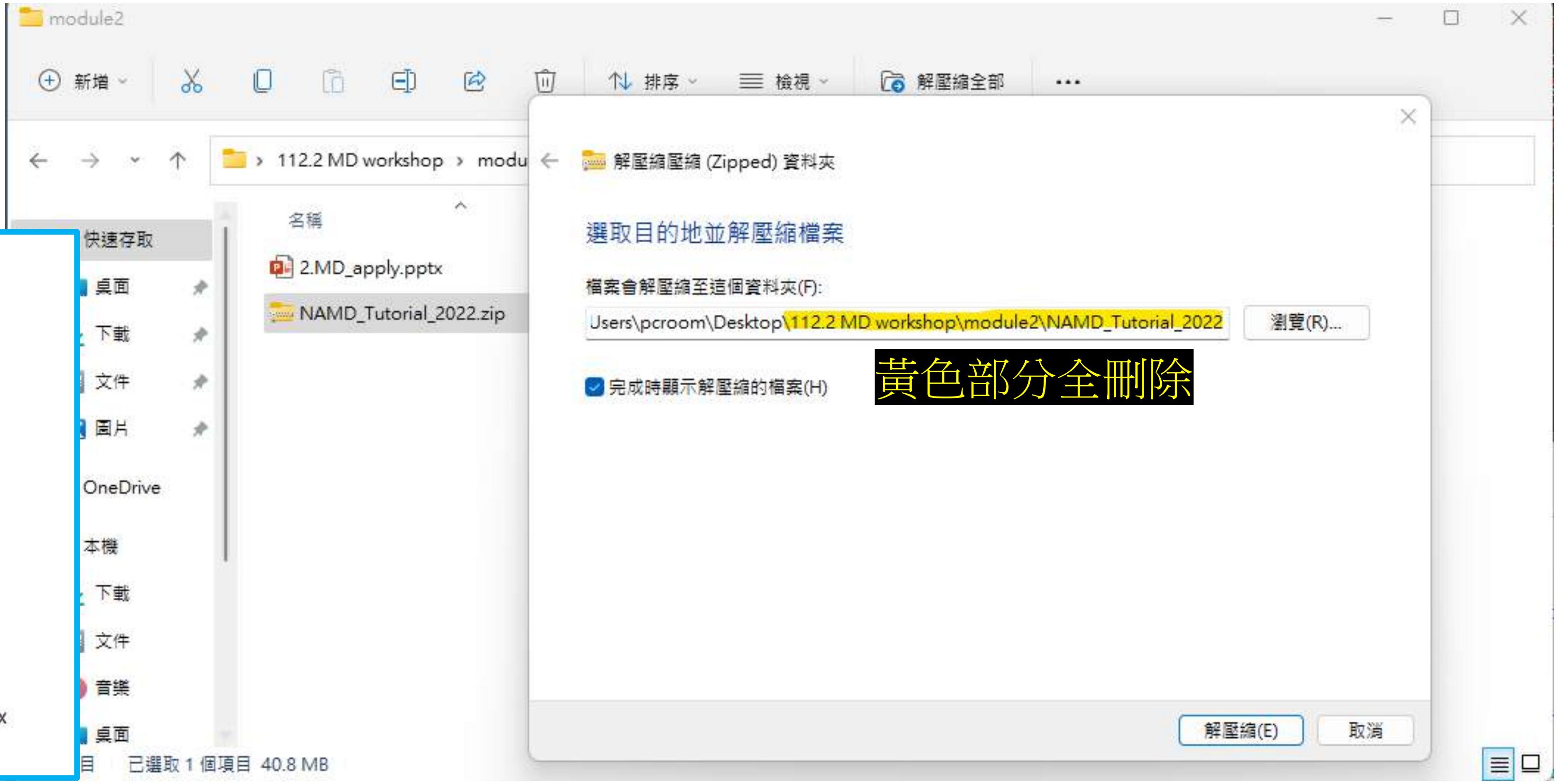
1. 從 113.2 MD workshop/module3 中，將 **NAMD_Tutorial_2025.zip** 資料夾拖曳至桌面
2. 按滑鼠右鍵 → 解壓縮檔案… → 目的地路徑 \.....\.....\Desktop\(此處全刪除)
3. 解壓縮

## 方法二：

或是直接從 **NAMD_Tutorial_2025.zip** 上按右鍵 → 解壓縮檔案…

桌面

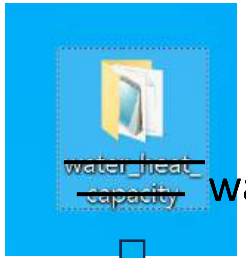



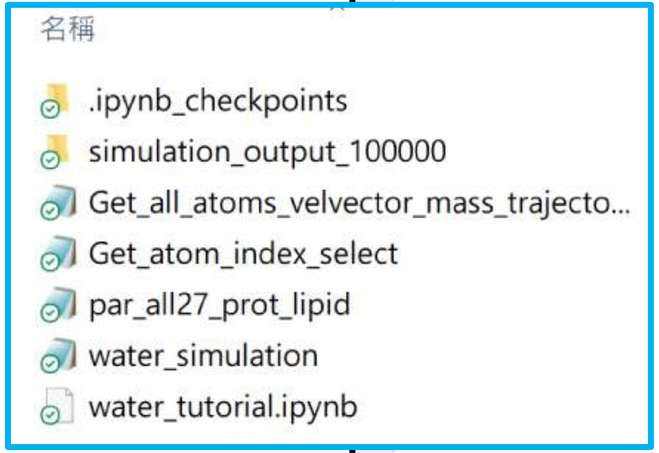
# Tutorial: 水盒模擬 (Water Box)

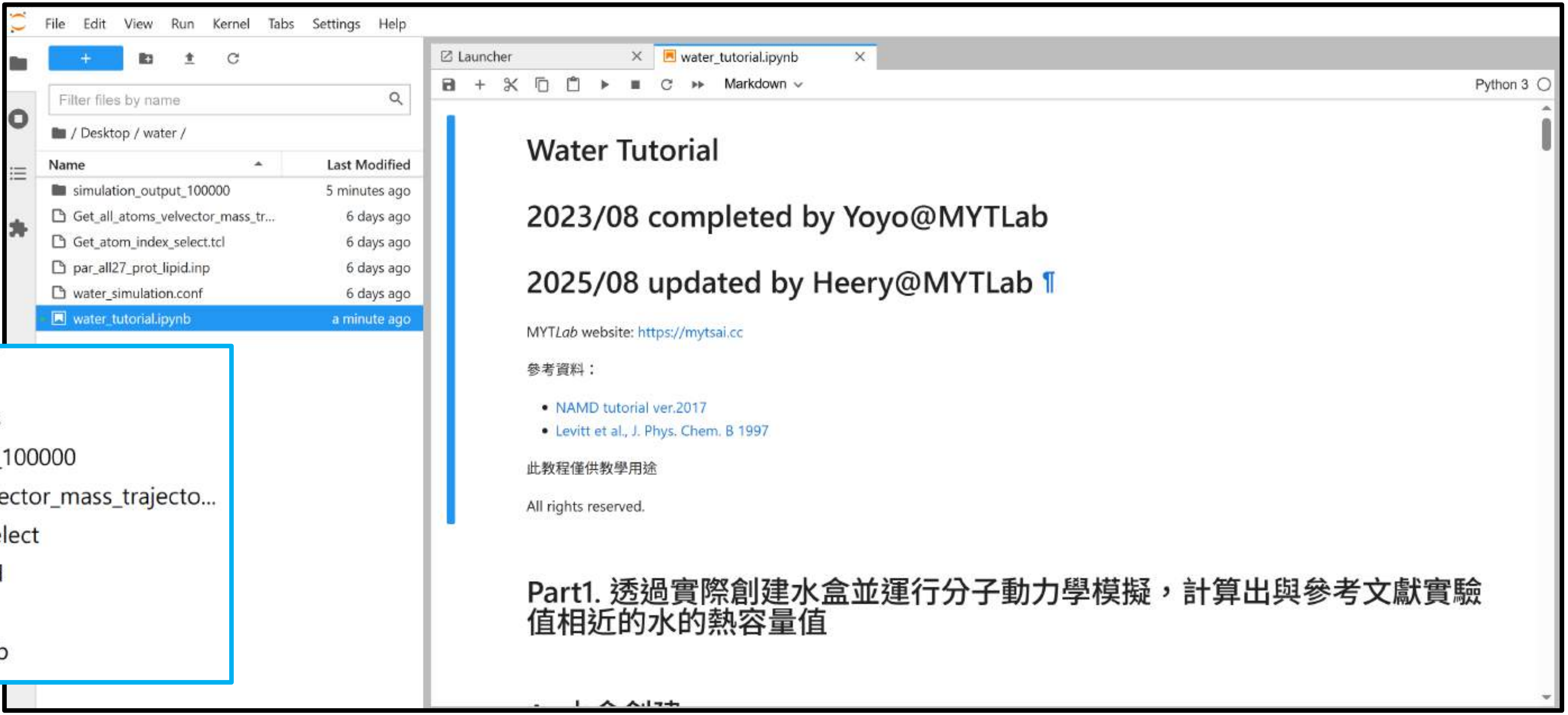
從 113.2 MD workshop/module2 中，將 **water** 資料夾拖曳至桌面

桌面

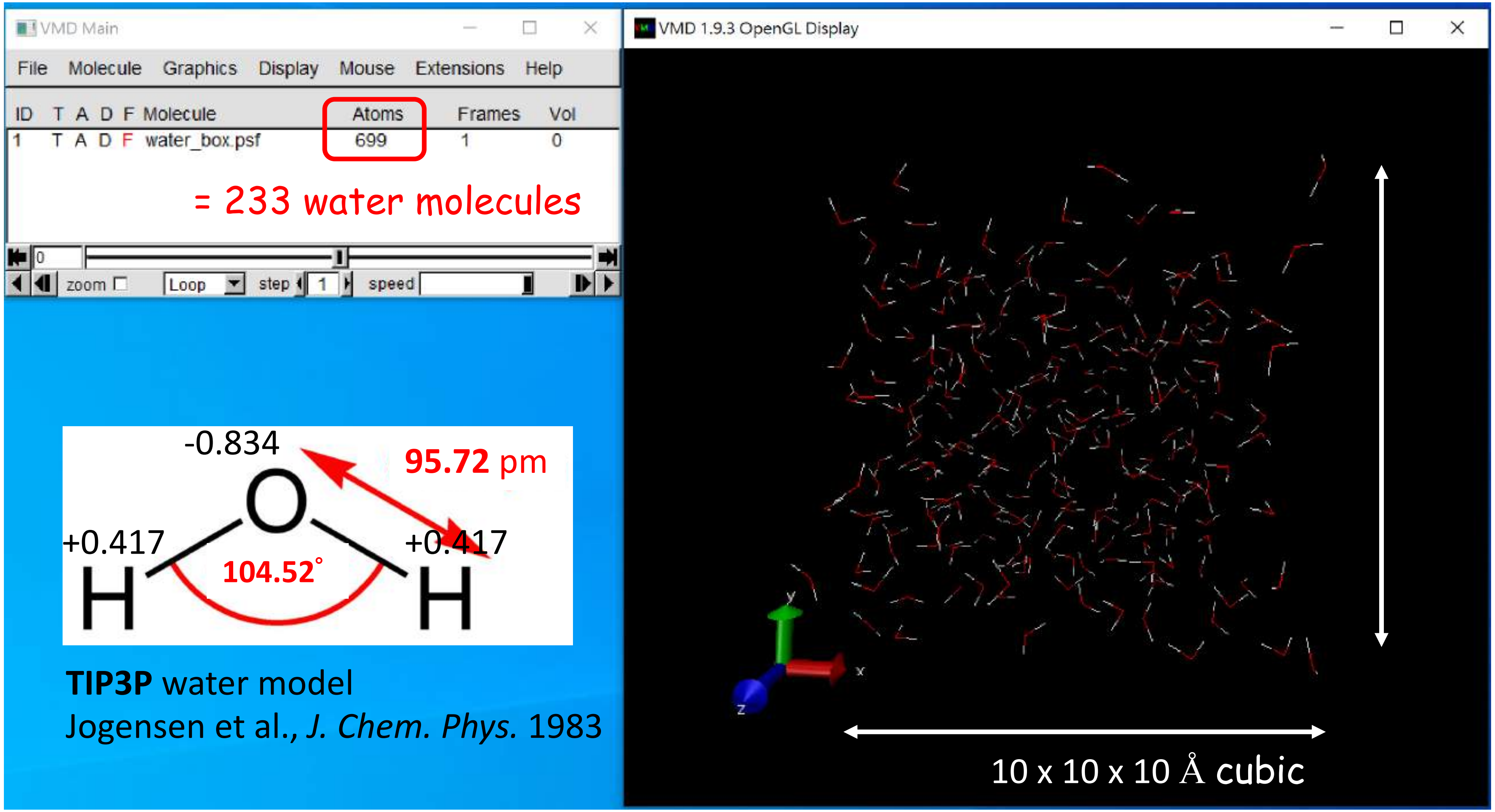
## 開啟 JupyterLab



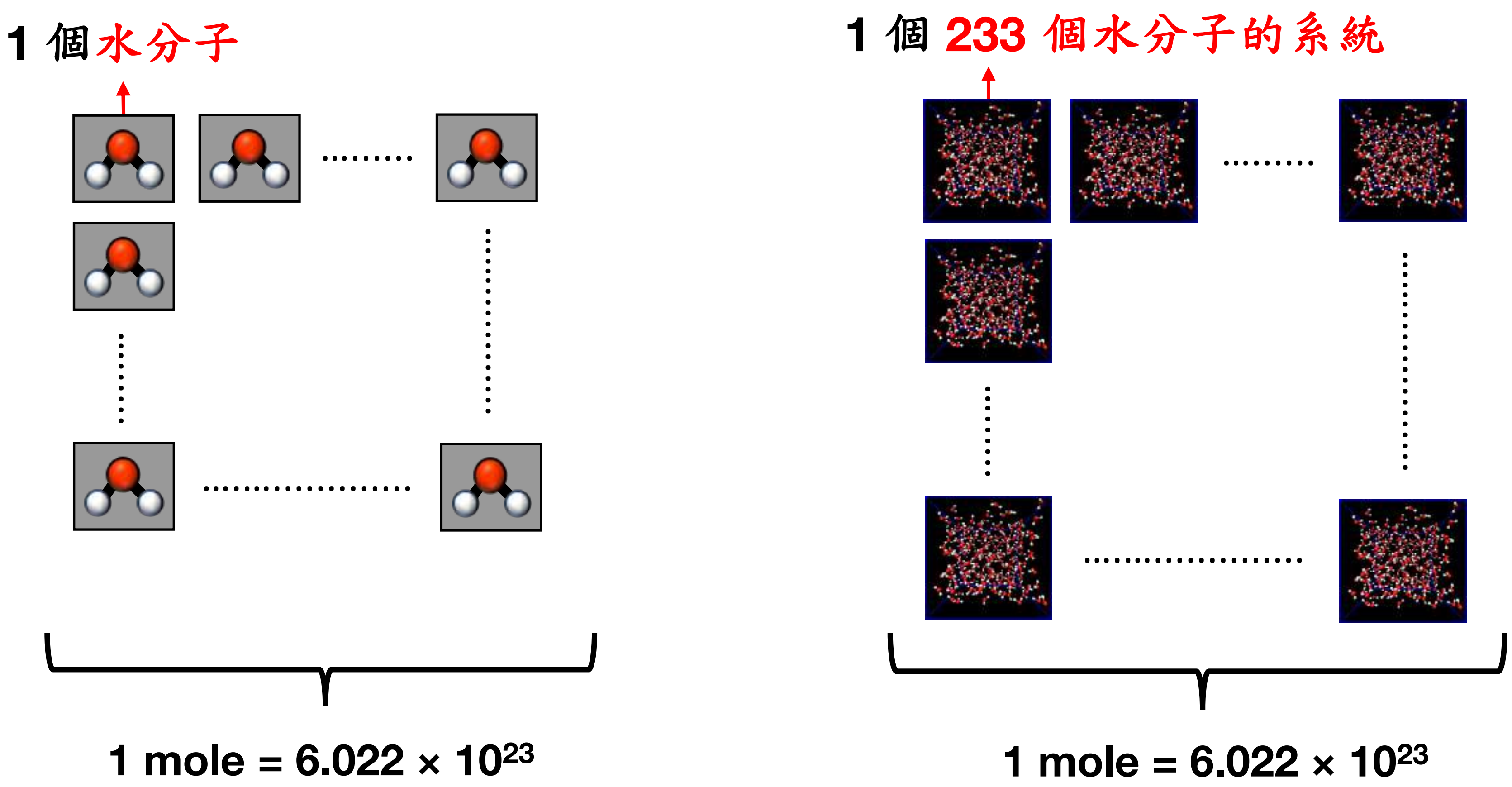



# Tutorial: 水盒模擬 (Water Box)




從水盒分子能量變化計算水的熱容量/比熱

# Tutorial: 水盒模擬 (Water Box)




Ex. 請注意，此計算出的數值是整個水盒(233 個水分子)的總熱容量，為了得到「單一水分子的熱容量」，必須用「總熱容量 / 233 」，這樣結果才能代表「單一水分子」的特性。

# Tutorial: 水盒模擬 (Water Box)

**請試著思考為什麼實際模擬計算出的單一水分子熱容量結果不等於 17 (cal/mol·K)?**

對於模擬，計算結果會與實驗結果有誤差，且由於每次模擬的獨立性，每次模擬的熱容量計算誤差會有所不同。

可能造成模擬誤差的原因有:

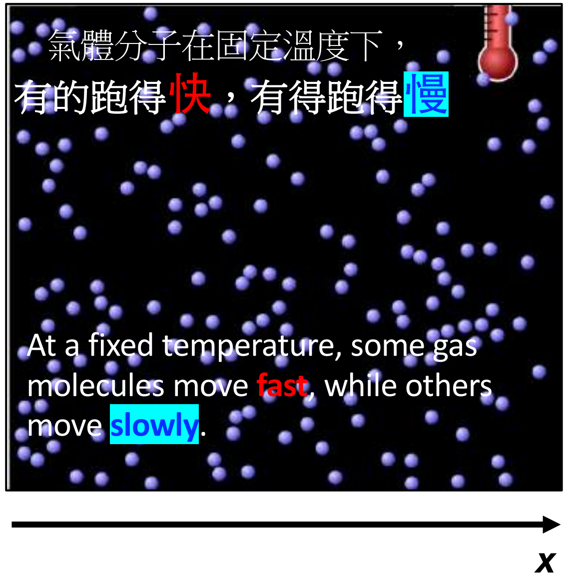
- 水模型本來就不完美: 模擬裡的水是用「簡化模型」表達的，不是真正的水，所以本來就會有誤差。
- 模擬的水太少: 由於系統粒子數有限，能量波動的統計不充分，導致熱容量計算存在有限尺寸效應與統計誤差，無法完全反映宏觀系統的真實行為。
- 模擬時間不夠長: 熱容量是靠能量的波動計算的，但模擬時間不夠長，這些波動還未收斂，就會算不準。

**因此若想計算出更準確的熱容量值，可以試著增加模擬步數使模擬系統穩定，並選取足夠收斂的能量部分再進行熱容量計算!**

# [補充] 統計熱力學

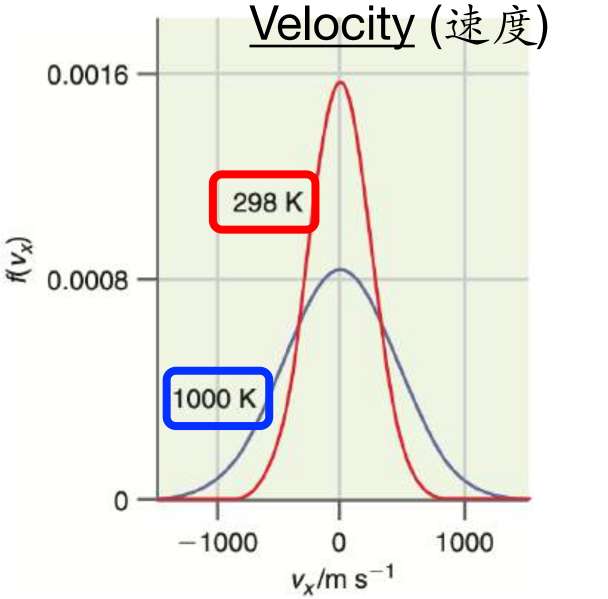
## Distribution function (分布函數)

**Ludwig E. Boltzmann**
(1844-1906) from wiki



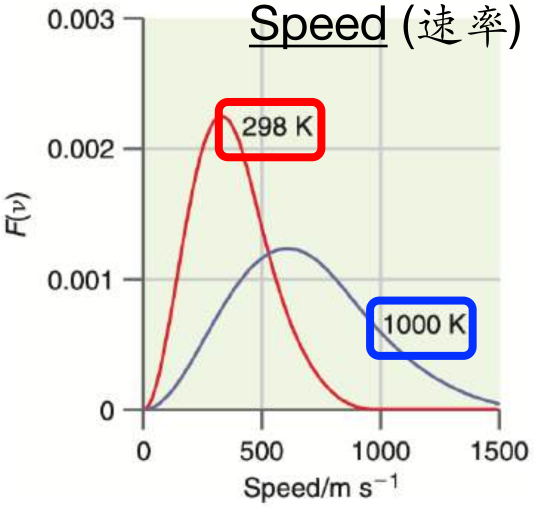

https://phet.colorado.edu/sims/html/gases-intro/latest/gases-intro_en.html

Velocity (速度)




$$f(v_j) = \left(\frac{M}{2\pi RT}\right)^{1/2} e^{(-Mv_j^2/2RT)}$$

Speed (速率)



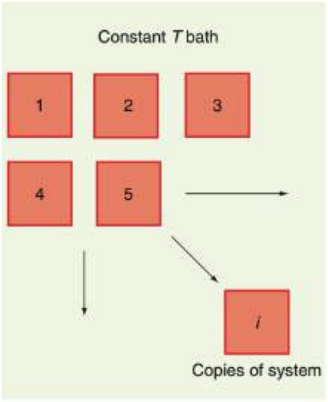

$$\boxed{F(v)\,dv = 4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} v^2 e^{-Mv^2/2RT}\,dv}$$

巨觀
**(macro)**

微觀(micro)

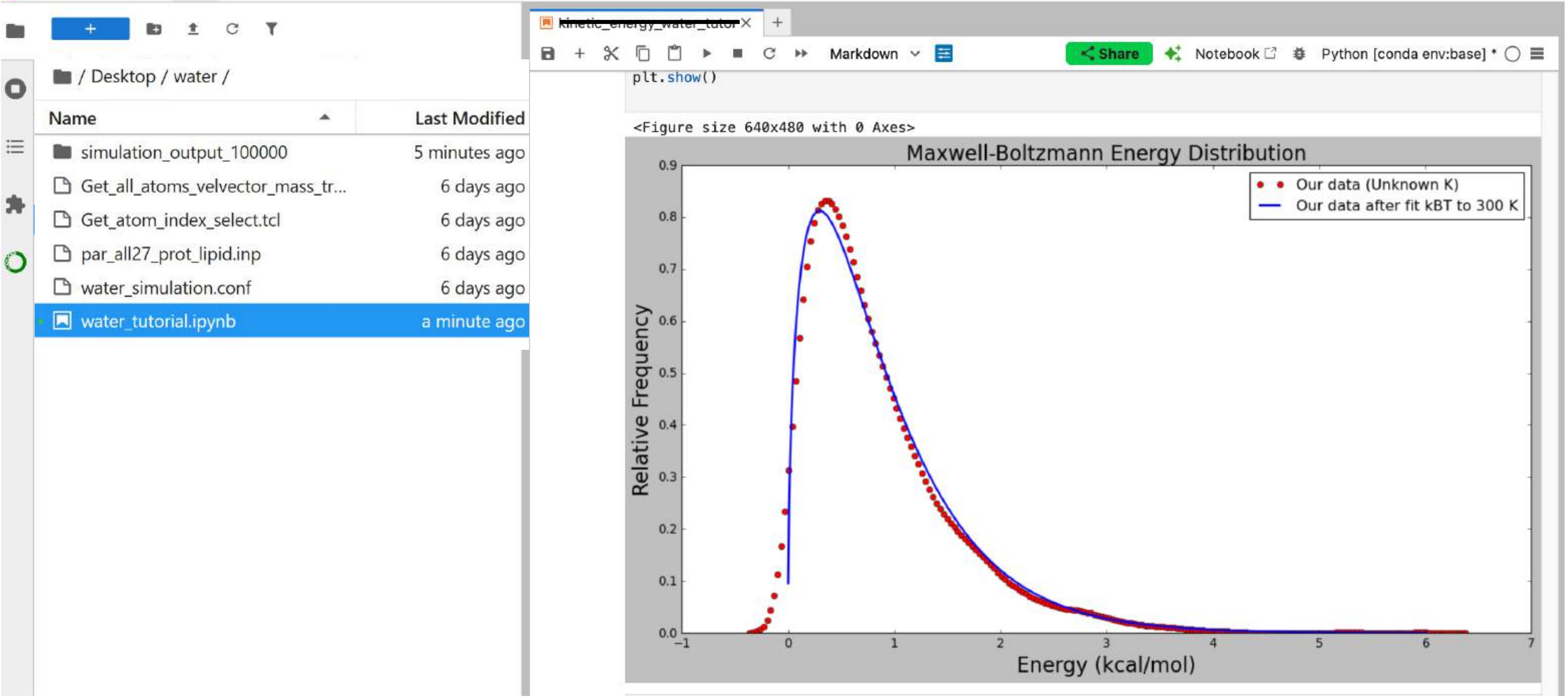
$$\boxed{Q = \sum_n e^{-\beta E_n}}$$

**Canonical partition function**
（正則配分函數）



# Tutorial: 數據分析—提取分子動能與能量分布

water_tutorial.ipynb



# 模擬結果動能計算流程

## VMD 內操作
(擷取原始模擬數據)

Get_all_atoms_velvector_mass_trajectory.tcl

計算系統個別原子速度向量和質量，產生:

→ my_atoms_velocity_vector_.txt

→ my_atoms_mass_.txt

Get_atom_index_select.tcl

計算選取範圍原子的 index，產生:

→ my_atom_indexs_select.txt

## JupyterLab 內操作
(原始模擬數據整理)

B.2.(b1) 讀取 my_atoms_velocity_vector_.txt

整理檔案內原子的速度向量

→ 以 my_velocity_vector_all_F 列表暫存

B.2.(b2) 讀取 my_atoms_mass_.txt

整理檔案內原子的質量

→ 以 my_mass_all_F 列表暫存

B.2.(b3) 讀取 my_atom_indexs_select.txt

整理檔案內選取原子的 index

→ 以 my_select_atom_index_all_F 列表暫存

## JupyterLab 內操作
(參照選取原子 index，選出速度向量和質量)

B.2.(c) 參照 my_select_atom_index_all_F

選取出列表 my_velocity_vector_all_F

對應的速度向量

→ 以 select_velocity_vectors 列表暫存

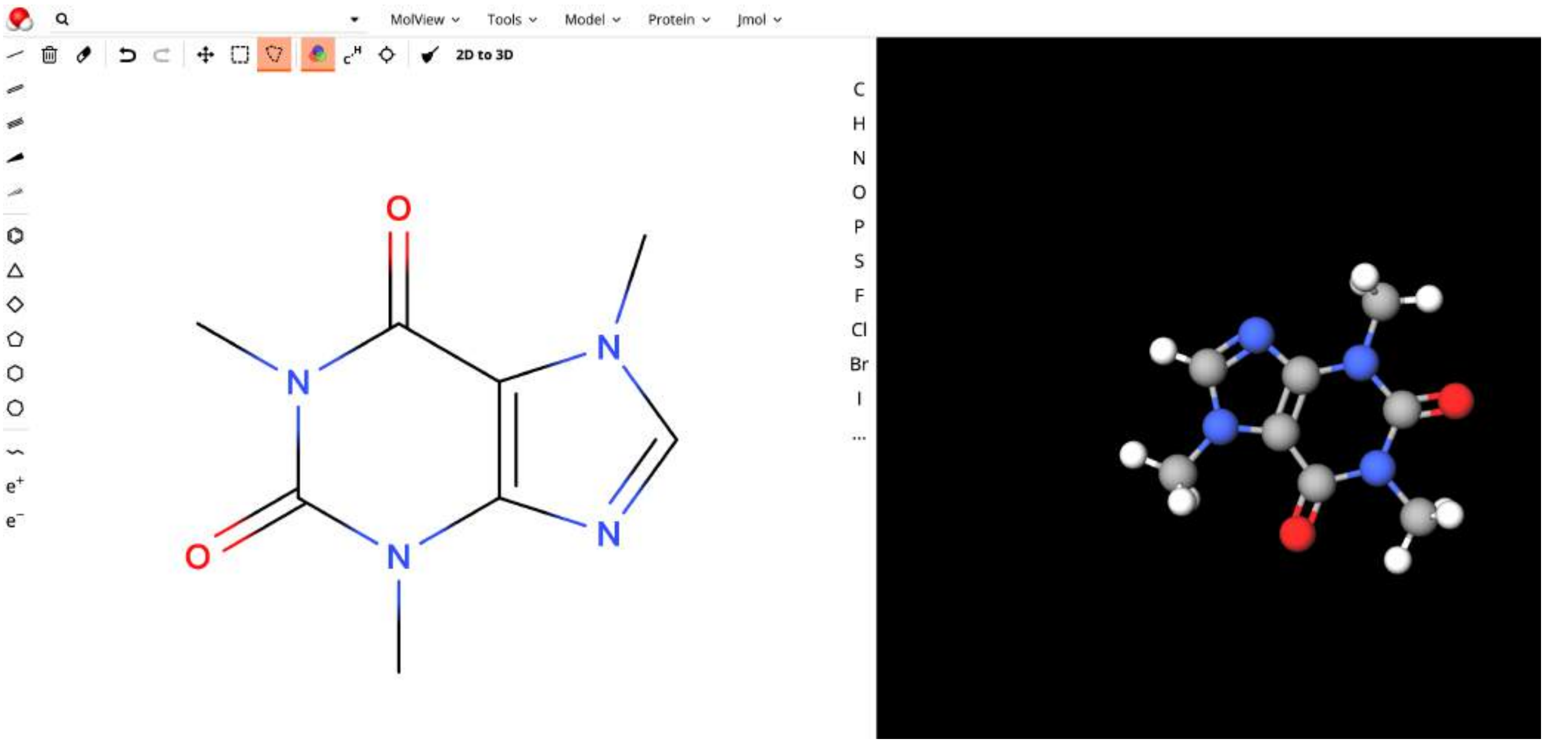
選取出列表 my_mass_all_F 對應的質量

→ 以 select_mass 列表暫存

## JupyterLab 內操作

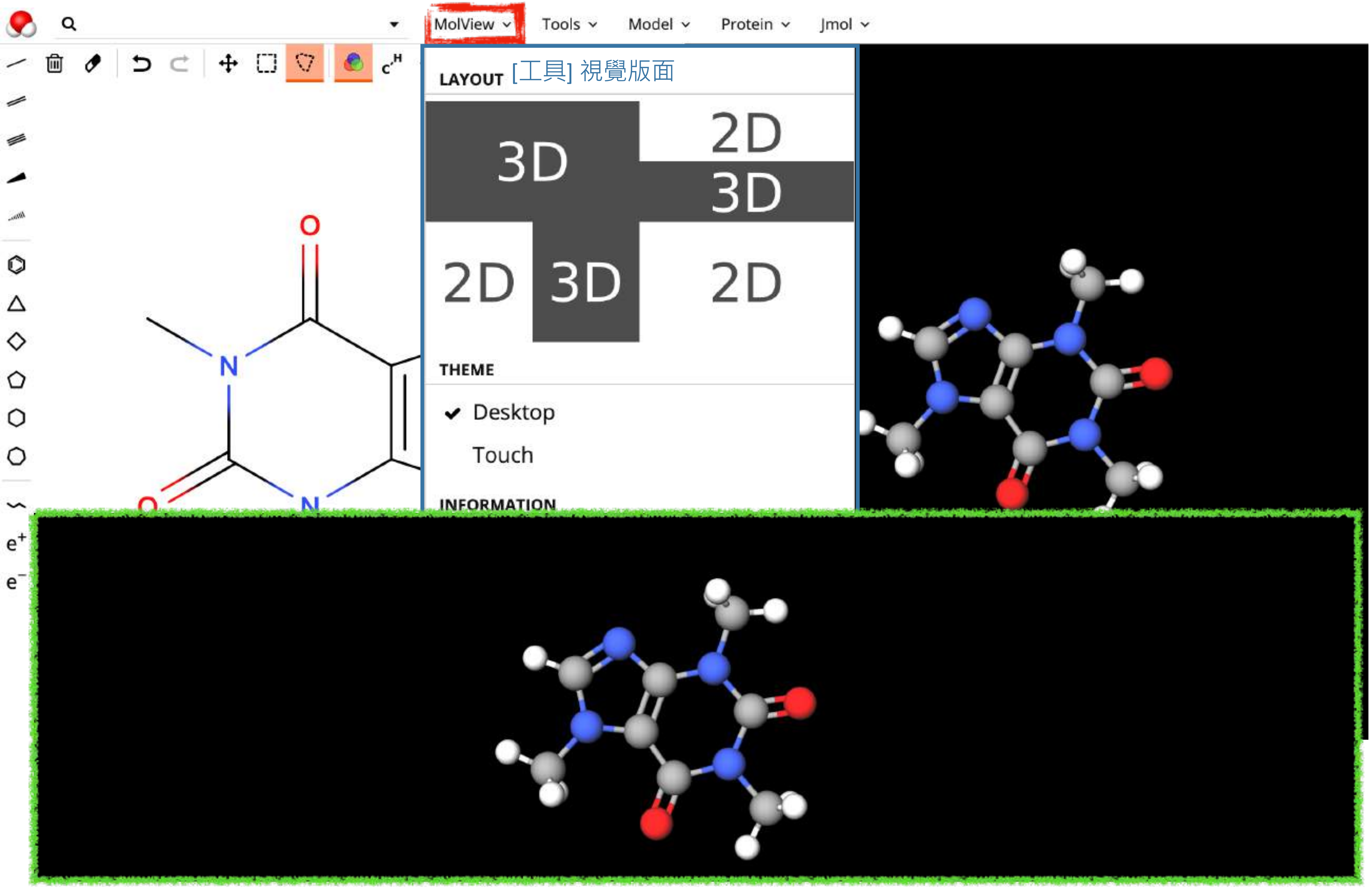
B.2.(d1~d4) 使用列表 select_velocity_vectors、select_mass 中速度以及原子質量資訊計算動能

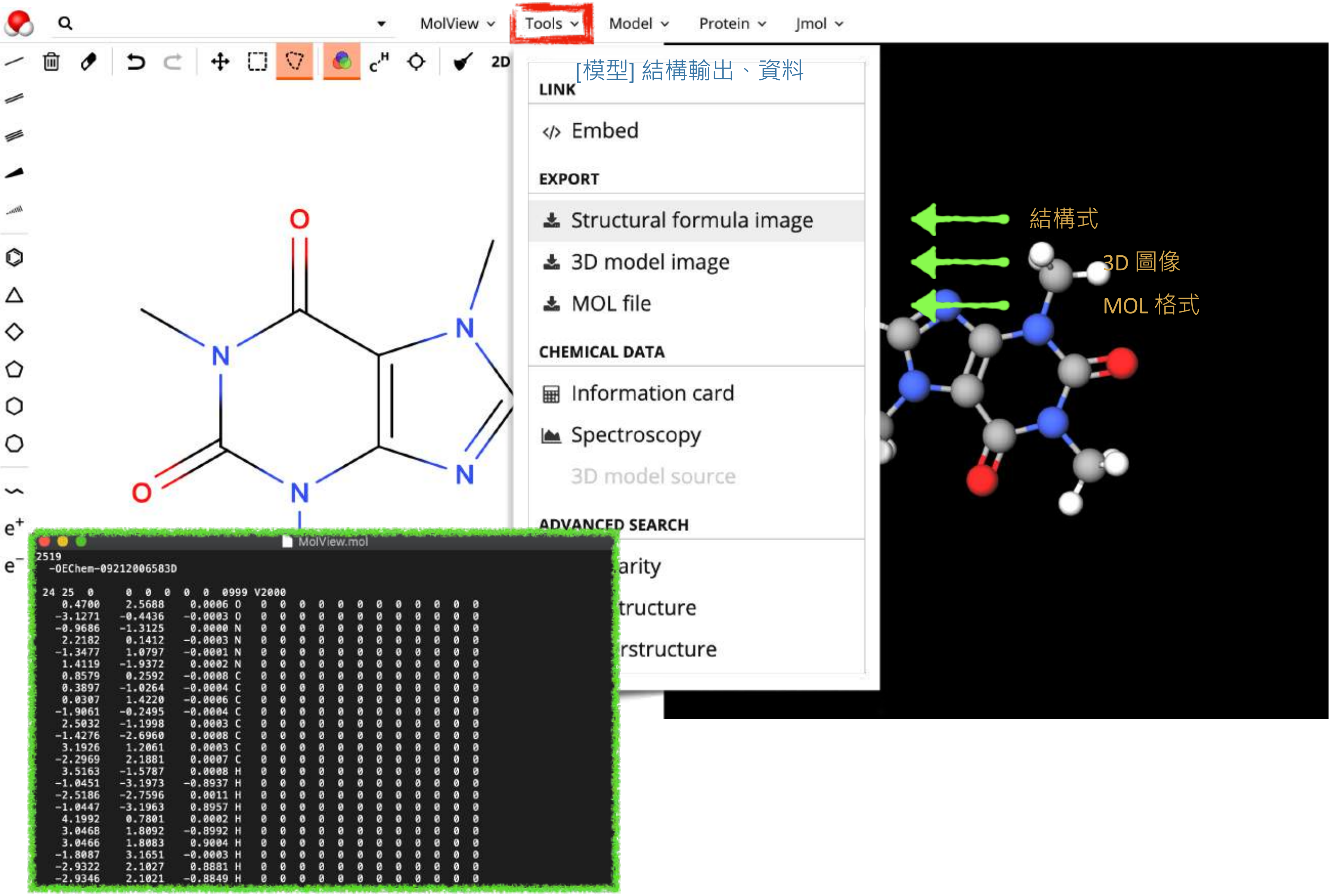
# MolView




https://molview.org/

- Video tutorial:
  https://www.youtube.com/watch?v=ZesFyuFj30s
  https://www.youtube.com/watch?v=af6PAAfVpRc


MolView
Tools
Model
Protein
Jmol
LAYOUT
[工具] 視覺版面
3D
2D
3D
2D
3D
2D
THEME
Desktop
Touch
INFORMATION


MolView
Tools
Model
Protein
Jmol
2D
[模型] 結構輸出、資料
LINK
Embed
EXPORT
Structural formula image
3D model image
MOL file
CHEMICAL DATA
Information card
Spectroscopy
3D model source
ADVANCED SEARCH
arity
tructure
rstructure
結構式
3D 圖像
MOL 格式
O
O
N
N
N
N
e+
e−
MolView.mol
2519
-OEChem-09212006583D
24 25 0 0 0 0 0 0 0999 V2000
0.4700 2.5688 0.0006 O 0 0 0 0 0 0 0 0 0 0 0 0
-3.1271 -0.4436 -0.0003 O 0 0 0 0 0 0 0 0 0 0 0 0
-0.9686 -1.3125 0.0000 N 0 0 0 0 0 0 0 0 0 0 0 0
2.2182 0.1412 -0.0003 N 0 0 0 0 0 0 0 0 0 0 0 0
-1.3477 1.0797 -0.0001 N 0 0 0 0 0 0 0 0 0 0 0 0
1.4119 -1.9372 0.0002 N 0 0 0 0 0 0 0 0 0 0 0 0
0.8579 0.2592 -0.0008 C 0 0 0 0 0 0 0 0 0 0 0 0
0.3897 -1.0264 -0.0004 C 0 0 0 0 0 0 0 0 0 0 0 0
0.0307 1.4220 -0.0006 C 0 0 0 0 0 0 0 0 0 0 0 0
-1.9061 -0.2495 -0.0004 C 0 0 0 0 0 0 0 0 0 0 0 0
2.5032 -1.1998 0.0003 C 0 0 0 0 0 0 0 0 0 0 0 0
-1.4276 -2.6960 0.0008 C 0 0 0 0 0 0 0 0 0 0 0 0
3.1926 1.2061 0.0003 C 0 0 0 0 0 0 0 0 0 0 0 0
-2.2969 2.1881 0.0007 C 0 0 0 0 0 0 0 0 0 0 0 0
3.5163 -1.5787 0.0008 H 0 0 0 0 0 0 0 0 0 0 0 0
-1.0451 -3.1973 -0.8937 H 0 0 0 0 0 0 0 0 0 0 0 0
-2.5186 -2.7596 0.0011 H 0 0 0 0 0 0 0 0 0 0 0 0
-1.0447 -3.1963 0.8957 H 0 0 0 0 0 0 0 0 0 0 0 0
4.1992 0.7801 0.0002 H 0 0 0 0 0 0 0 0 0 0 0 0
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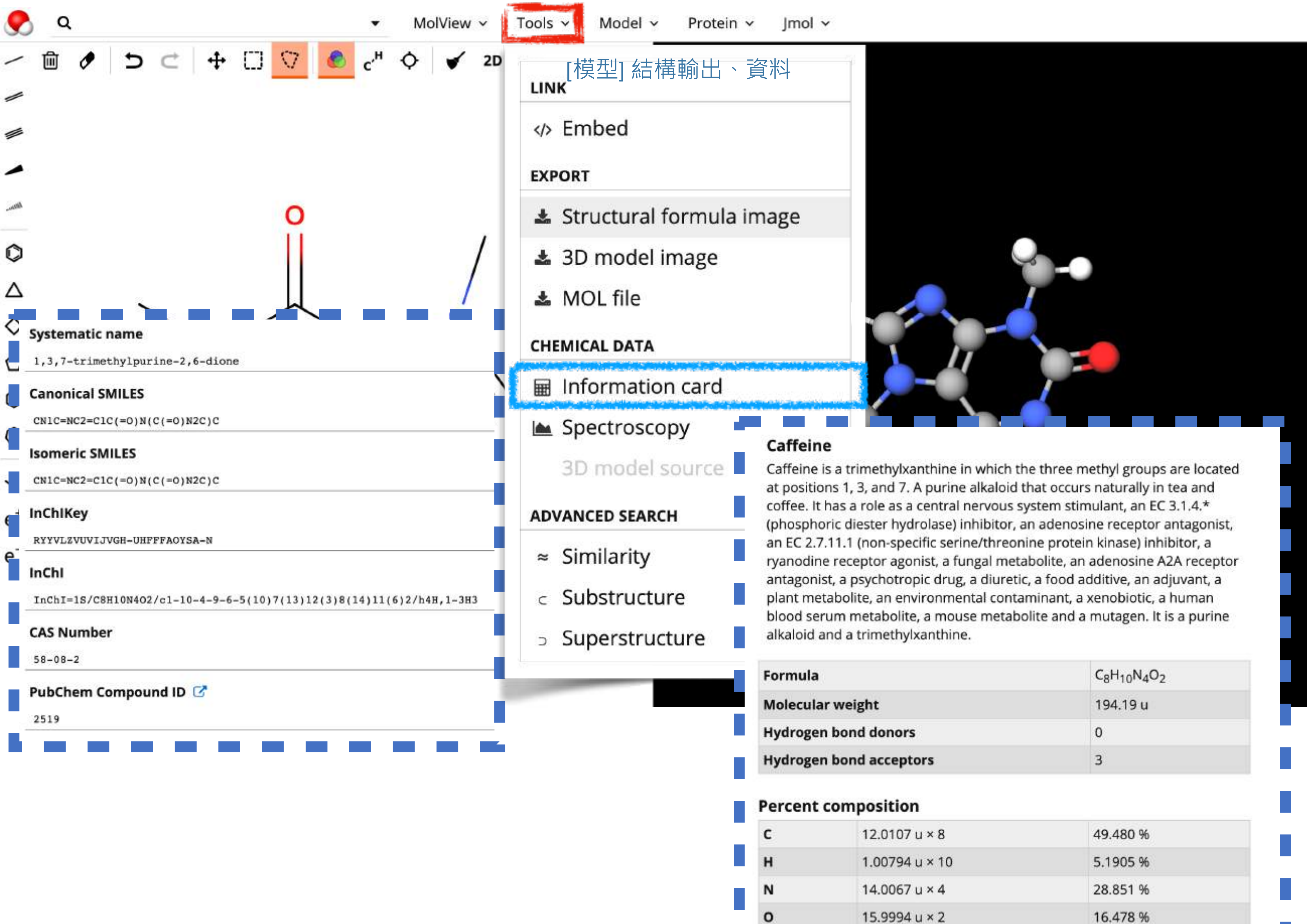
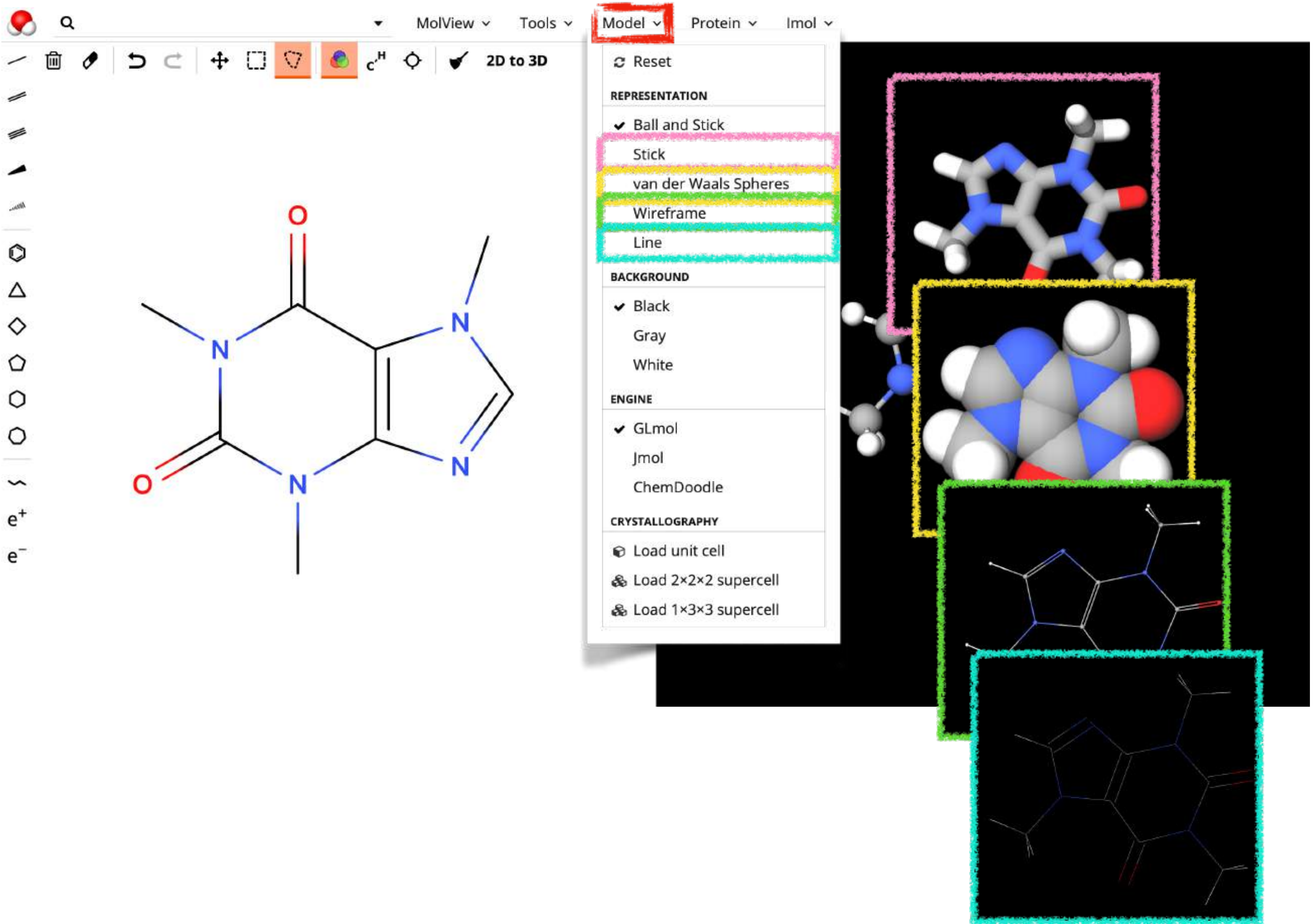
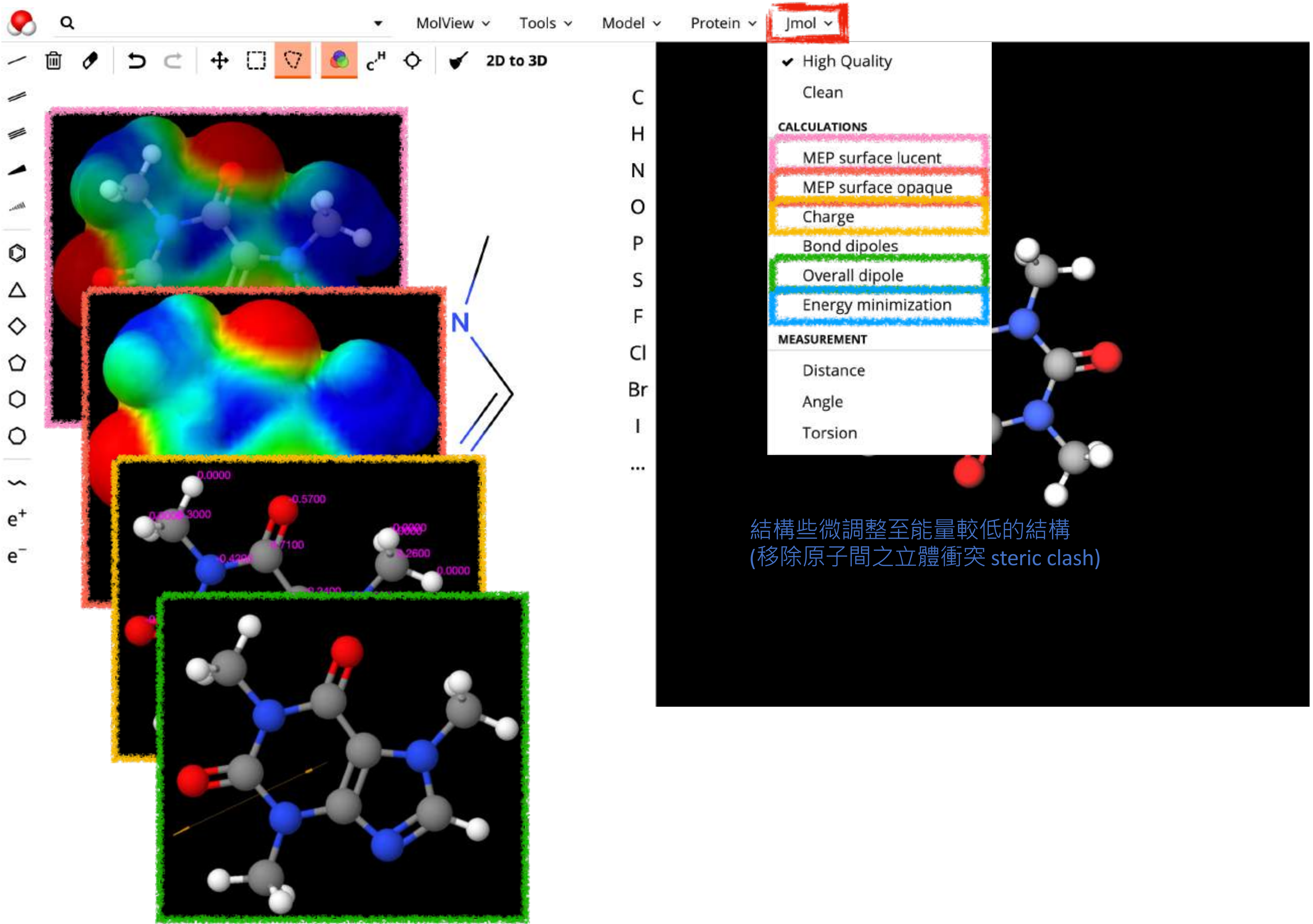
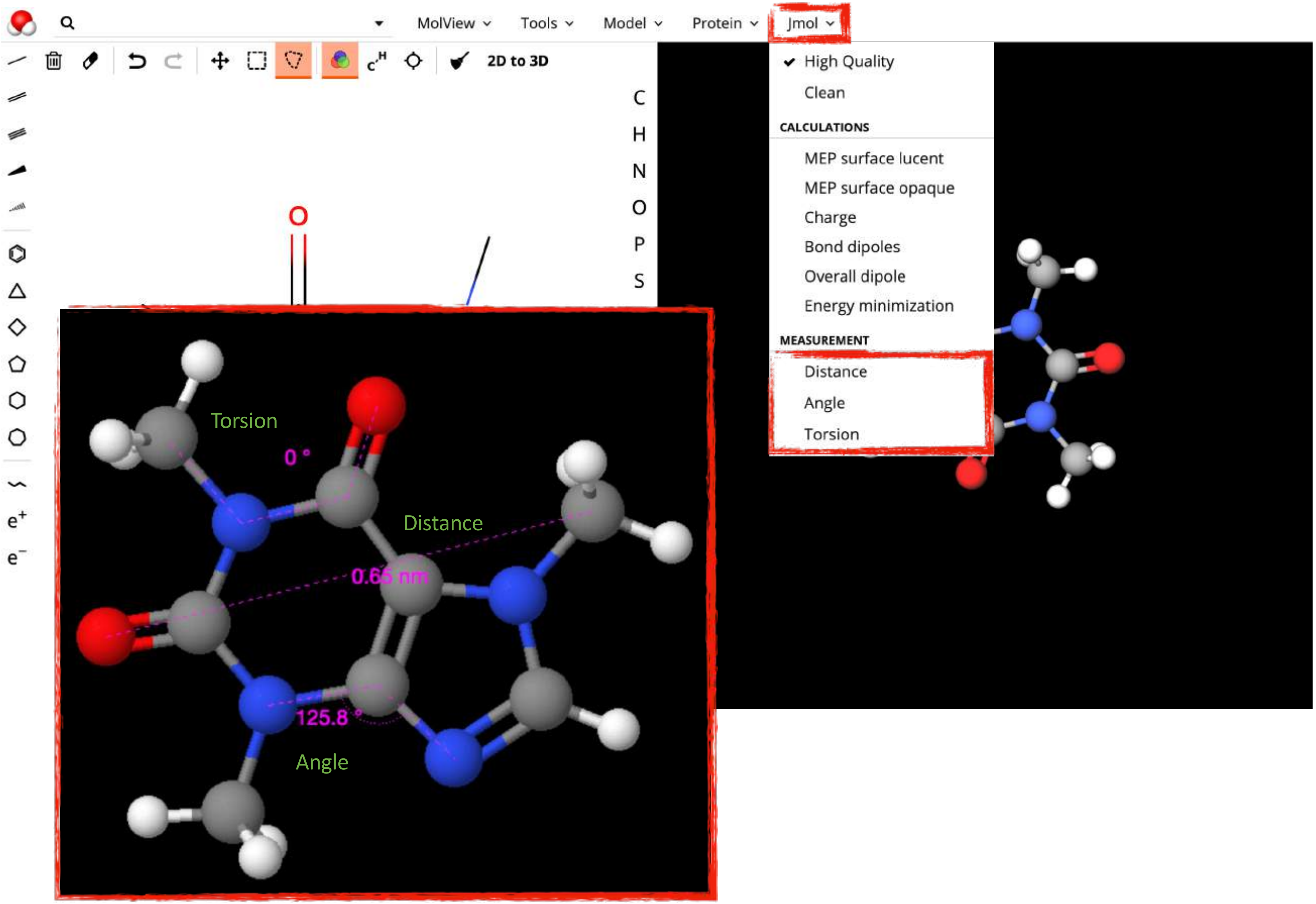


MolView
Tools
Model
Protein
Jmol
2D
[模型] 結構輸出、資料
LINK
Embed
EXPORT
Structural formula image
3D model image
MOL file
CHEMICAL DATA
Information card
Spectroscopy
3D model source
ADVANCED SEARCH
Similarity
Substructure
Superstructure
O
Systematic name
1,3,7-trimethylpurine-2,6-dione
Canonical SMILES
CN1C=NC2=C1C(=O)N(C(=O)N2C)C
Isomeric SMILES
CN1C=NC2=C1C(=O)N(C(=O)N2C)C
InChIKey
RYYVLZVUVIJVGH-UHFFFAOYSA-N
InChI
InChI=1S/C8H10N4O2/c1-10-4-9-6-5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3
CAS Number
58-08-2
PubChem Compound ID
2519
Caffeine
Caffeine is a trimethylxanthine in which the three methyl groups are located at positions 1, 3, and 7. A purine alkaloid that occurs naturally in tea and coffee. It has a role as a central nervous system stimulant, an EC 3.1.4.* (phosphoric diester hydrolase) inhibitor, an adenosine receptor antagonist, an EC 2.7.11.1 (non-specific serine/threonine protein kinase) inhibitor, a ryanodine receptor agonist, a fungal metabolite, an adenosine A2A receptor antagonist, a psychotropic drug, a diuretic, a food additive, an adjuvant, a plant metabolite, an environmental contaminant, a xenobiotic, a human blood serum metabolite, a mouse metabolite and a mutagen. It is a purine alkaloid and a trimethylxanthine.
Formula $C_8H_{10}N_4O_2$
Molecular weight 194.19 u
Hydrogen bond donors 0
Hydrogen bond acceptors 3
Percent composition
C 12.0107 u × 8 49.480 %
H 1.00794 u × 10 5.1905 %
N 14.0067 u × 4 28.851 %
O 15.9994 u × 2 16.478 %

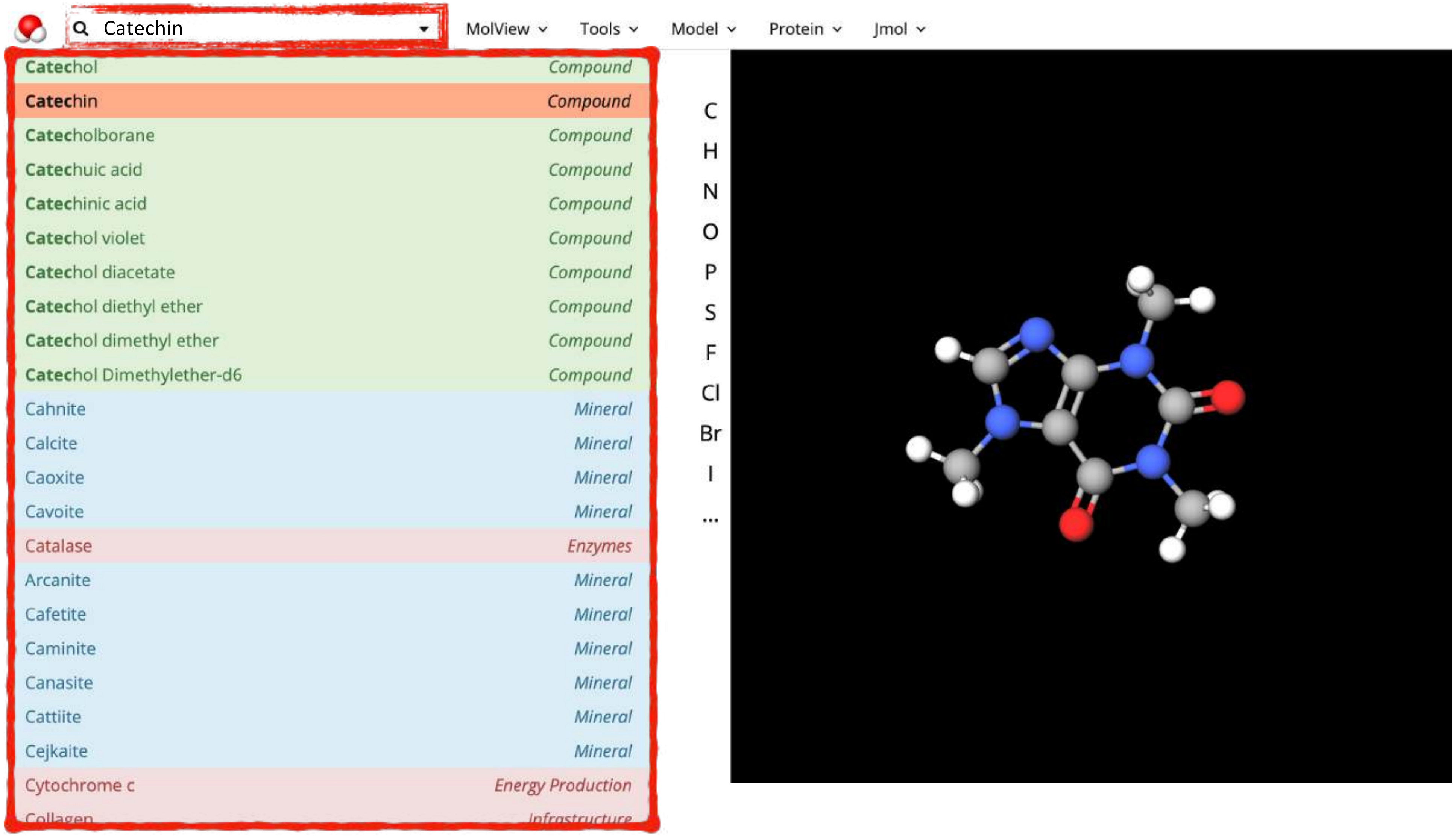


MolView
Tools
Model
Protein
Jmol
2D to 3D
Reset
REPRESENTATION
Ball and Stick
Stick
van der Waals Spheres
Wireframe
Line
BACKGROUND
Black
Gray
White
ENGINE
GLmol
Jmol
ChemDoodle
CRYSTALLOGRAPHY
Load unit cell
Load 2×2×2 supercell
Load 1×3×3 supercell
O
N
N
N
N
O


MolView
Tools
Model
Protein
Jmol
2D to 3D
High Quality
Clean
CALCULATIONS
MEP surface lucent
MEP surface opaque
Charge
Bond dipoles
Overall dipole
Energy minimization
MEASUREMENT
Distance
Angle
Torsion
結構些微調整至能量較低的結構
(移除原子間之立體衝突 steric clash)



MolView
Tools
Model
Protein
Jmol
2D to 3D
C
H
N
O
P
S
High Quality
Clean
CALCULATIONS
MEP surface lucent
MEP surface opaque
Charge
Bond dipoles
Overall dipole
Energy minimization
MEASUREMENT
Distance
Angle
Torsion
Torsion
0 °
Distance
0.65 nm
125.8 °
Angle


Catechin
MolView
Tools
Model
Protein
Jmol
Catechol Compound
Catechin Compound
Catecholborane Compound
Catechuic acid Compound
Catechinic acid Compound
Catechol violet Compound
Catechol diacetate Compound
Catechol diethyl ether Compound
Catechol dimethyl ether Compound
Catechol Dimethylether-d6 Compound
Cahnite Mineral
Calcite Mineral
Caoxite Mineral
Cavoite Mineral
Catalase Enzymes
Arcanite Mineral
Cafetite Mineral
Caminite Mineral
Canasite Mineral
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Cejkaite Mineral
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Cl
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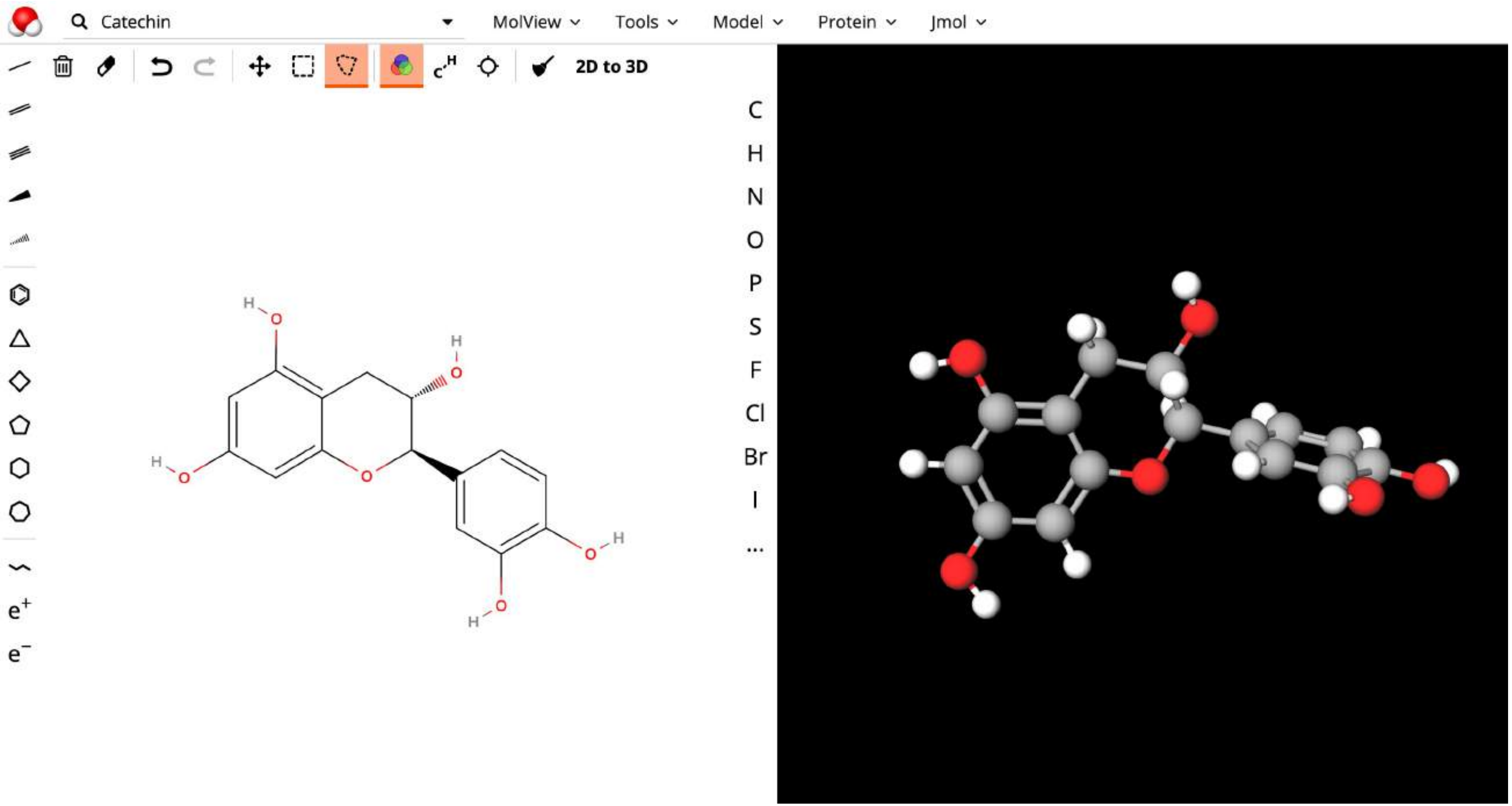
I
...

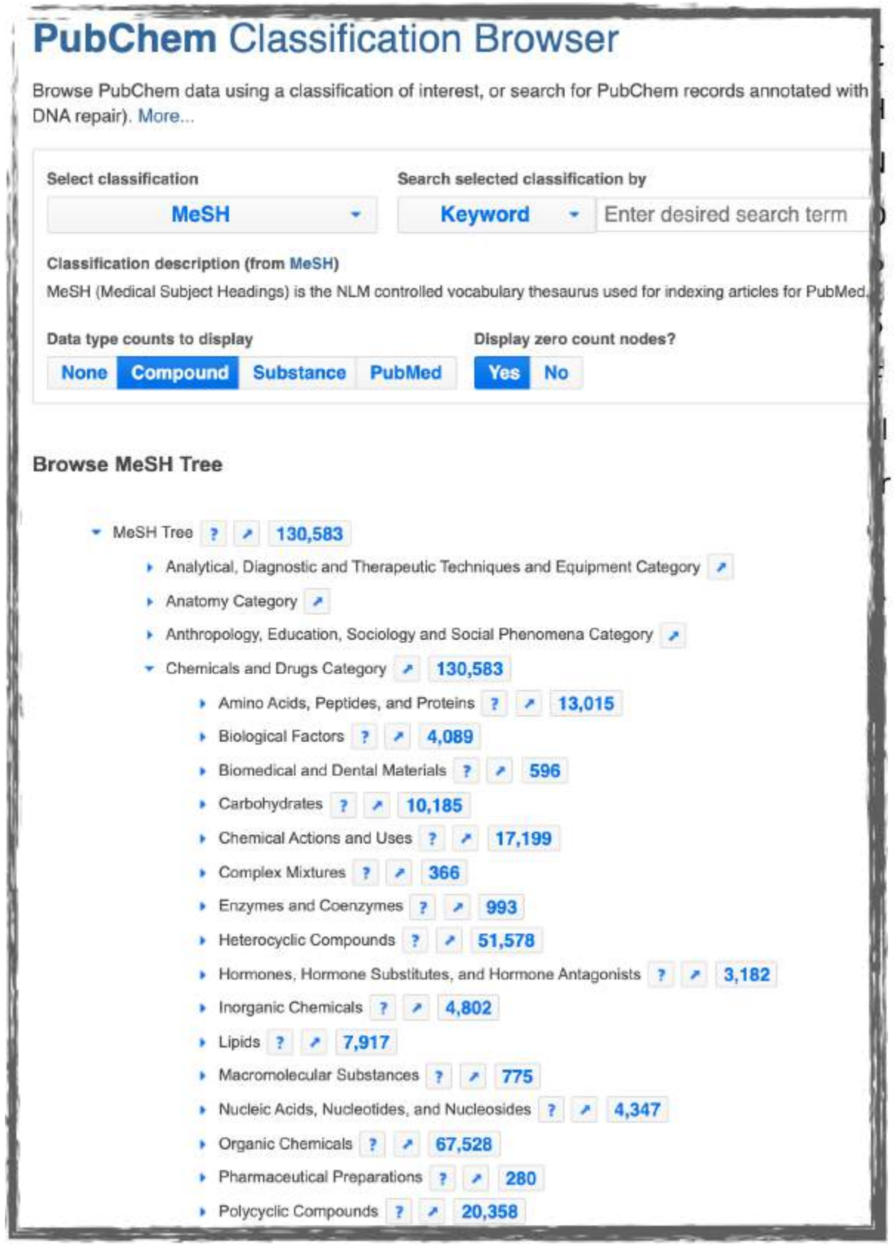

Catechin
MolView
Tools
Model
Protein
Jmol
2D to 3D
C
H
N
O
P
S
F
Cl
Br
I
...
e+
e−

# Your Turn!

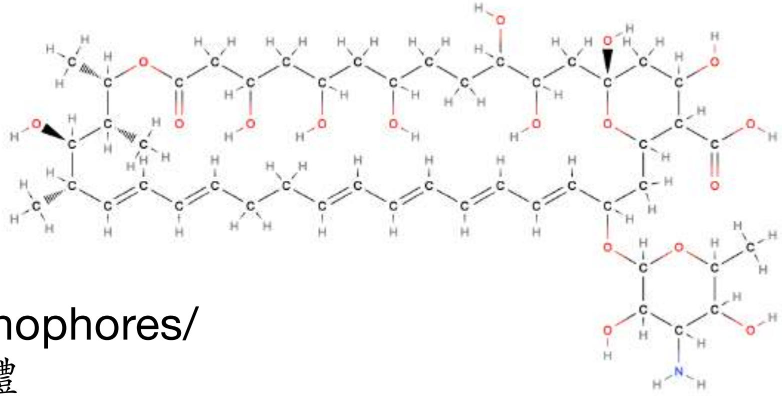
找尋日常生活會用到的小分子藥物



- 搜尋PubChem compounds
- 名稱/分類/介紹
- 結構式 & 3D影像下載
- MOL格式
- 量測分子大小長度（近似即可）截圖或下載
- 將分子用surface表示後截圖或下載



# Nystatine as an example



- 名稱/分類/介紹
  Nystatine/Antibiotics, antifungal, Macrolide,Ionophores/
  抗黴菌素/抗生素, 抗真菌, 大環內酯類, 離子載體

  Macrolide antifungal antibiotic complex produced by Streptomyces noursei, S. aureus, and other Streptomyces species. The biologically active components of the complex are nystatin A1, A2, and A3.
  由諾氏鏈黴菌、金黃色葡萄球菌和其他鏈黴菌產生的大環內酯類抗真菌抗生素複合物。 該複合物的生物活性成分是製黴菌素 A1、A2 和 A3。

- 結構式 & 3D影像下載

- MOL格式

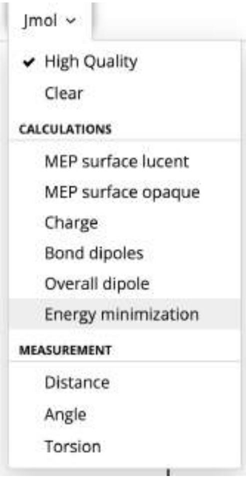
- 量測分子大小長度（近似即可）截圖或下載

- 將分子用surface表示後截圖或下載

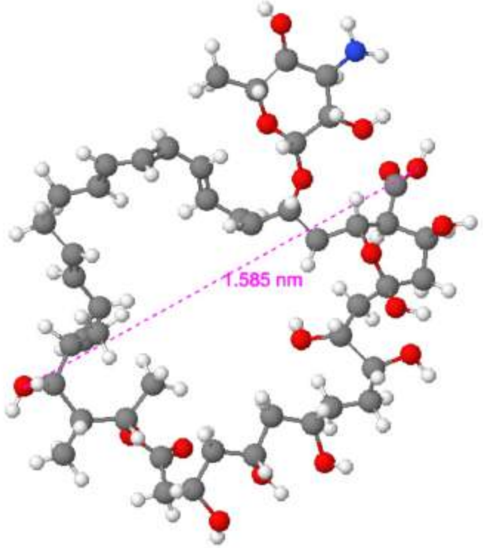
```
C47H75NO17
APtclcactv10122200512D 0   0.00000     0.00000

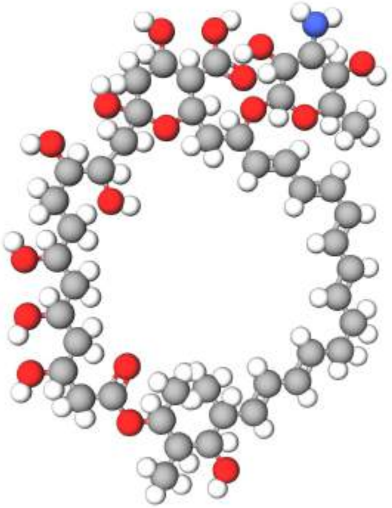
140142  0  0  1  0  0  0  0  0999 V2000
   12.3100    0.6569    0.0000 O   0  0  0  0  0  0  0  0  0  0  0  0
   12.3100    1.6569    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
   12.1495    2.2558    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
   11.3100    1.6569    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
   10.7274    1.4449    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
   11.4177    1.0463    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
   10.8100    2.5230    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
   11.4300    2.5230    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
    9.8100    2.5230    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    9.5000    1.9860    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
    9.3100    3.3890    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    9.4177    3.9996    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
    8.3100    3.3890    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    8.0000    2.8520    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
    7.8100    4.2550    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    8.1200    4.7919    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
    6.8100    4.2550    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    6.5000    4.7919    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
    6.3100    3.3890    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    6.6200    2.8520    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
    5.3100    3.3890    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    5.0000    3.9259    0.0000 H   0  0  0  0  0  0  0  0  0  0  0  0
```

# Nystatine as an example

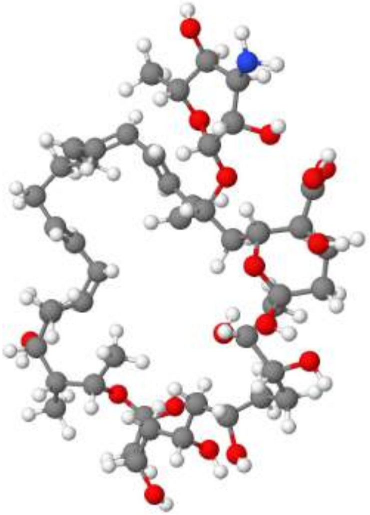


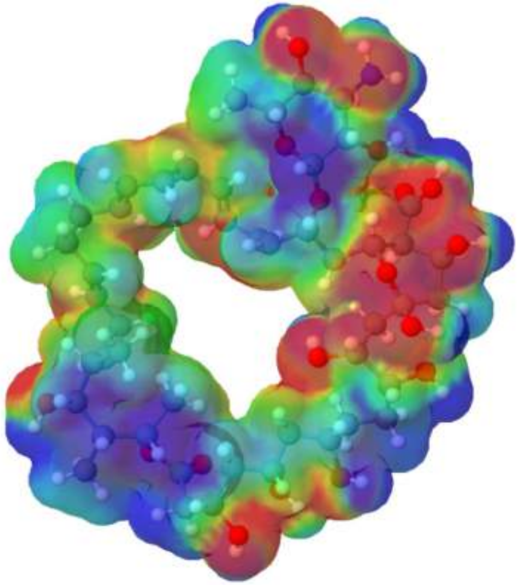

注意分子結構會動!



**Energy minimized**



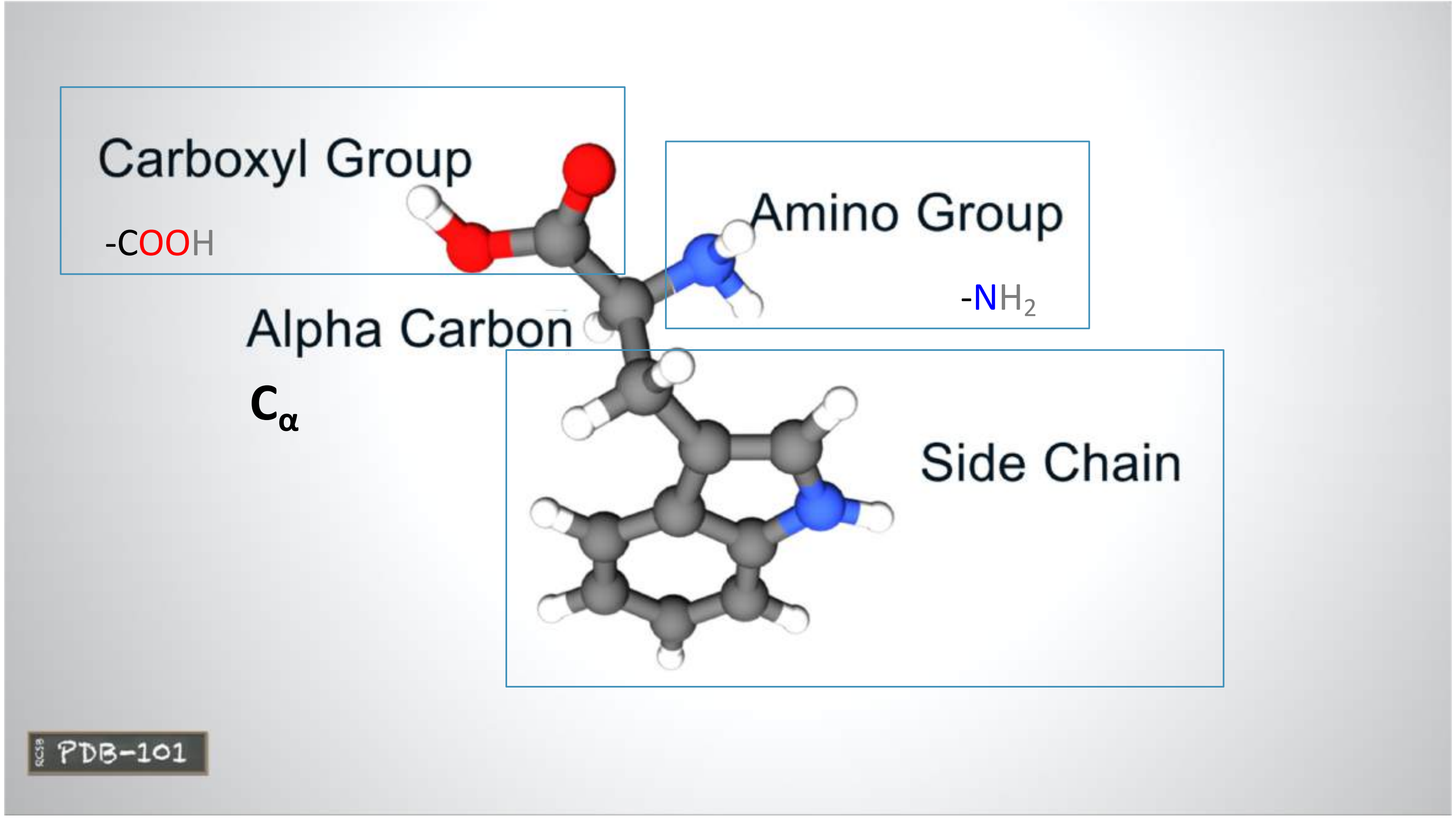


# 生活常見藥物

- Acetaminophen 乙醯胺酚(普拿疼有效成分)
- Amphetamine 安非他命
- Methamphetamine 冰毒 --> 絕命毒師主要製毒
- Remdesivir 瑞德西韋 --> 老藥新用來治療Covid-19
- Biopterin 生物喋呤 (link) —> 缺乏會導致苯丙酮尿症(罕見疾病)
- Cephalosporin 抗生素(link) —> https://www.nejs.app/2018/10/cephalosporins.html
- Aspirin 退燒
- Artemisinin 青蒿素 (屠呦呦發現，獲2015諾貝爾生理醫學獎)
- Barbital 巴比妥(安眠藥 link。)
- Bilirubin 膽紅素 link (血紅素的主要代謝產物；膽汁的主要色素；肝功能的重要指標；Can 'toxic' bilirubin treat a variety of illnesses? )
- Vancomycin (萬古黴素) --> 治療細菌感染
- Clindamycin (克林黴素) --> 治療細菌感染(痘痘、挫瘡)
- Tamsulosin (坦索羅辛) --> 受體拮抗劑 (調整泌尿到機能、使尿路平滑肌鬆弛，幫助排出腎結石、治療前列腺肥大症)
- Flopropione --> 調整胰膽道、泌尿道、機能。緩解消化管平滑肌、胰膽道及尿路系統平滑肌收縮之痙攣作用

# The Building Blocks of Proteins

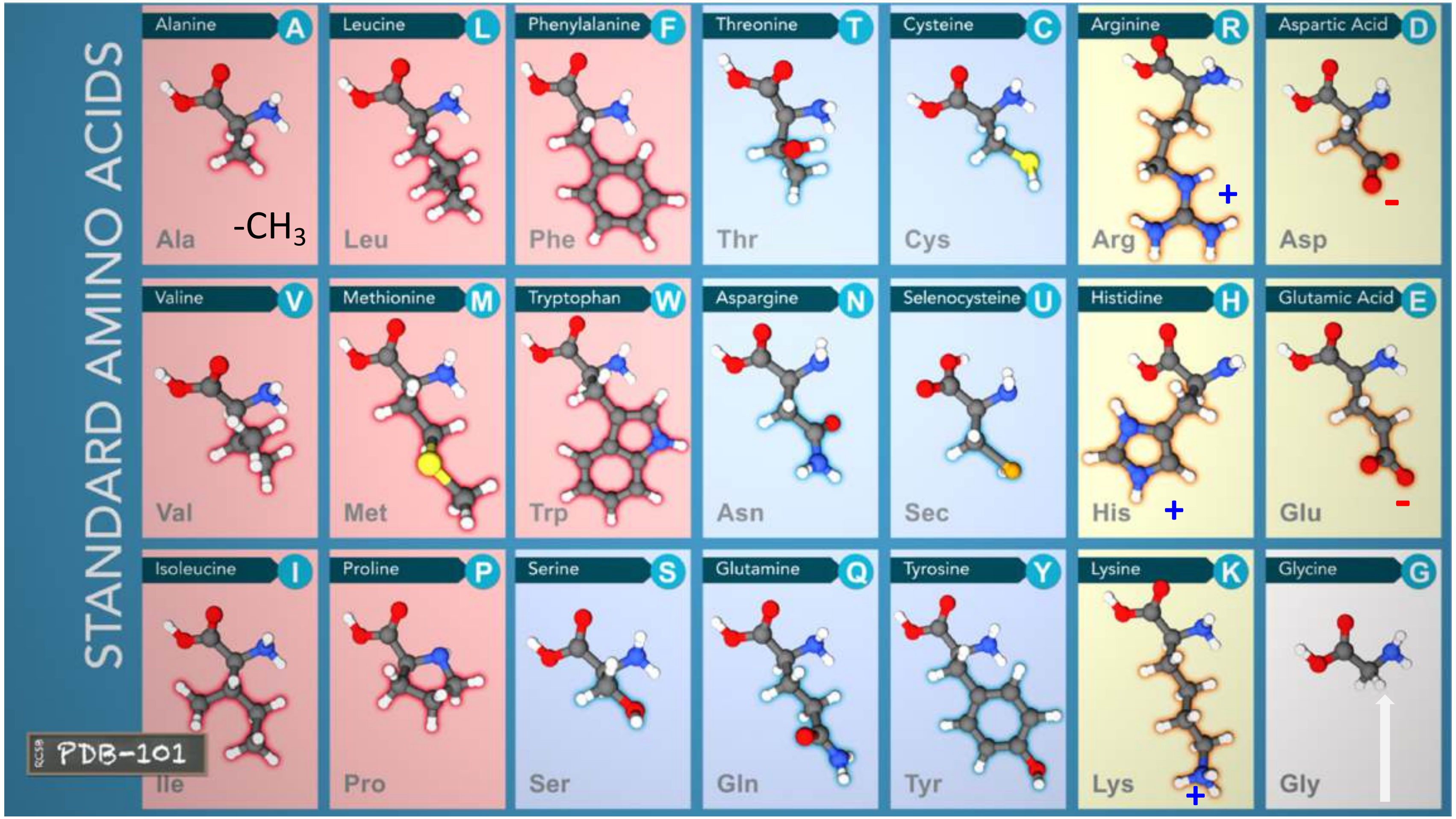





The atoms of each amino acid form an **amino group**, a **carboxyl group**, and a **side chain** attached to a central carbon atom.

# The Building Blocks of Proteins

pdb101.rcsb.org

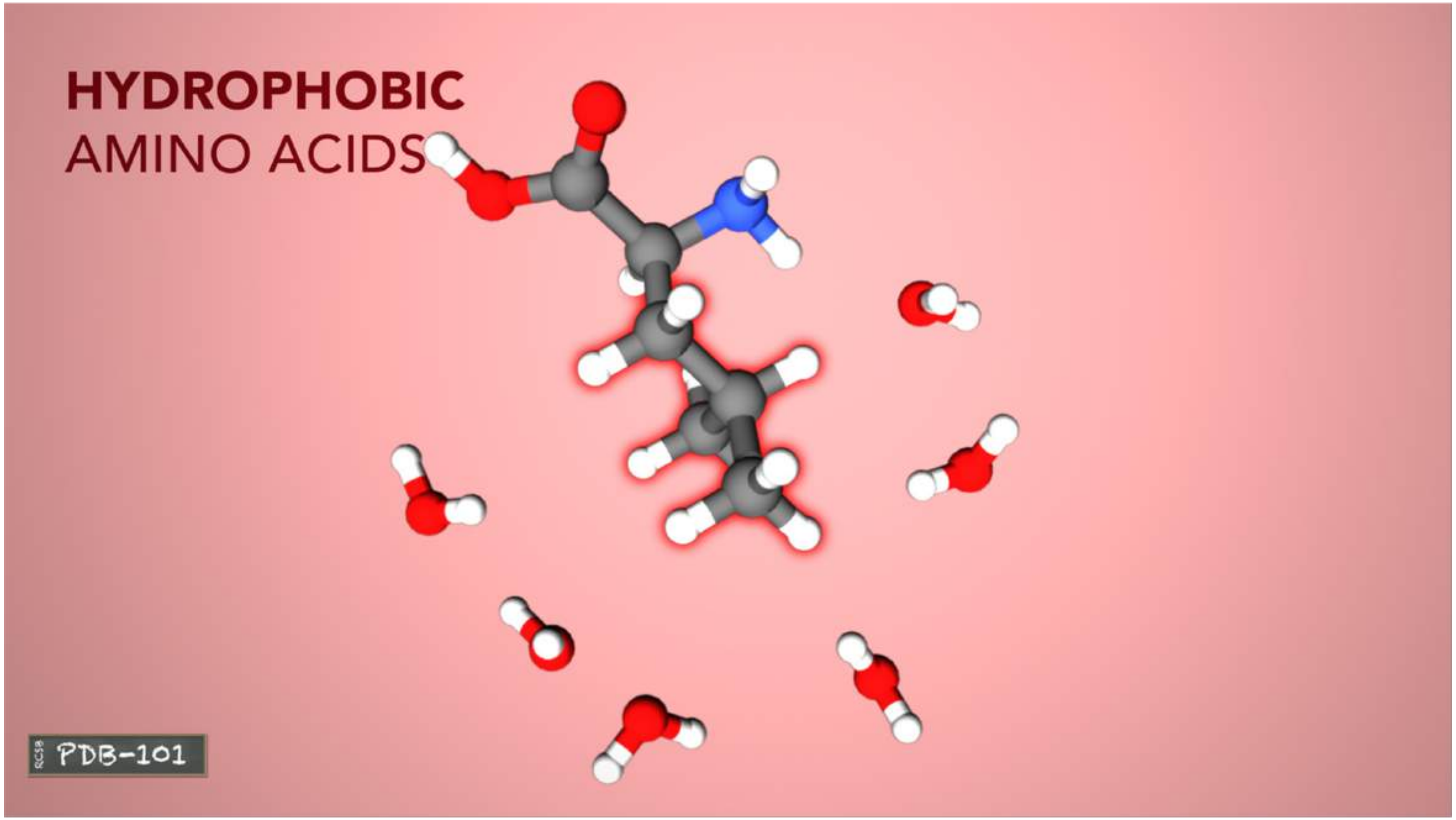



Hydrophobic | Hydrophilic or polar | Charged | No side chain

The side chain is the only part that varies from amino acid to amino acid and determines its chemical properties.

# The Building Blocks of Proteins

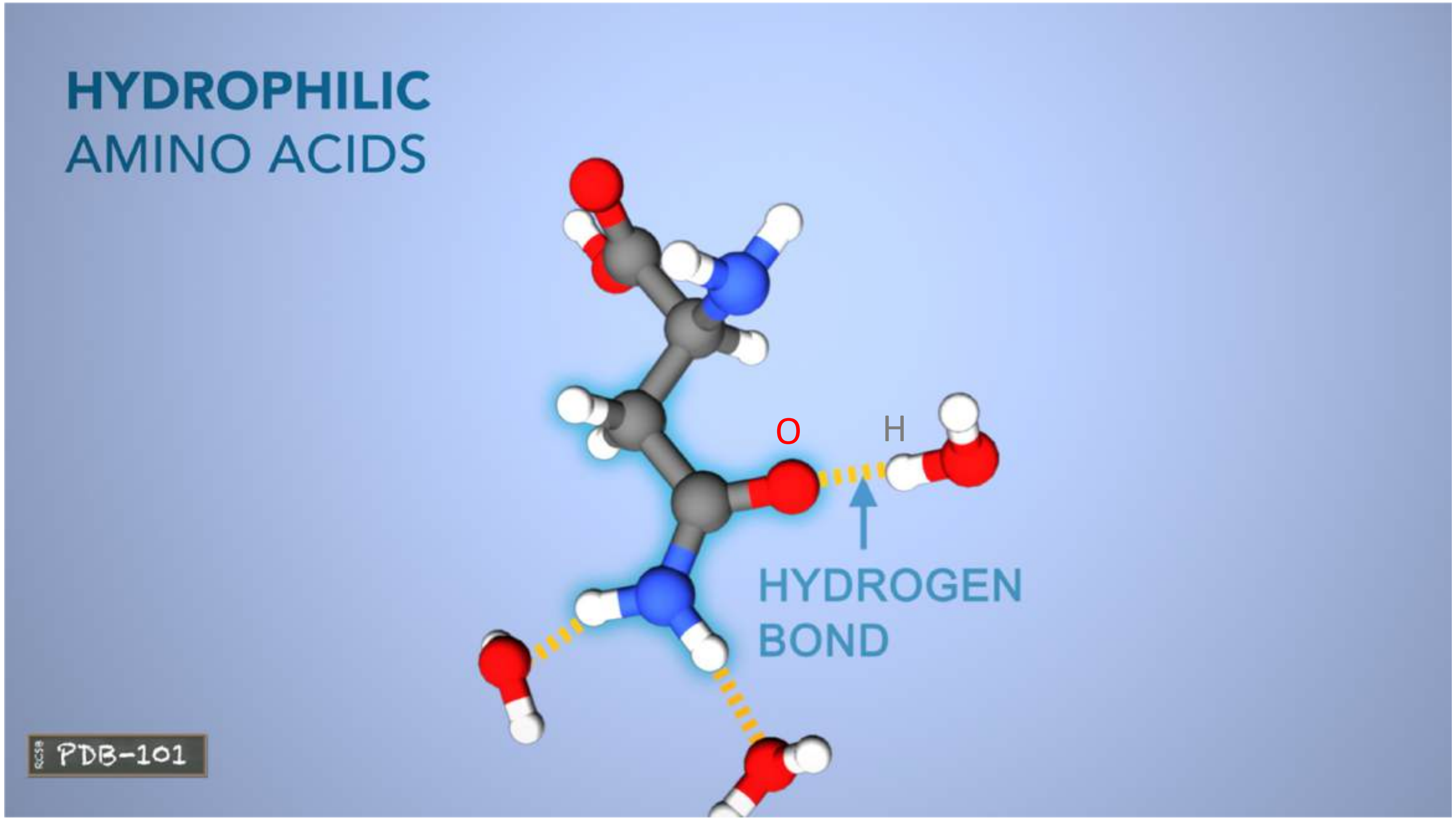
pdb101.rcsb.org




**Hydrophobic** amino acids have carbon-rich side chains, which don't interact well with water.

# The Building Blocks of Proteins

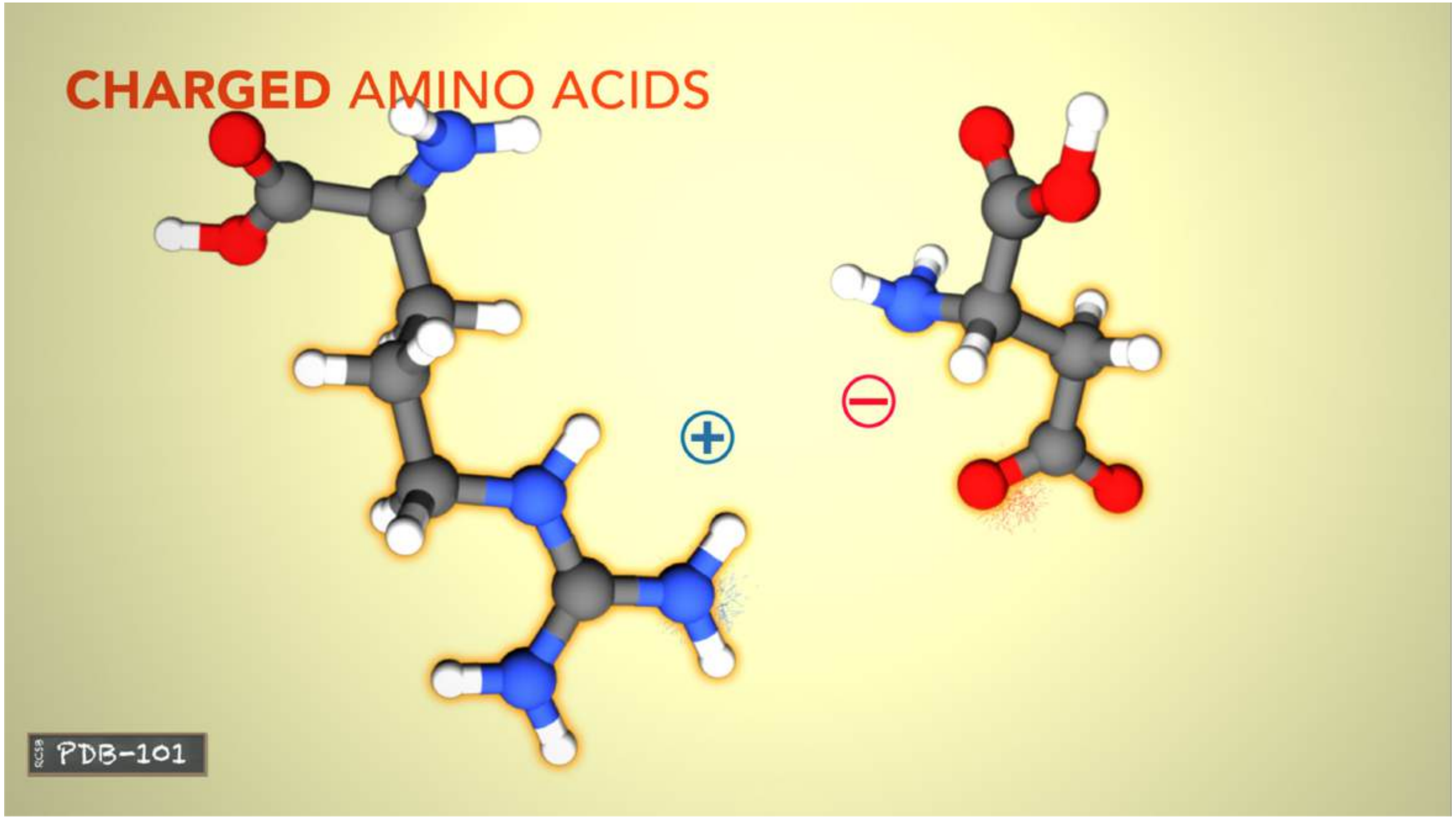
pdb101.rcsb.org



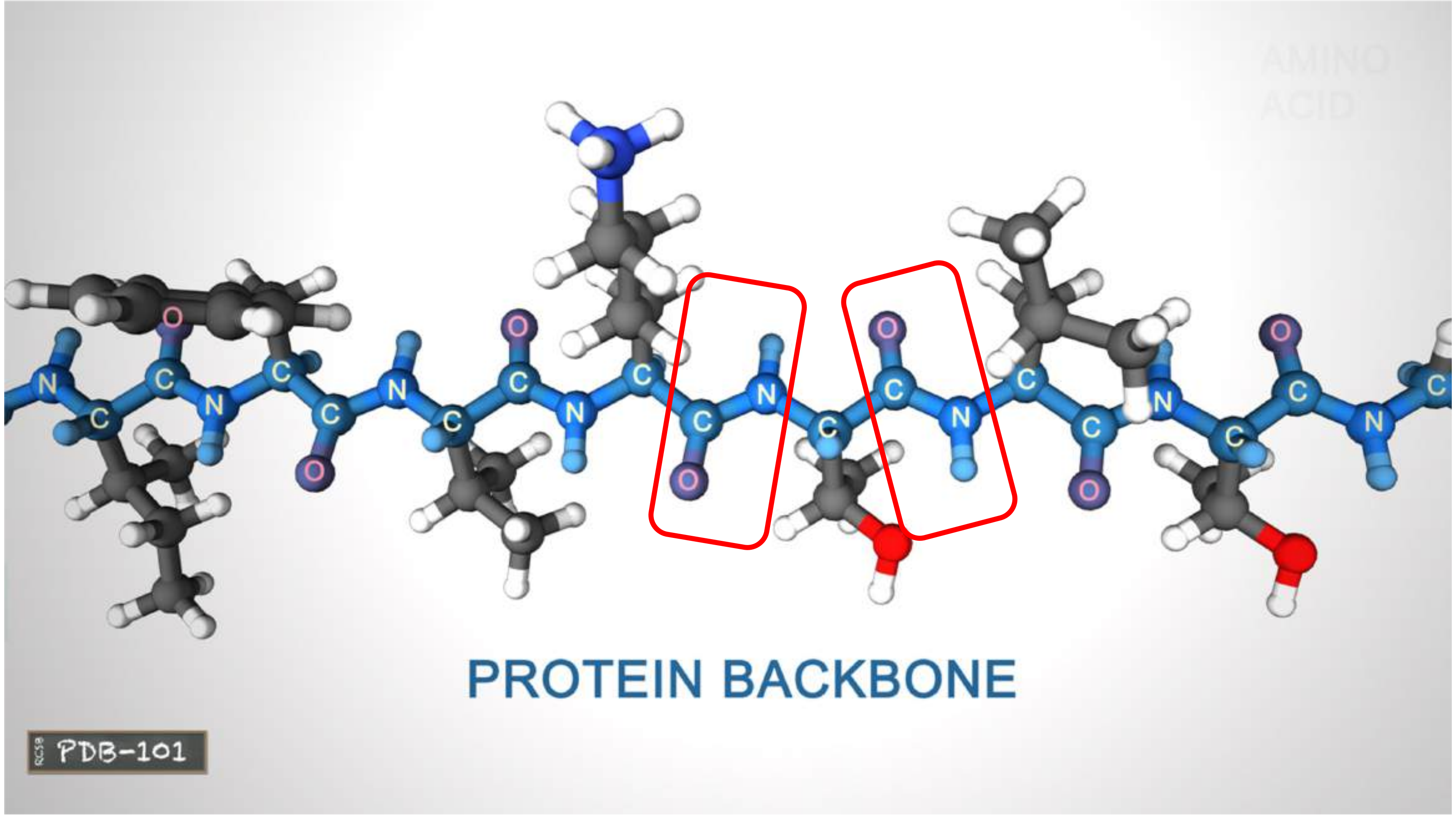

**Hydrophilic** or **polar** amino acids interact well with water.

# The Building Blocks of Proteins





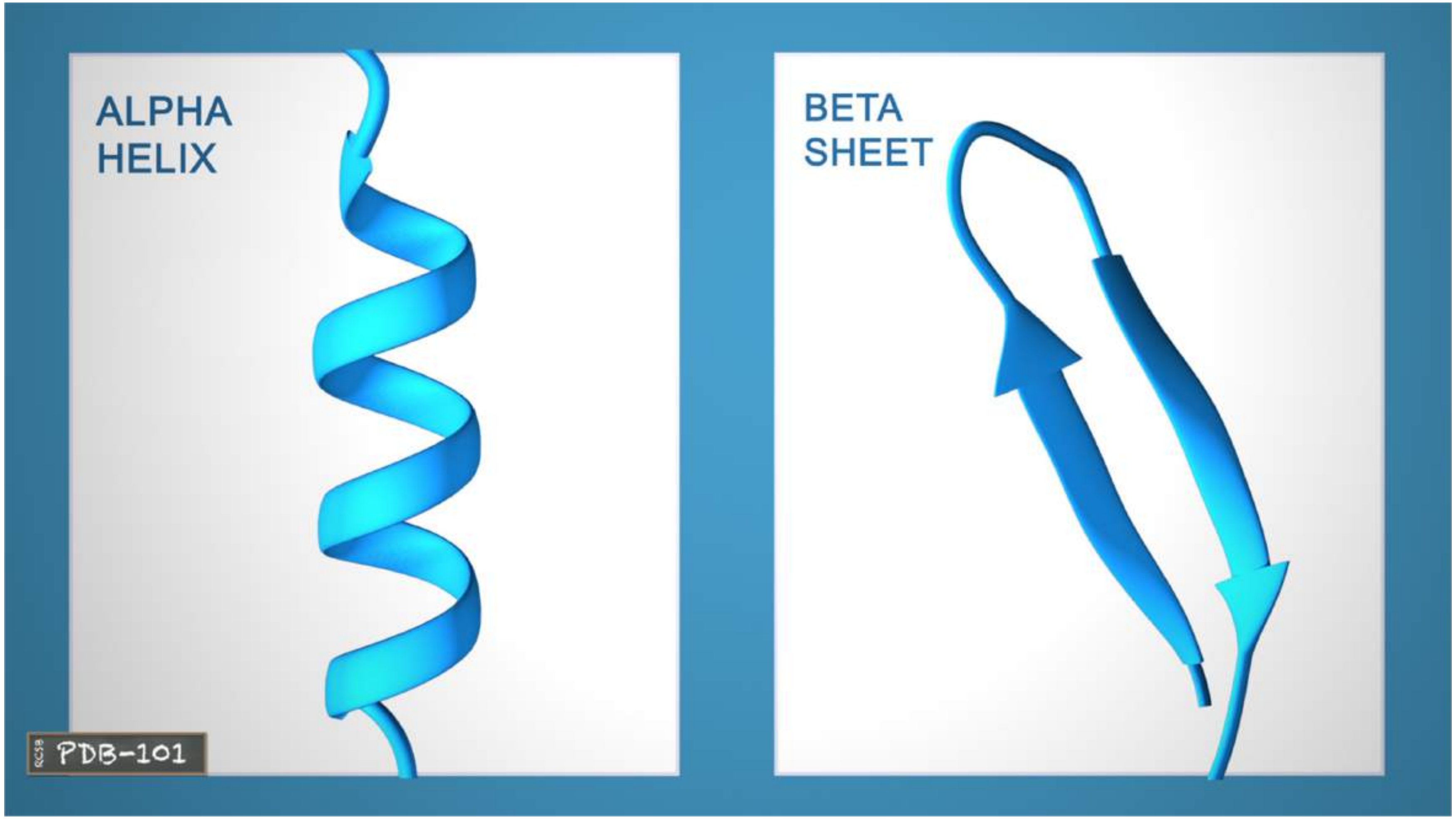

**Charged** amino acids interact with oppositely charged amino acids or other molecules.




The linked series of carbon, nitrogen, and oxygen atoms make up the **protein backbone**.

# Protein Structure: Secondary Structure

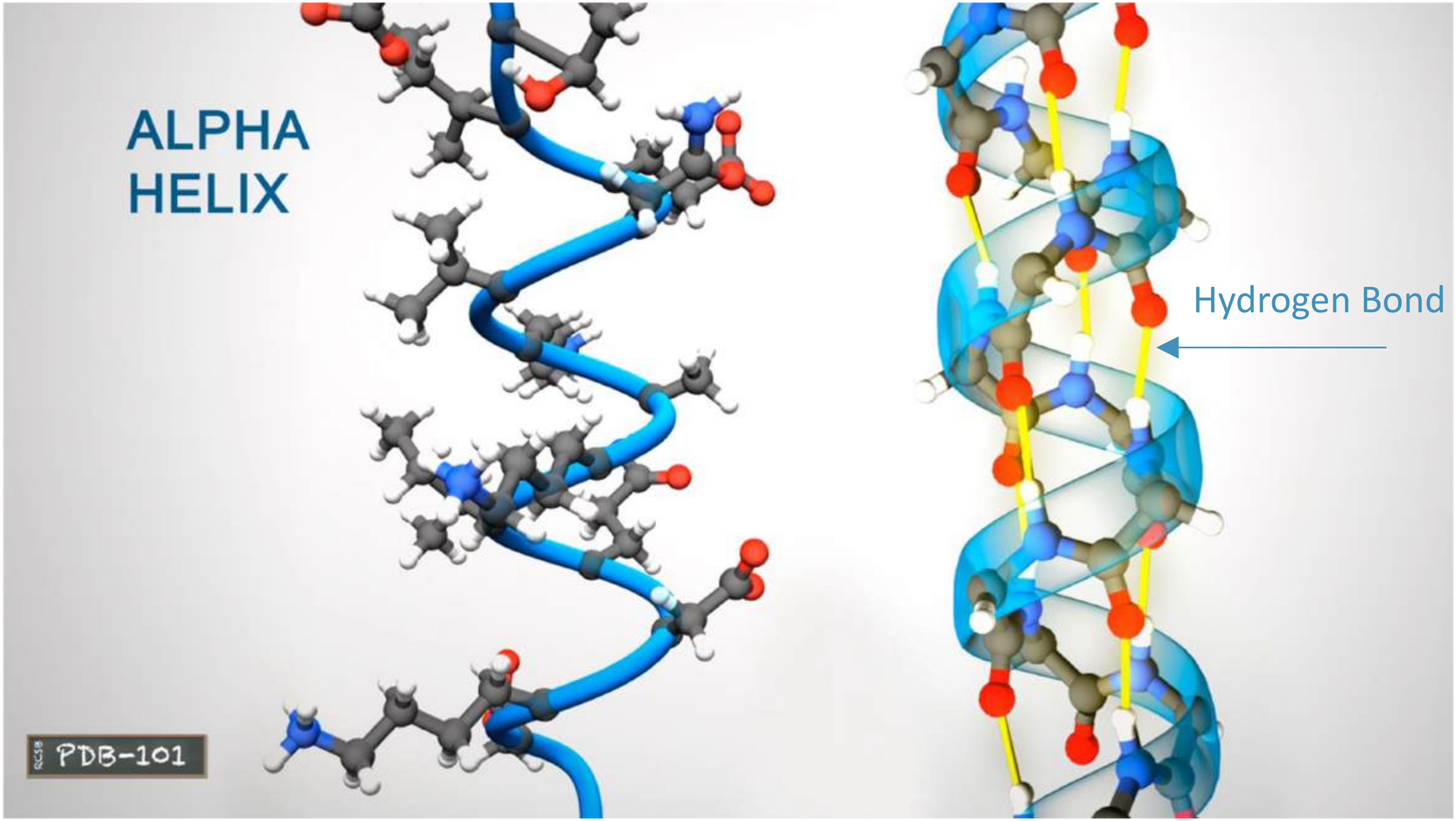
pdb101.rcsb.org




The protein chains often fold into two types of secondary structures: **alpha helices**, or **beta sheets**.

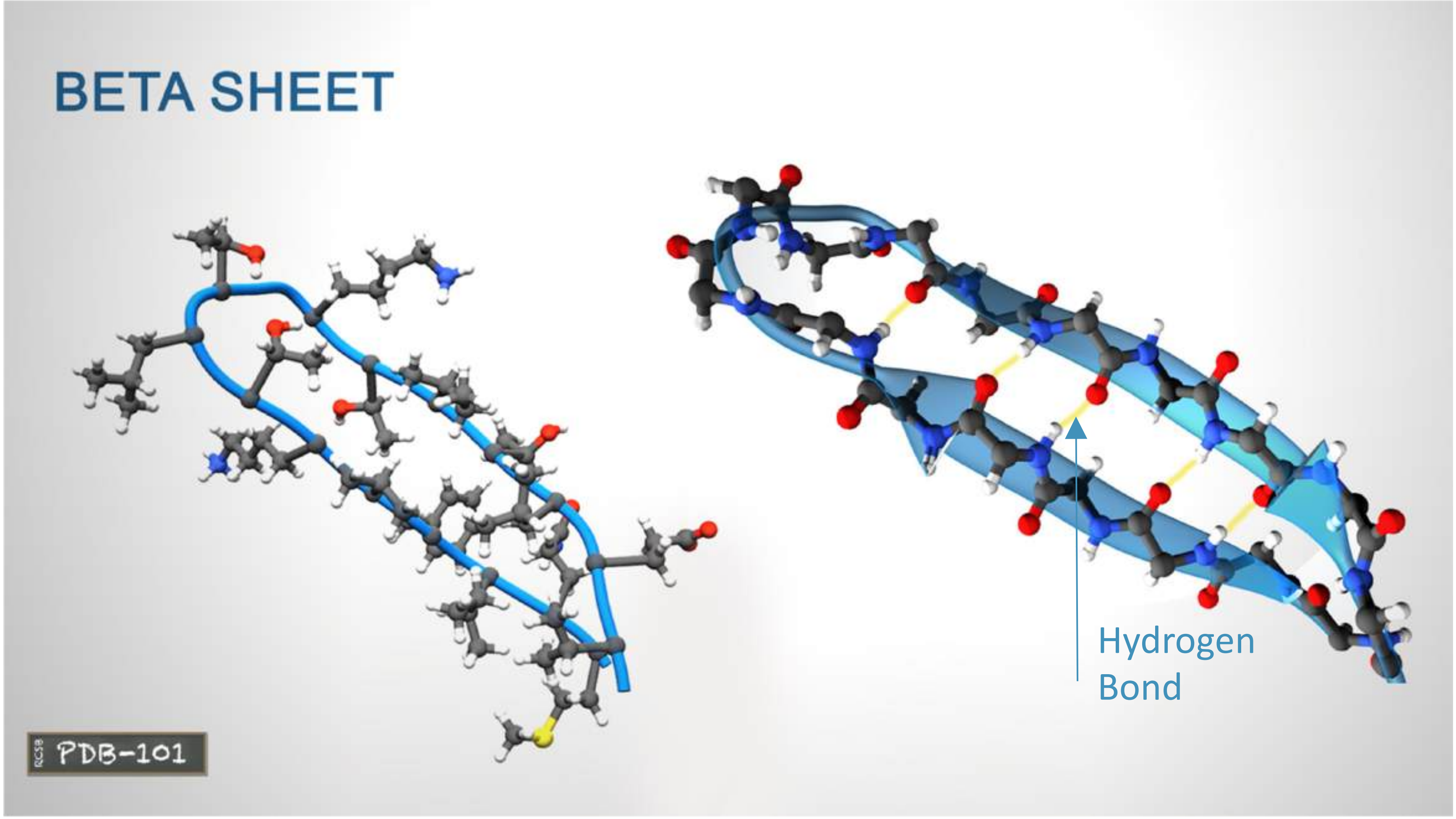
# Protein Structure: Secondary Structure

pdb101.rcsb.org




An alpha helix is a **right-handed coil** stabilized by **hydrogen bonds** between the amine and carboxyl groups of nearby amino acids

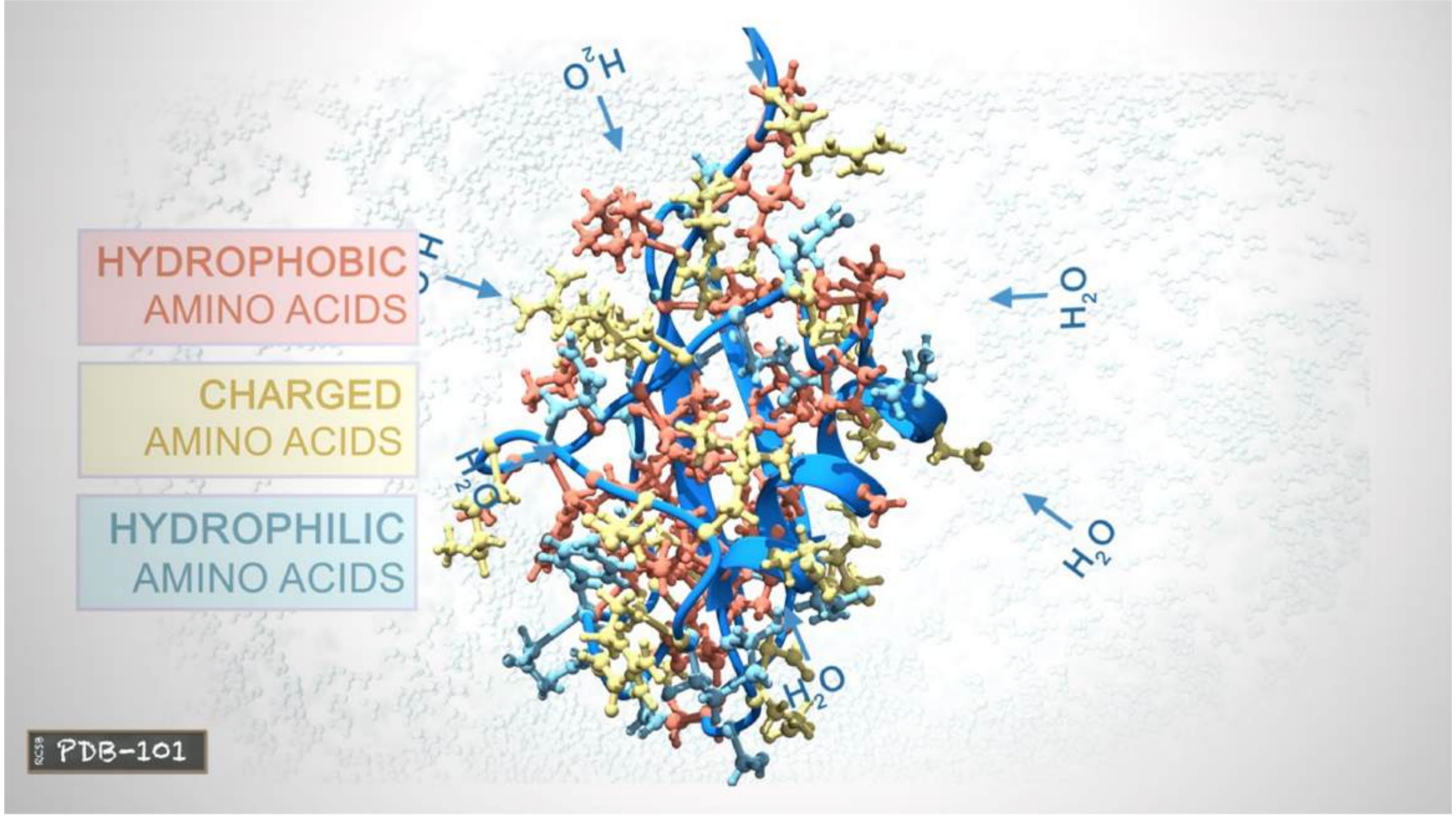
Protein Structure: **Secondary Structure** pdb101.rcsb.org




Beta-sheets are formed when hydrogen bonds stabilize **two or more adjacent strands**.

# Protein Structure: Tertiary Structure

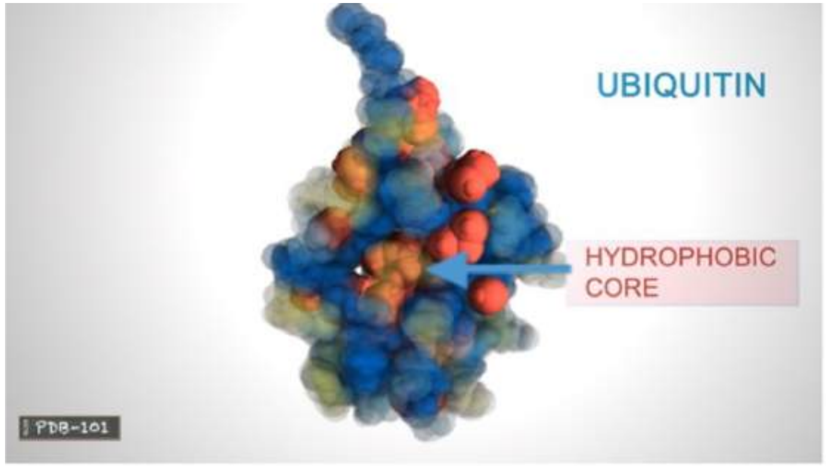
pdb101.rcsb.org

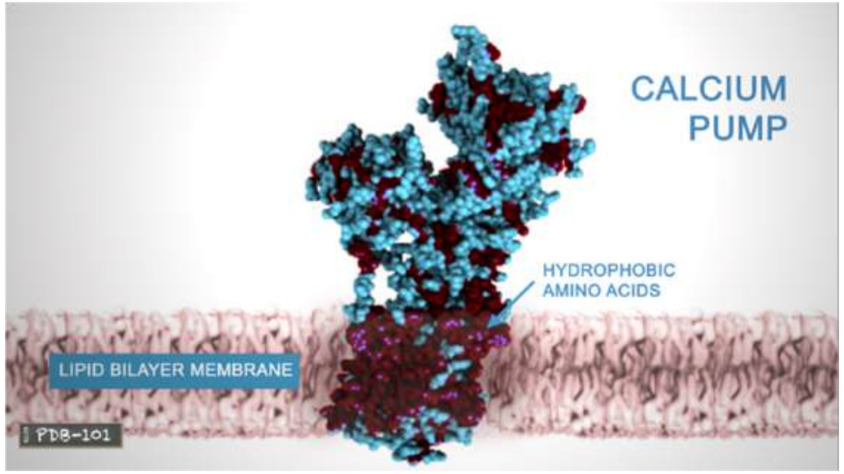



The tertiary structure of a protein is the **three-dimensional shape of the protein chain**. This shape is determined by the characteristics of the amino acids making up the chain.

# Protein Structure: Tertiary Structure

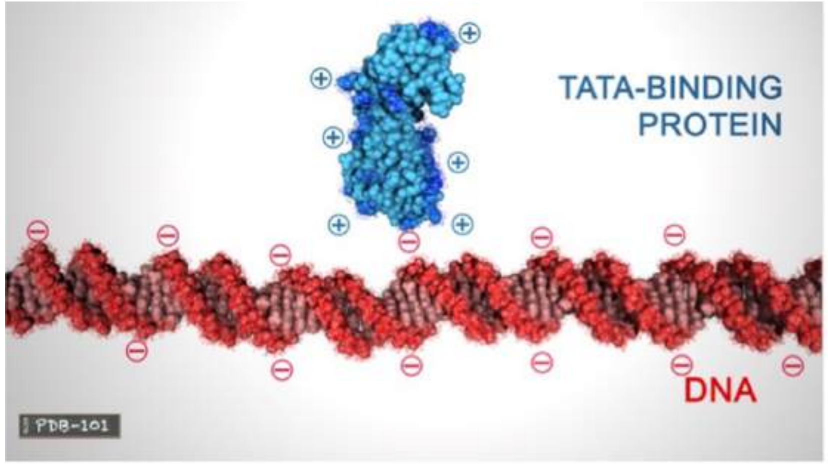
pdb101.rcsb.org

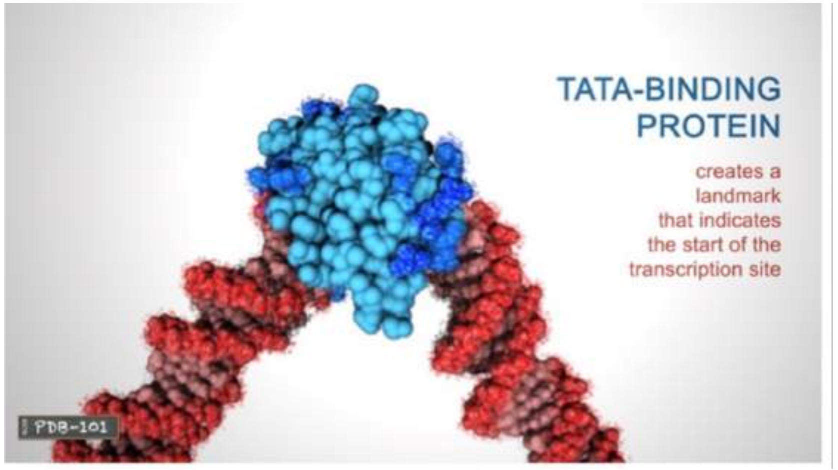



Many proteins form globular shapes with hydrophobic side chains sheltered inside, away from the surrounding water.

Membrane-bound proteins have hydrophobic residues clustered together on the outside, so that they can interact with the lipids in the membrane.

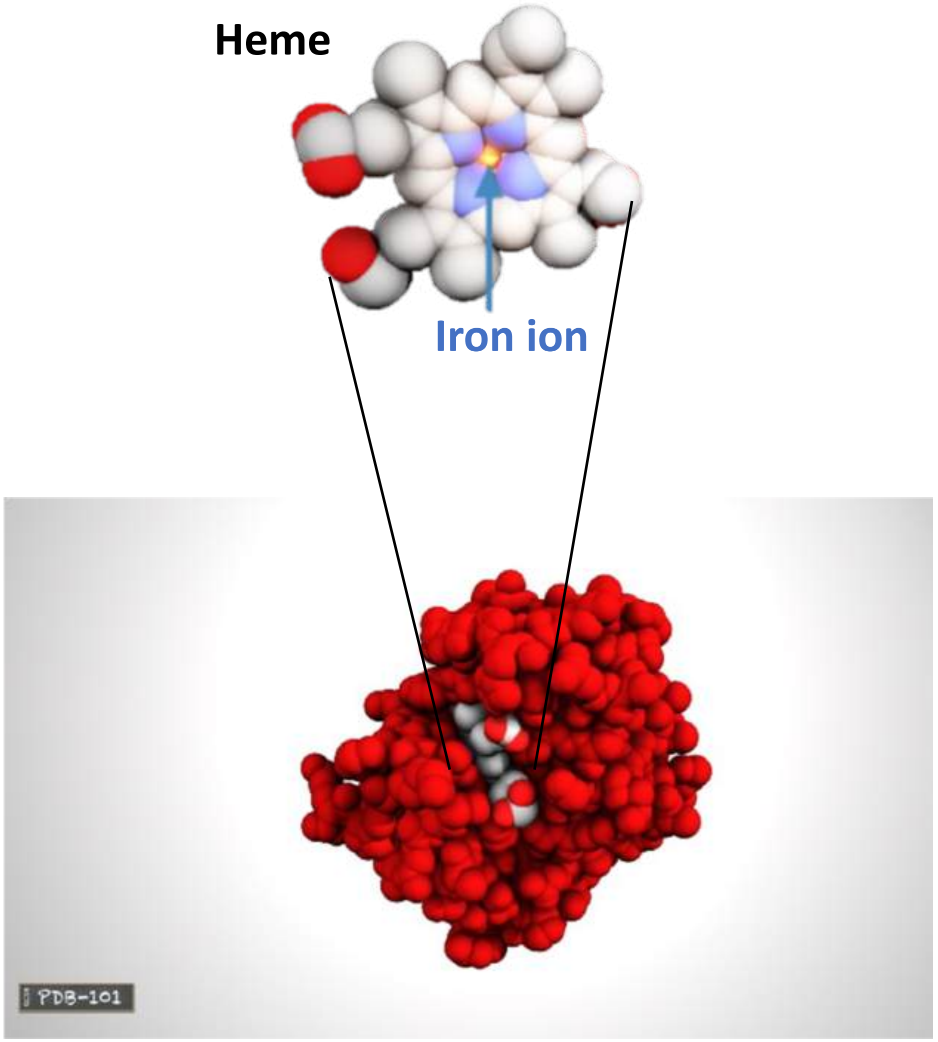



Charged amino acids allow proteins to interact with molecules that have complementary charges.

# Protein Structure: Tertiary Structure

pdb101.rcsb.org

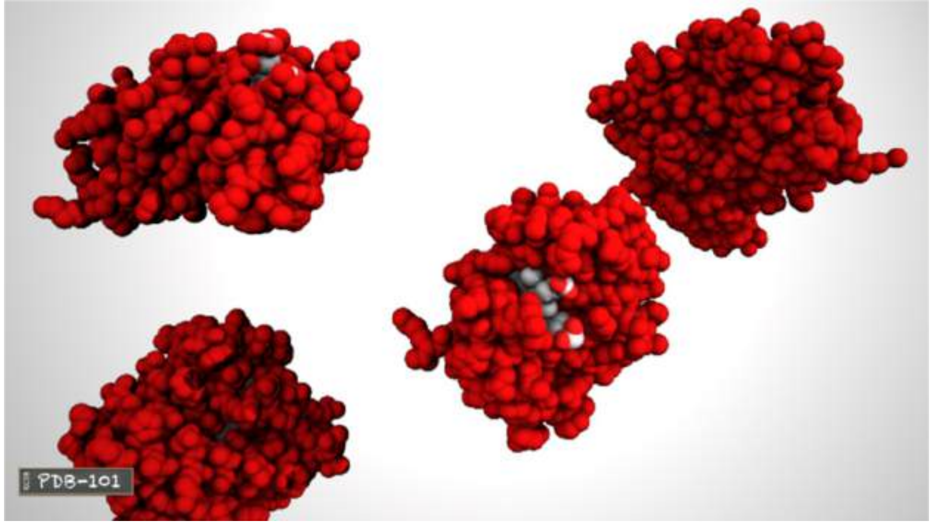


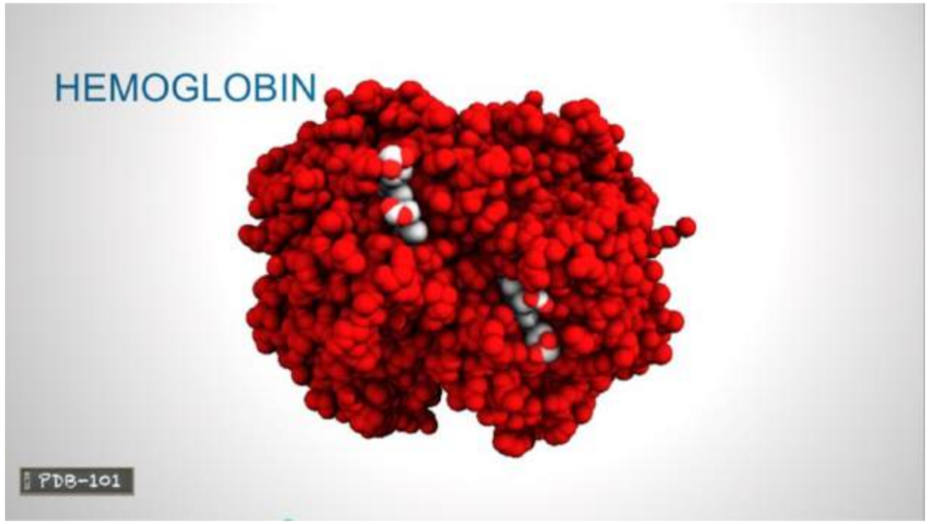

The functions of many proteins rely on their three-dimensional shapes.

For example, **hemoglobin** forms a pocket to hold **heme**, a small molecule with an **iron** atom in the center that binds oxygen.

# Protein Structure: Quaternary Structure

pdb101.rcsb.org

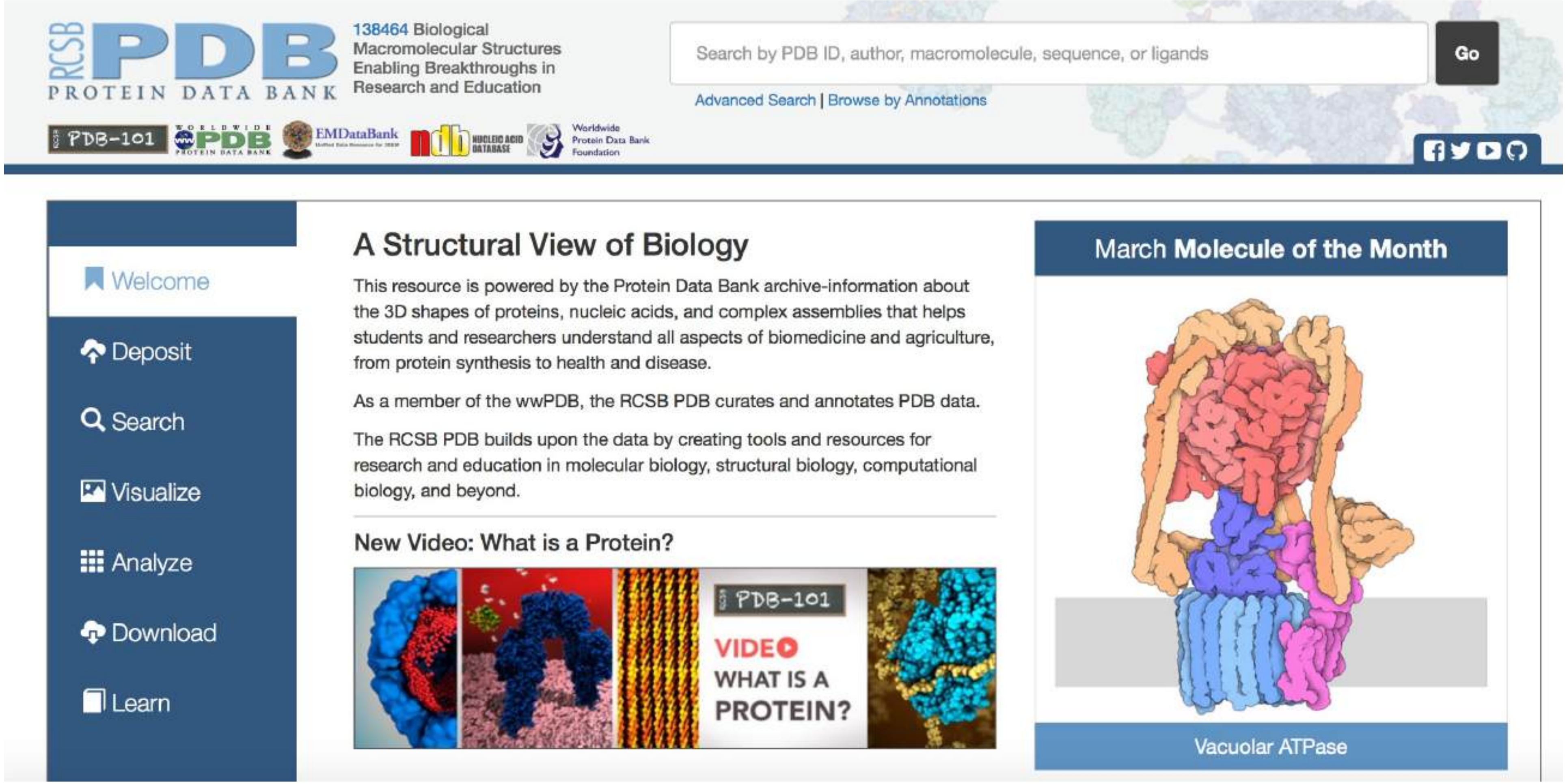


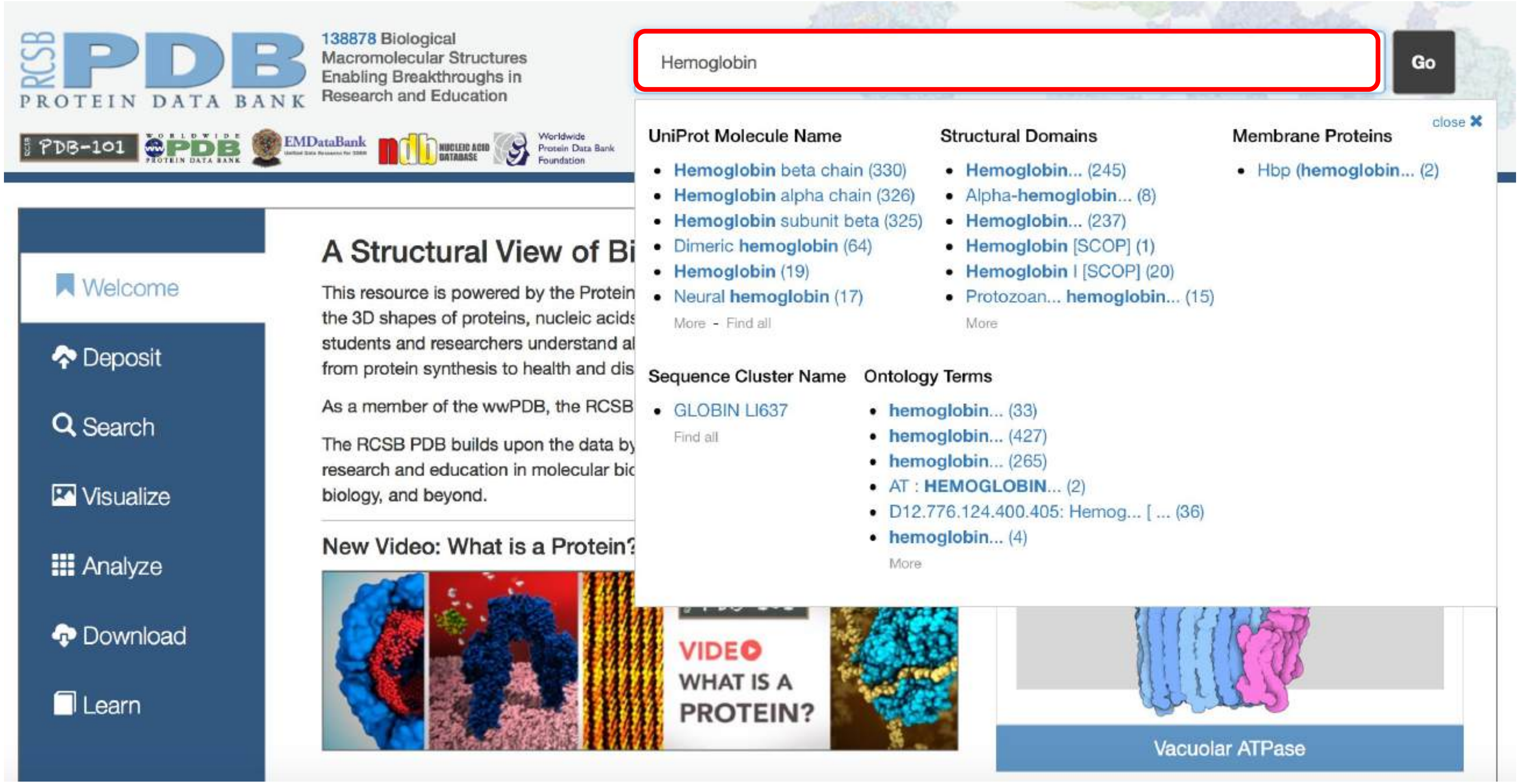

**Two or more polypeptide chains can come together** to form one functional molecule with several subunits.

For example, the four subunits of hemoglobin cooperate so that the complex can pick up more oxygen in the lungs and release it in the body.

# Protein Data Bank (PDB)



# Protein Data Bank (PDB)



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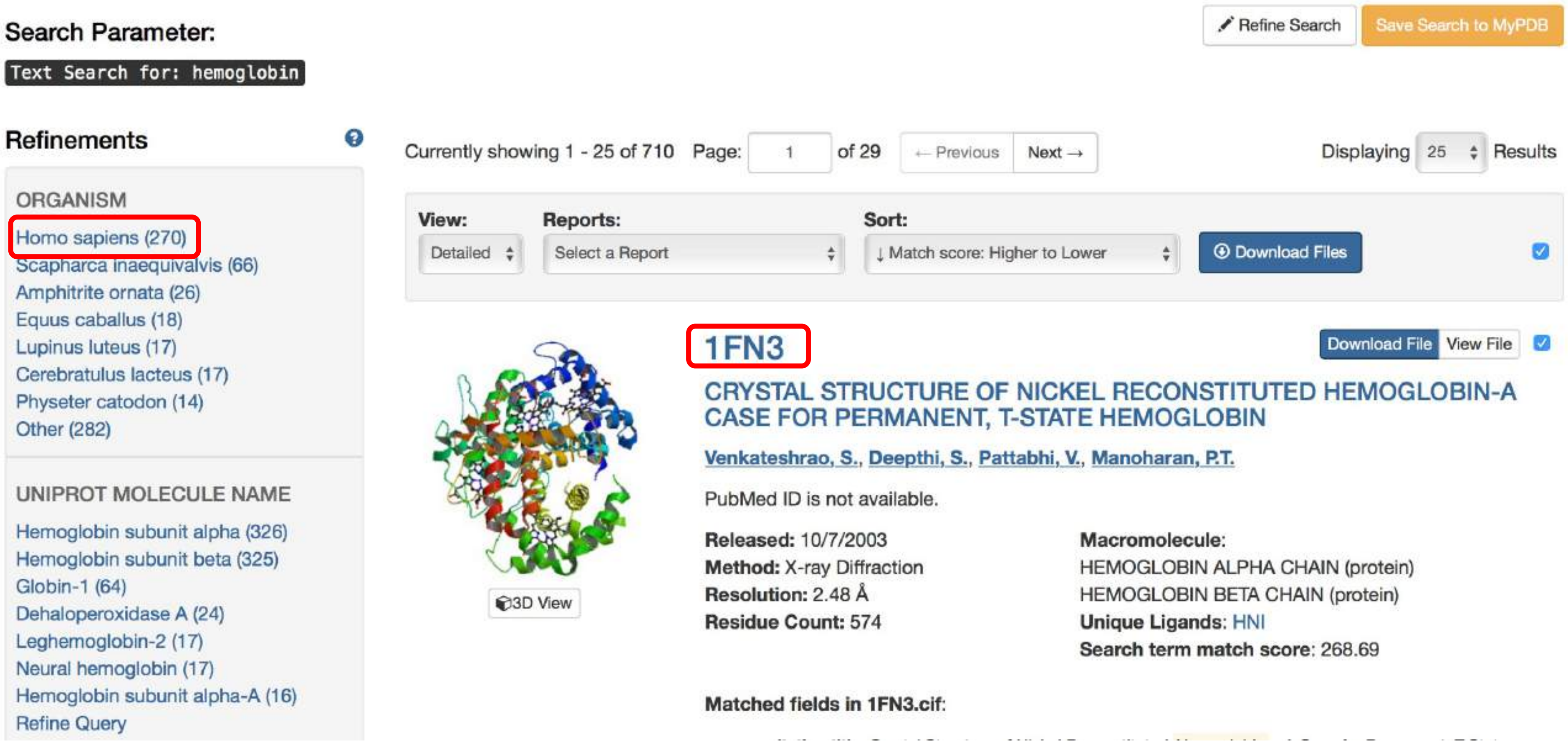


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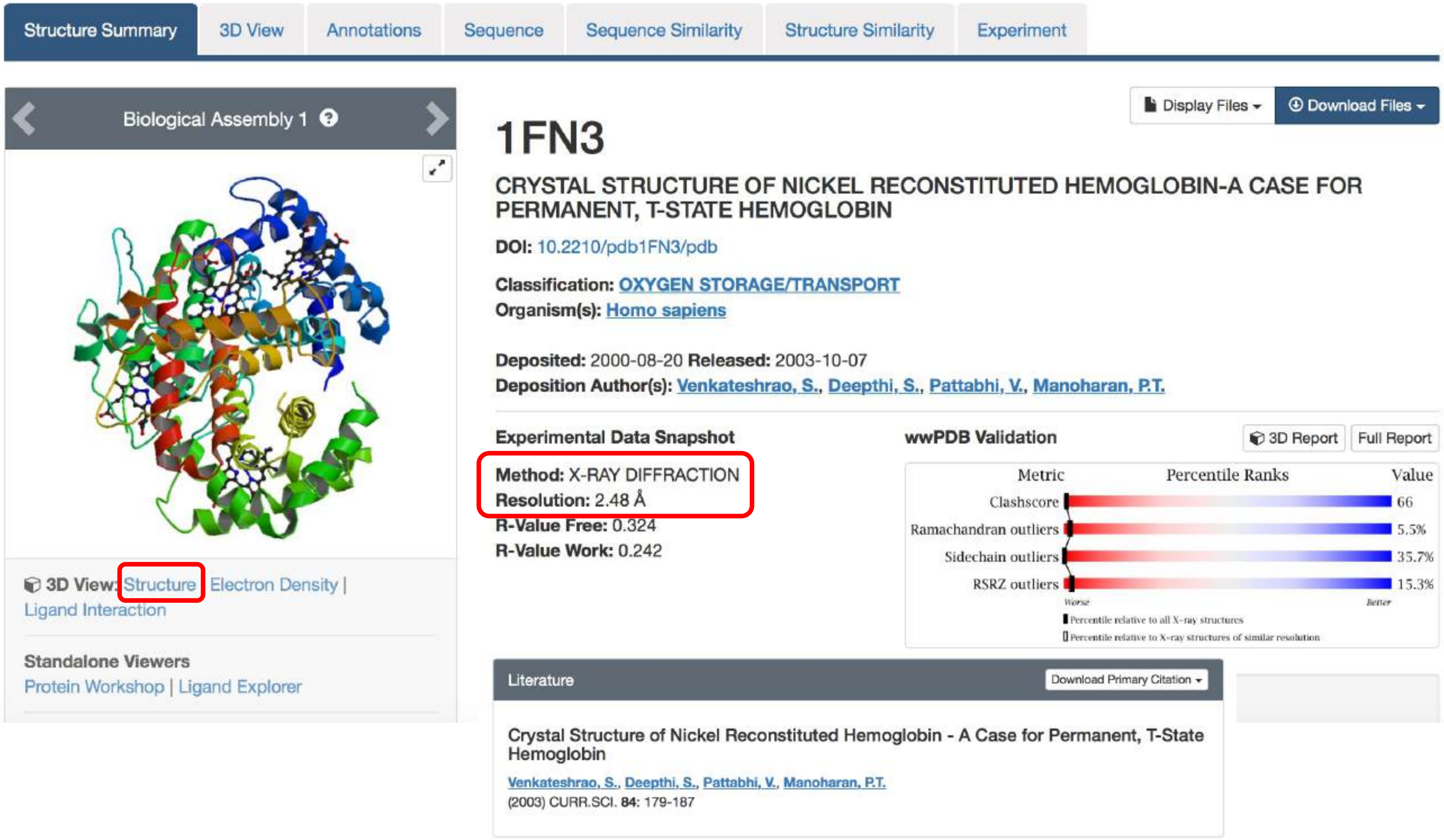
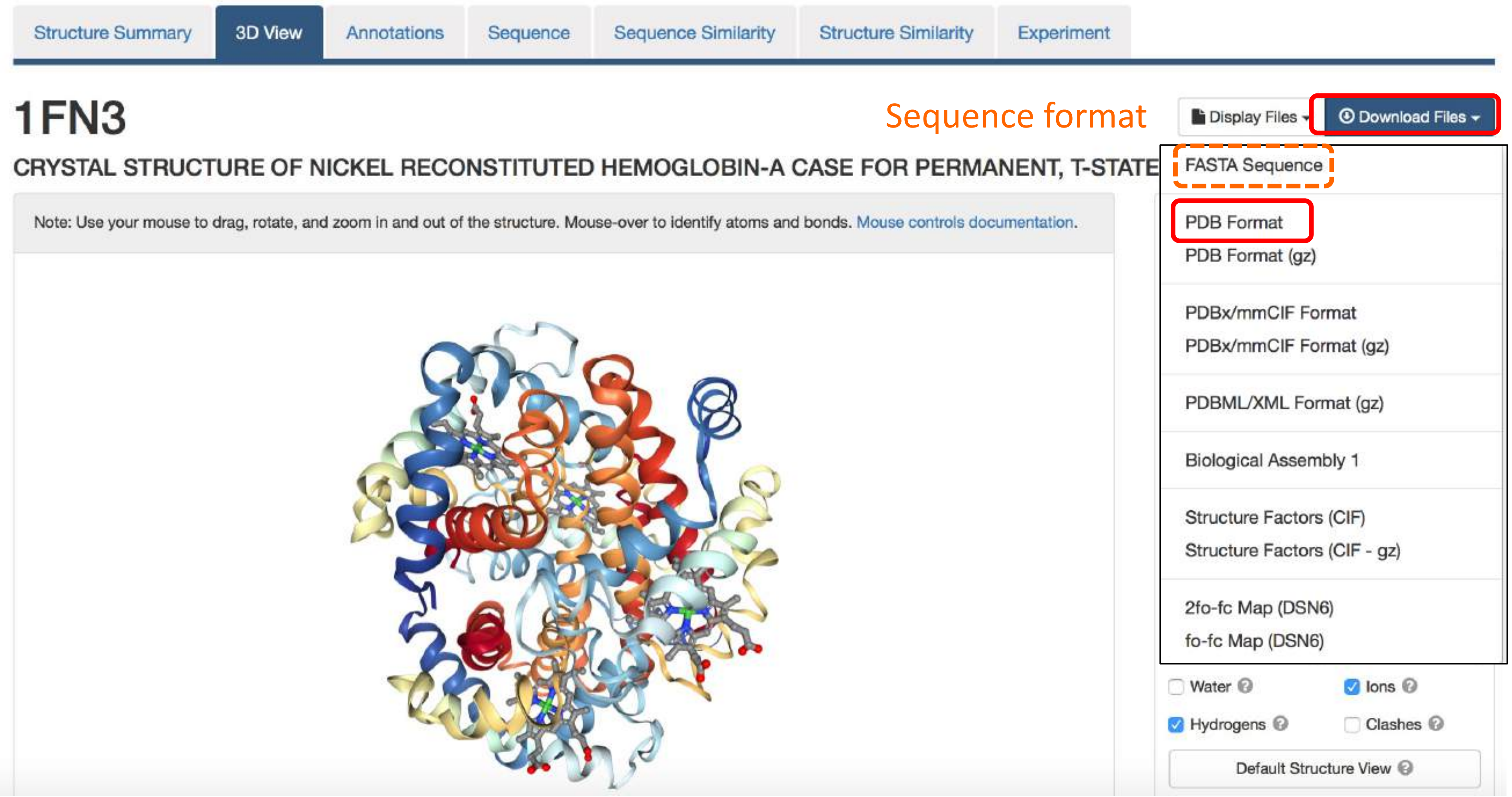
# Protein Data Bank (PDB)



# Protein Data Bank (PDB)



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TITLE    2 PERMANENT, T-STATE HEMOGLOBIN
                               .
                               .
                               .
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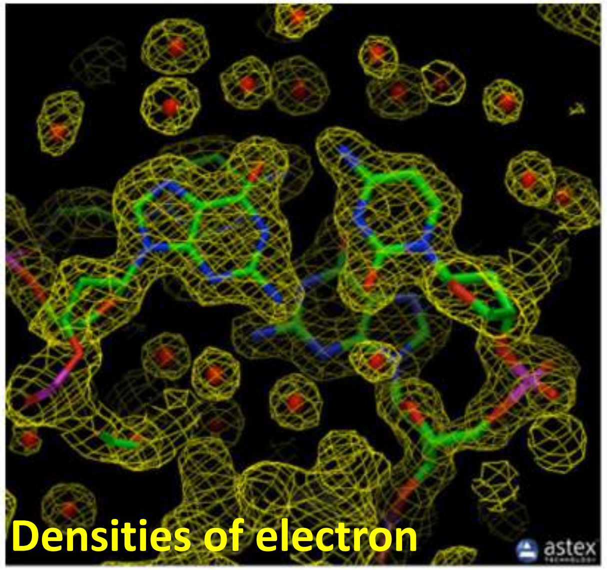
SOURCE   2 ORGANISM_SCIENTIFIC: HOMO SAPIENS;
SOURCE   3 ORGANISM_COMMON: HUMAN;
SOURCE   4 ORGANISM_TAXID: 9606;
                               .
                               .
KEYWDS    PERMANENT T-STATE, METAL ION COORDINATION, SUBUNIT INEQUIVALENCE,
KEYWDS   2 SPECTROSCOPY, CRYSTALLOGRAPHY., OXYGEN STORAGE-TRANSPORT COMPLEX
EXPDTA    X-RAY DIFFRACTION
AUTHOR    S.VENKATESHRAO,S.DEEPTHI,V.PATTABHI,P.T.MANOHARAN
                               .
                               .
REMARK   2 RESOLUTION.    2.48 ANGSTROMS.
                               .
                               .
ATOM      1  N   VAL A   1      42.219  19.998   3.819  1.00 28.12           N
ATOM      2  CA  VAL A   1      43.152  19.932   2.662  1.00 30.16           C
ATOM      3  C   VAL A   1      42.742  20.917   1.574  1.00 26.03           C
ATOM      4  O   VAL A   1      41.698  20.768   0.947  1.00 29.47           O
ATOM      5  CB  VAL A   1      43.197  18.504   2.062  1.00 33.08           C
ATOM      6  CG1 VAL A   1      41.808  18.088   1.572  1.00 36.44           C
ATOM      7  CG2 VAL A   1      44.230  18.448   0.938  1.00 34.80           C
ATOM      8  N   LEU A   2      43.570  21.931   1.360  1.00 23.04           N
ATOM      9  CA  LEU A   2      43.289  22.947   0.349  1.00 17.29           C
ATOM     10  C   LEU A   2      44.042  22.680  -0.952  1.00 12.46           C
                           .
                           .
                           .
```

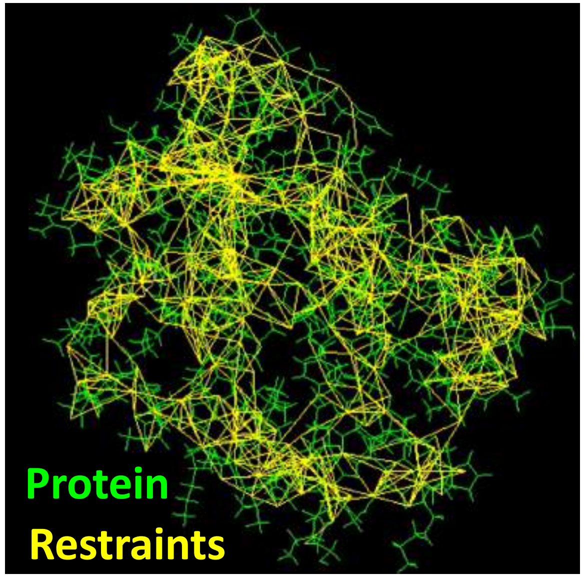
Atomic coordinates

# Methods for Determining Atomic Structures

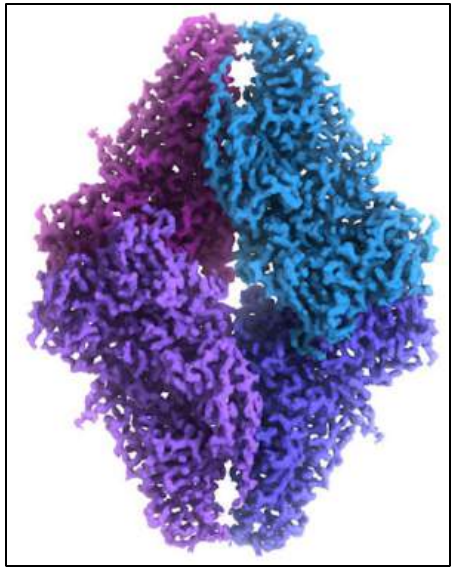
| X-ray Crystallography | NMR Spectroscopy | cryo-Electron Microscopy |
|---|---|---|
| diffraction pattern | distance between atoms | image of the overall shape |




- 需要純化與結晶
- 只能解析重原子
- PDB 資料最多!
- 誤差 1-2埃




- 溶液或固態 (無需結晶)
- 多種結構(系綜)
- 誤差2-4埃

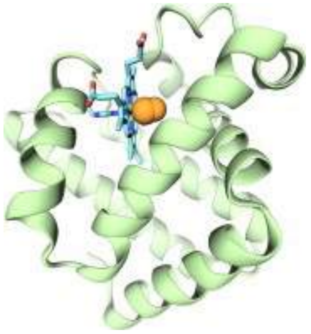


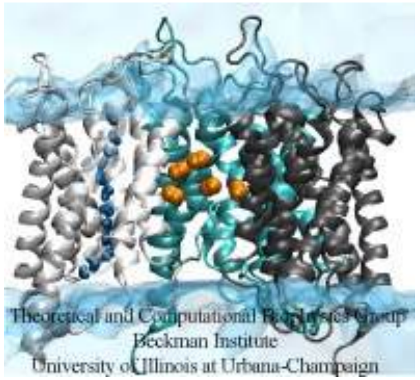
- 專解巨型大分子
- 解析度稍低
- 多種方法混搭使用能完善解析度
- 誤差2-9埃

**http://pdb101.rcsb.org**

http://www.ks.uiuc.edu/Research/vmd/

VMD is designed for **visualization**, **analysis**, and modeling of biological systems such as proteins, nucleic acids, lipid bilayer assemblies, etc.
VMD can read standard Protein Data Bank (**PDB**) files and display the contained structure.






- Support all major computer platforms (ex. MacOS X, Windows, Linux)
- Many molecular rendering and coloring methods

# 新世代 VMD 2正推出中 …(先觀望…)

# A New Era in Molecular Visualization

After nearly 30 years of evolution, VMD 2.0 is here with a brand-new user interface, making molecular visualization more intuitive and powerful than ever. With enhanced GPU acceleration, streamlined workflows, and advanced analysis tools, VMD 2.0 is designed to push the boundaries of molecular modeling and simulation.

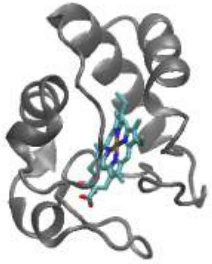
Quick Guide | Download VMD 2

VMD2
Visual Molecular Dynamics

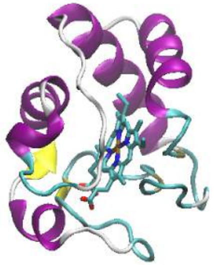
https://www.ks.uiuc.edu/Research/vmd/vmd2intro/index.html

# What Can We Do with VMD?

- Visualize the structures that have been solved and reported

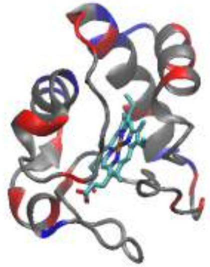


Cartoon diagram



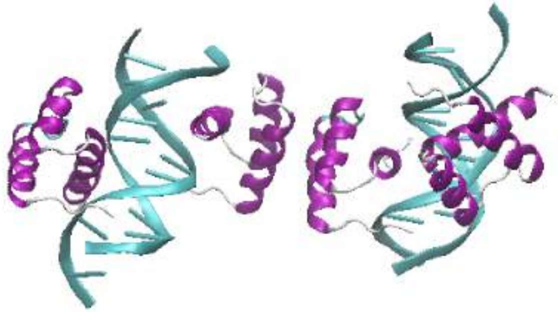
Secondary structure

PDB: **1hrc**



Positively charged
Negatively charged

蛋白質 + 小分子



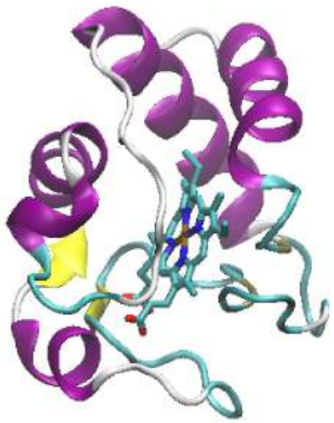
Multiple protein assembly

PDB: **4qtr**

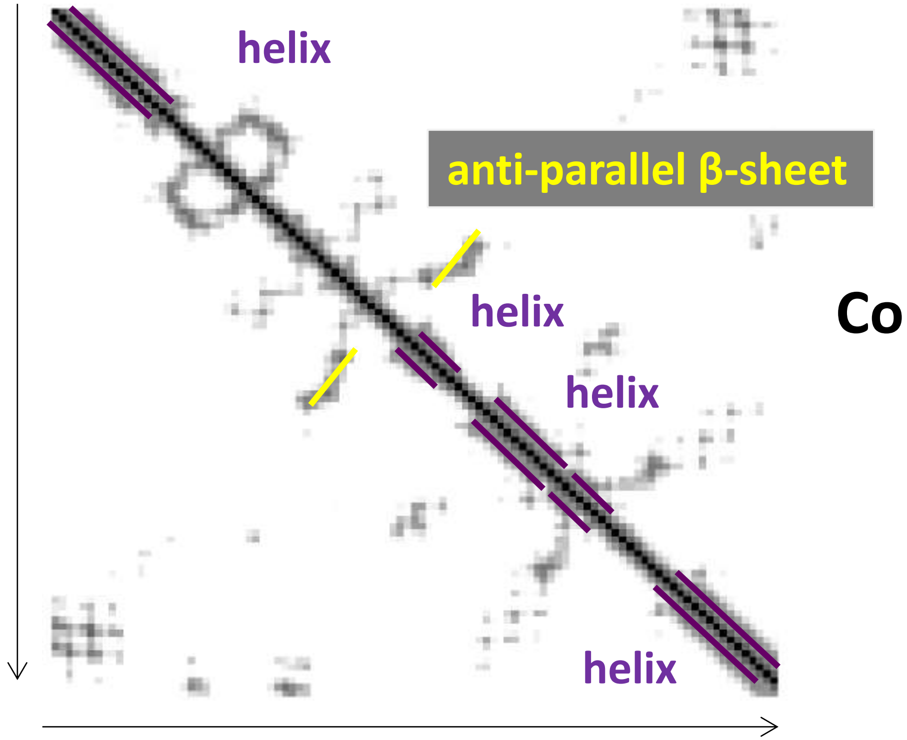
蛋白質 + 核酸
(複雜組裝)

# What Can We Do with VMD?

- Structural analysis (Contact map)



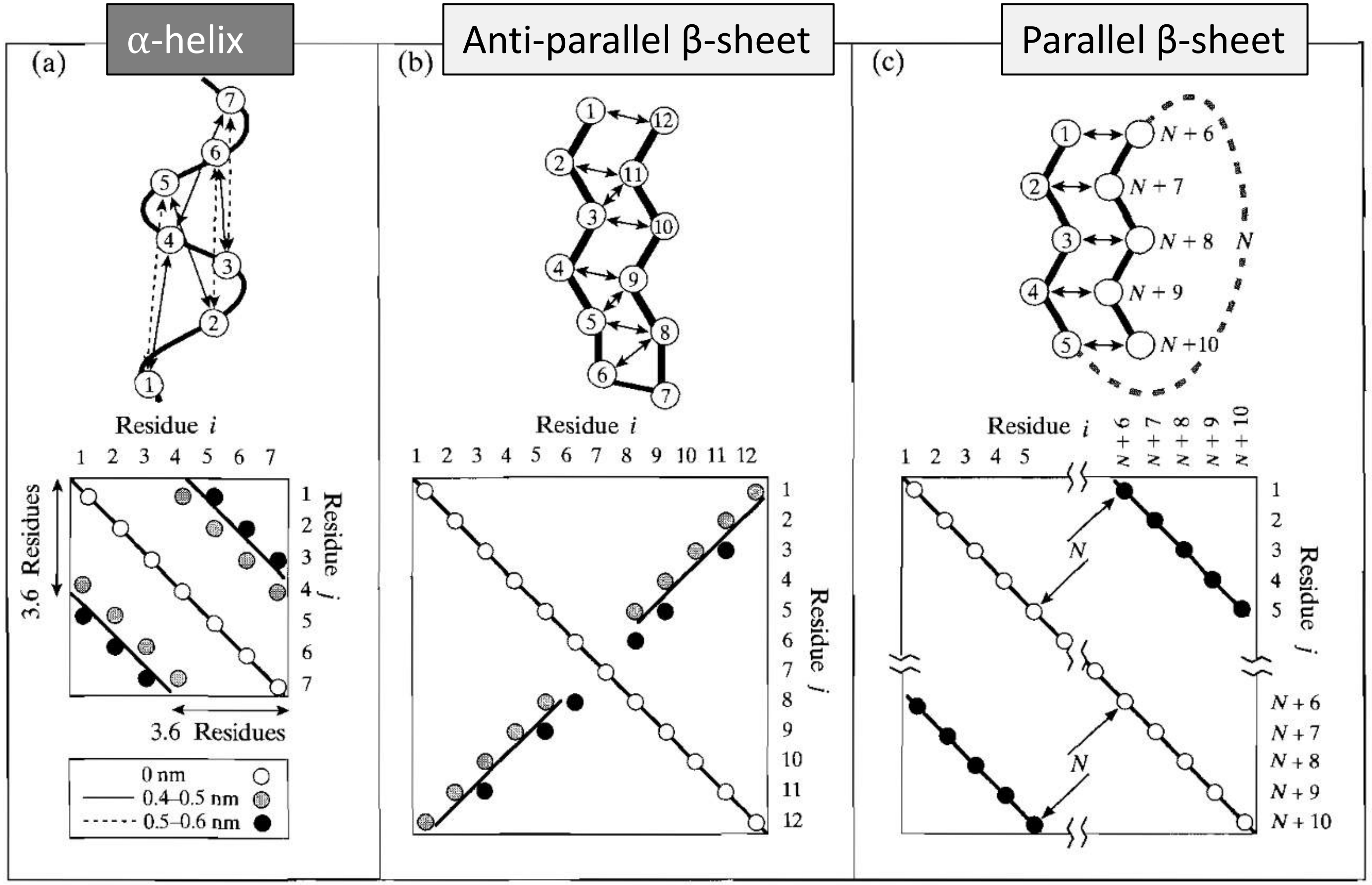
PDB: 1hrc




**Contact map**

# What Can We Do with VMD?

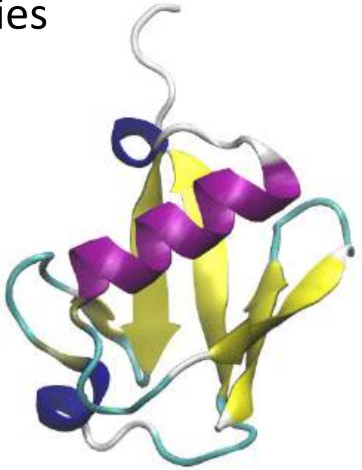
- Structural analysis (Contact plots for α-helix and β-sheets)



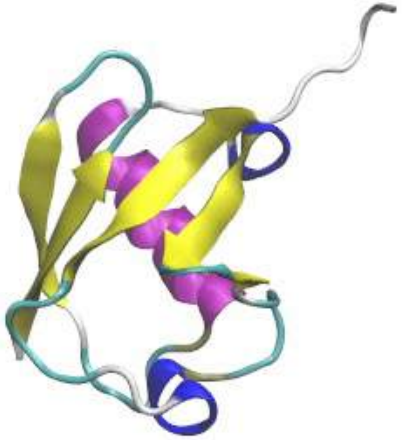

van Holde, Johnson, Ho, *Principles of Physical Biochemistry* 2nd edition, Pearson, New Jersey

# What Can We Do with VMD?

- Make movies

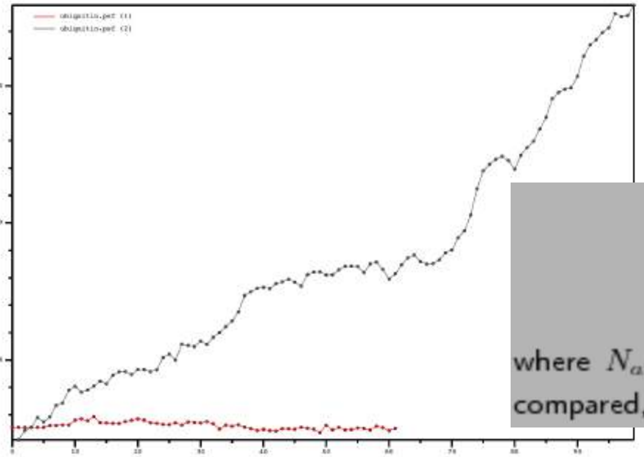


PDB: 1ubq



- Trajectory analysis

RMSD



Time frame

**<u>R</u>oot <u>M</u>ean <u>S</u>quare <u>D</u>eviation**

$$RMSD = \sqrt{\frac{\sum_{i=1}^{N_{atoms}} (r_i(t_1) - r_i(t_2))^2}{N_{atoms}}} \qquad (1)$$

where $N_{atoms}$ is the number of atoms whose positions are being compared, and $r_i(t)$ is the position of atom $i$ at time $t$.

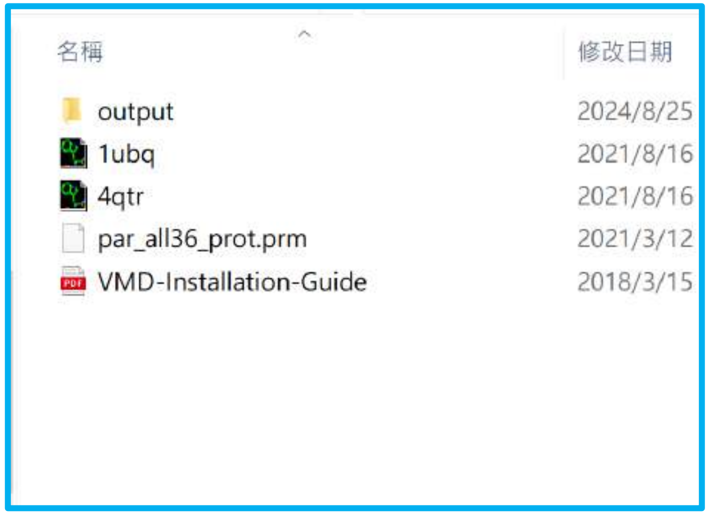
# Now You Can Try to Launch VMD

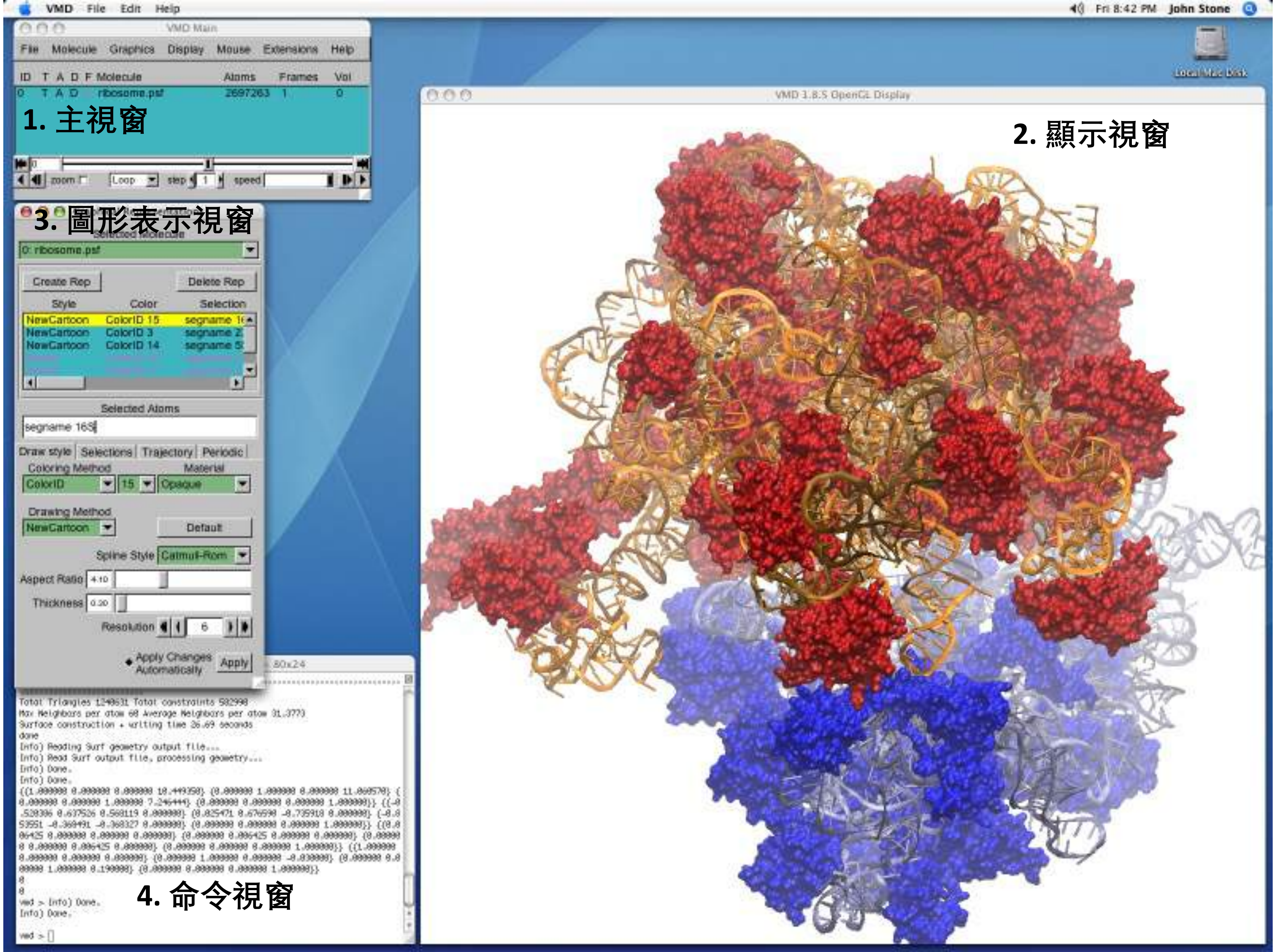
桌面
(Desktop)


**vmd**

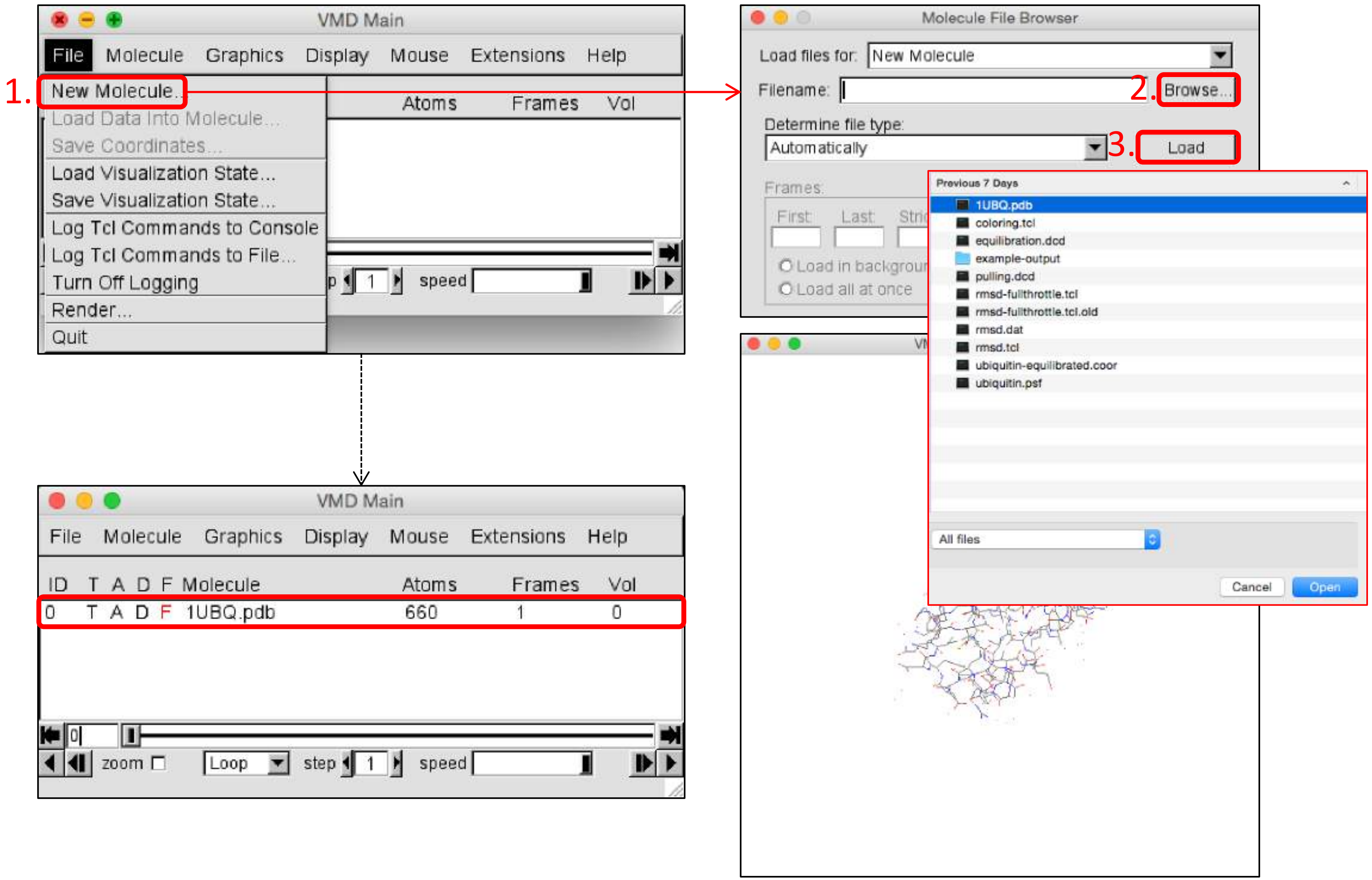
進入113.2 MD workshop\\<u>module2</u> 資料夾，將 **vmd** 移至桌面 (Desktop)

See *VMD-Installation-Guide.pdf* if you want to install it on your computer

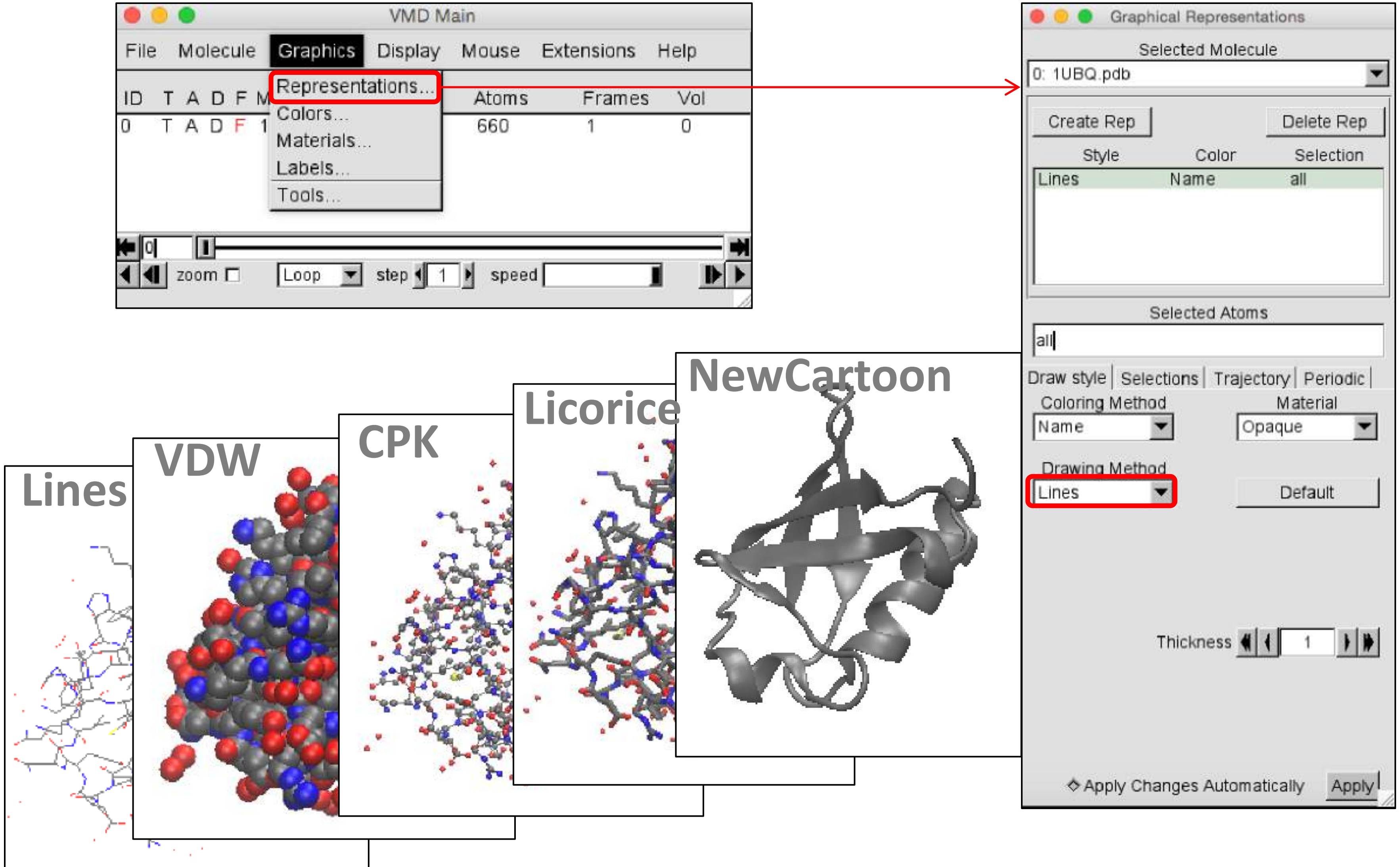



1. 主視窗
2. 顯示視窗
3. 圖形表示視窗
4. 命令視窗

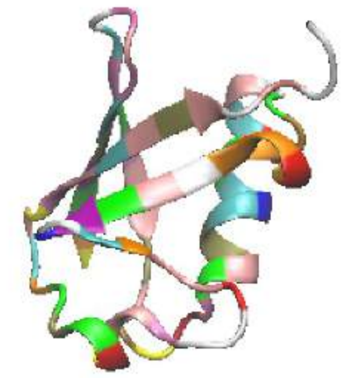
# **Loading** a *Molecule*, **Exploring** it and Choosing Representations



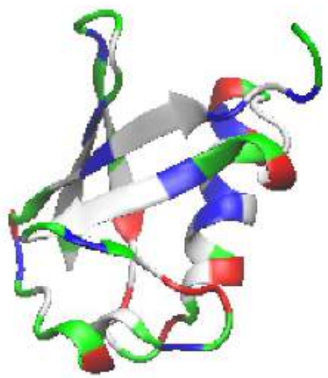
# Loading a Molecule, Exploring it and **Choosing** ***Representations***



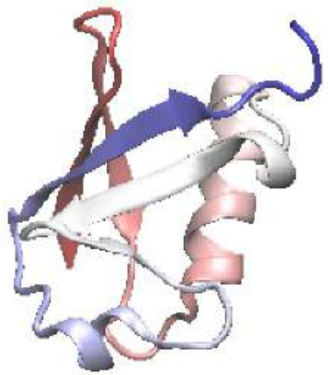
# Loading a Molecule, Exploring it and Choosing *Representations*



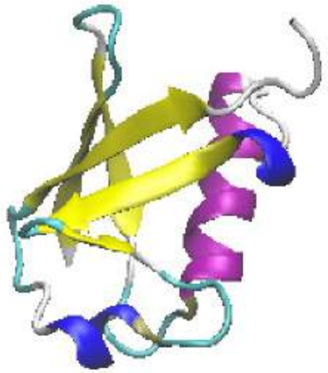
ResName



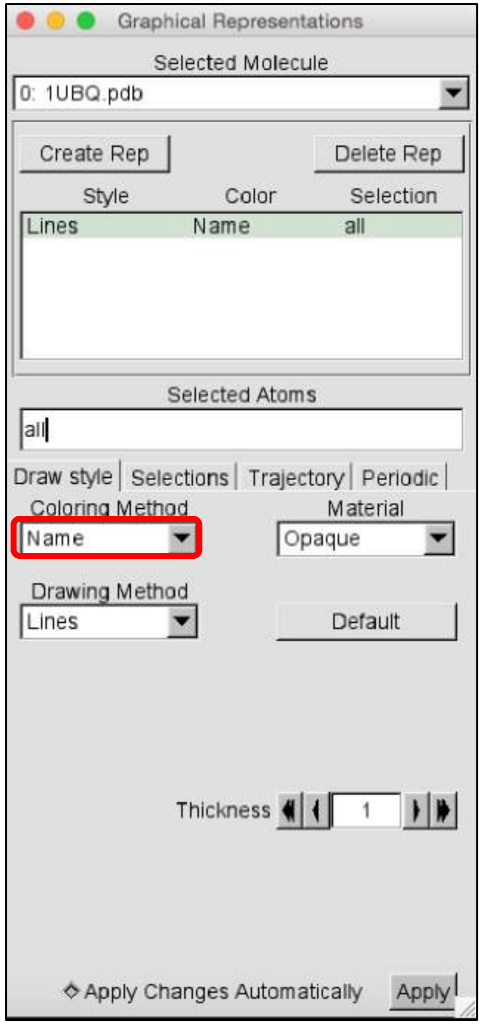
ResType



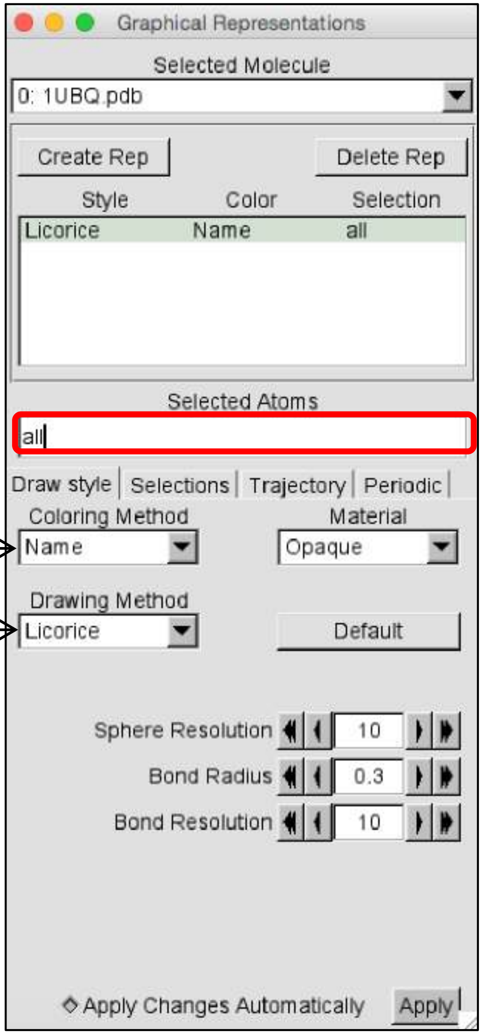
Index



Secondary Structure



# Exercise: Selected Atoms




**Name**

**Licorice**

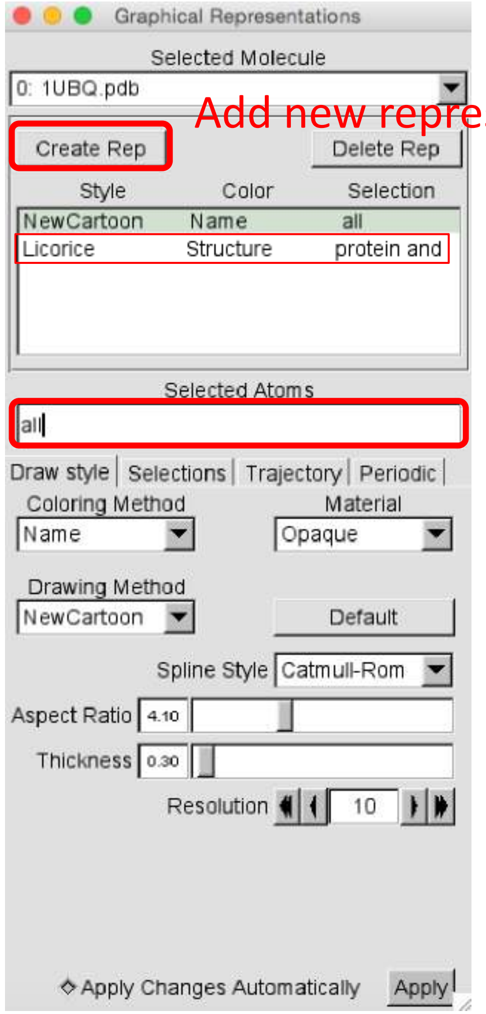
| Selection | Action |
|---|---|
| `protein` | Shows the Protein |
| `resid 1` | The first residue |
| `(resid 1 76) and (not water)` | The first and last residues |
| `(resid 23 to 34) and (protein)` | The $\alpha$ -helix |

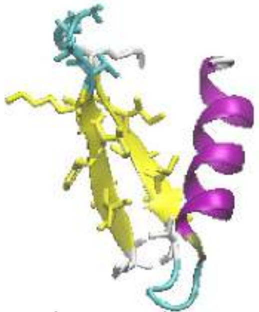
In **Drawing Method**, switch to “NewCartoon” to see the helix

Also try,

```
protein and resname LYS ARG
protein and resname GLU ASP
```

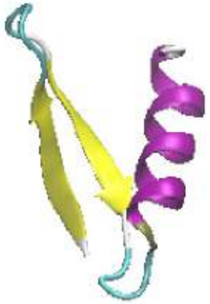
# Displaying Different Selections



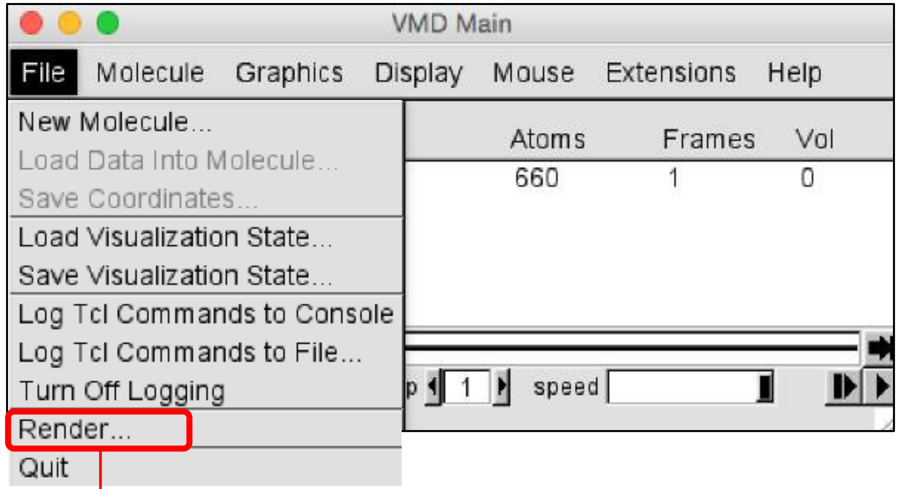
"protein and *resid 1 to 17*" with **Licorice**

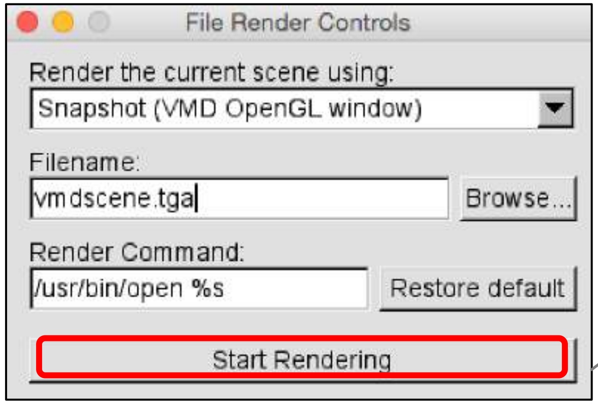
"protein and *resid 1 to 17*"

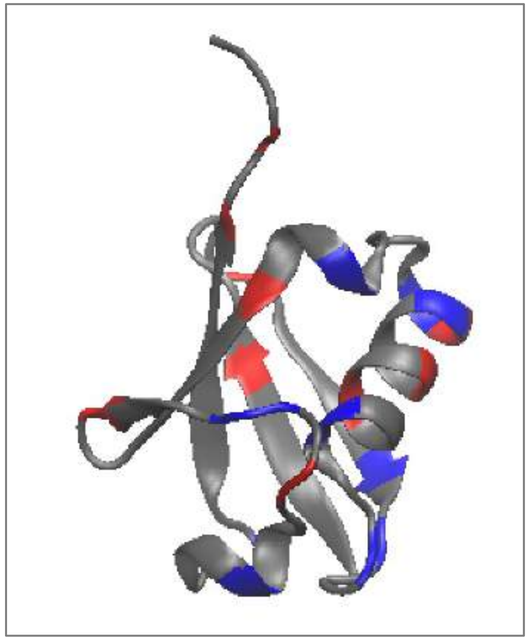


"protein and *resid 1 to 36*"

# Rendering an Image



File: vmdscene.tga

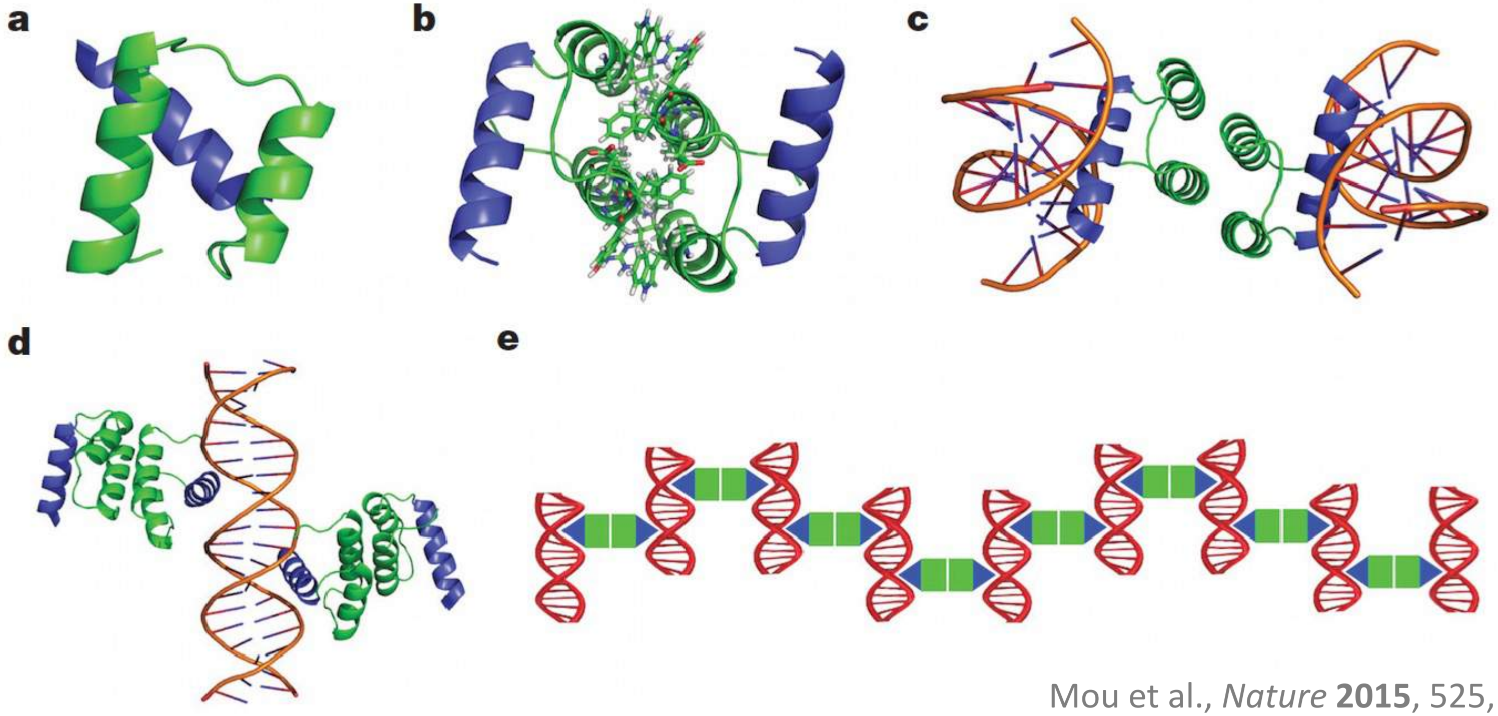
LETTER

doi:10.1038/nature14874

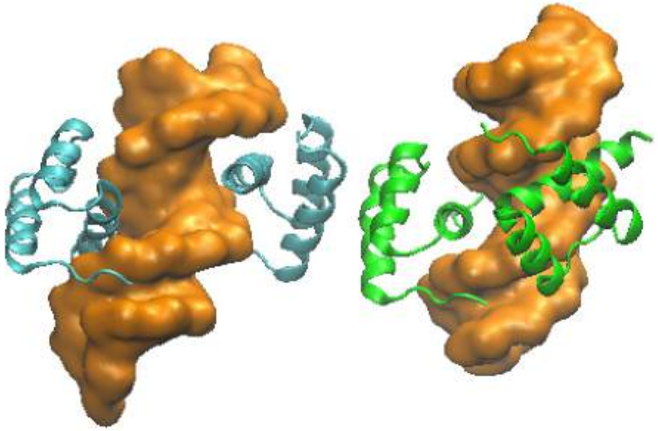
# Computational design of co-assembling protein–DNA nanowires

Yun Mou[1], Jiun-Yann Yu[2], Timothy M. Wannier[2], Chin-Lin Guo[3] & Stephen L. Mayo[1,2]

230 | NATURE | VOL 525 | 10 SEPTEMBER 2015




Mou et al., *Nature* **2015**, 525, 230



**Co-crystal structure for dualENH complexed with dsDNA**
(PDB ID: **4QTR**)

**Demo**

- Measure the distance between two atoms (Dimension of the molecular assembly)
- Visualize the distribution of charged residues on proteins
- Extract the structure of protein binding interface
- Calculate the total energy of the binding interface

**Extended Data Table 1 | Sequences of wild-type ENH and dualENH**

| | |
|---|---|
| ENH (WT) | EKRPRTAFSSEQLARLKREFNENRYLTERRRQQLSSELGLNEAQIKIWFQNKRAKIKKST |
| dualENH | ---------E--KKA-DLA-YFD----PEW-RY--QR----------------------- |

# Change Directory to Home “Desktop“

```
() 32 % pwd
/
() 33 % cd ~
(mytsai) 34 % pwd
/Users/mytsai
(mytsai) 35 % cd Desktop/
(Desktop) 36 % pwd
/Users/mytsai/Desktop
(Desktop) 37 %    cd vmd
(vmd) 38  %
```

→ Check where you are in the path

→ Change to ‘home’ directory

→ Check the path again

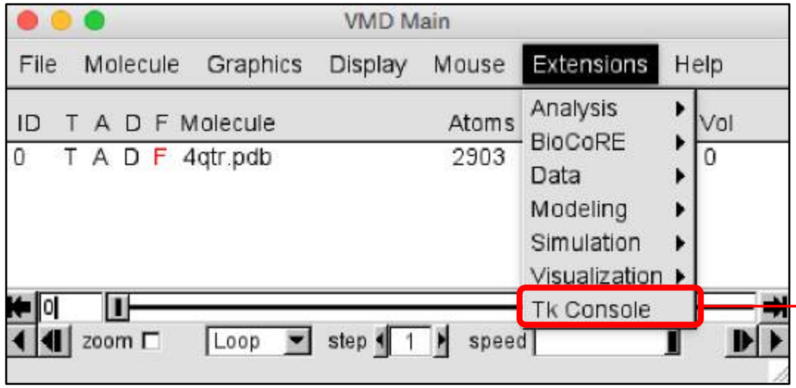
→ Change to ‘home’ Desktop

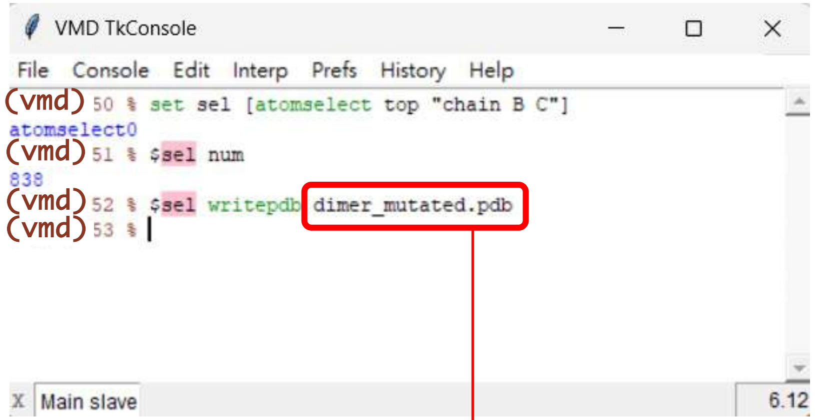
→ Check the path again

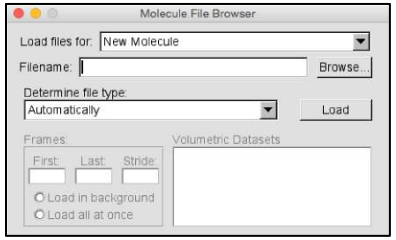
→ OK!

“~” (curly brace): SHIFT + the key next to “1”

# Extract the Structure at Binding Interface

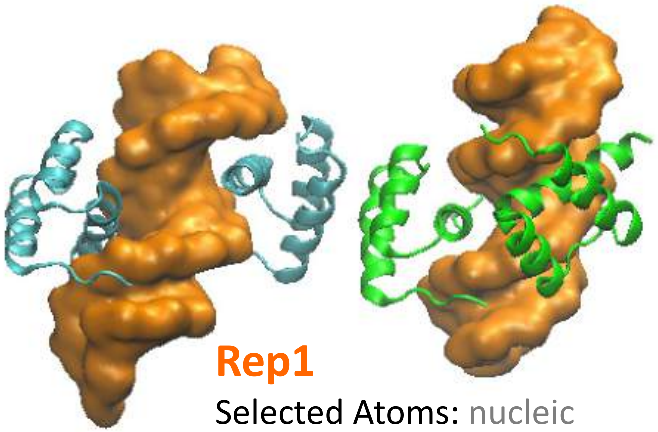



Load with

*File → New Molecule*



Rep3

… Chain C D

… ColorID → 7

… NewCartoon

Rep1

Selected Atoms: nucleic

Coloring Method: ColorID → 3

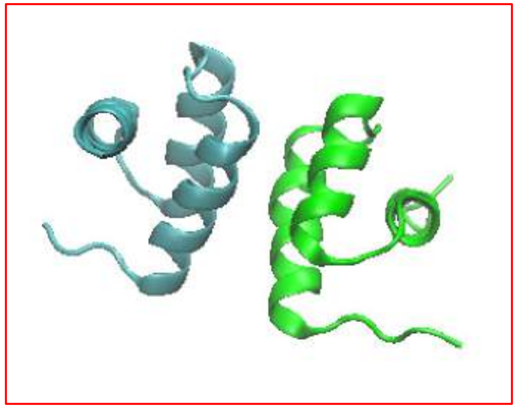
Drawing Method: QuickSurf

Rep2

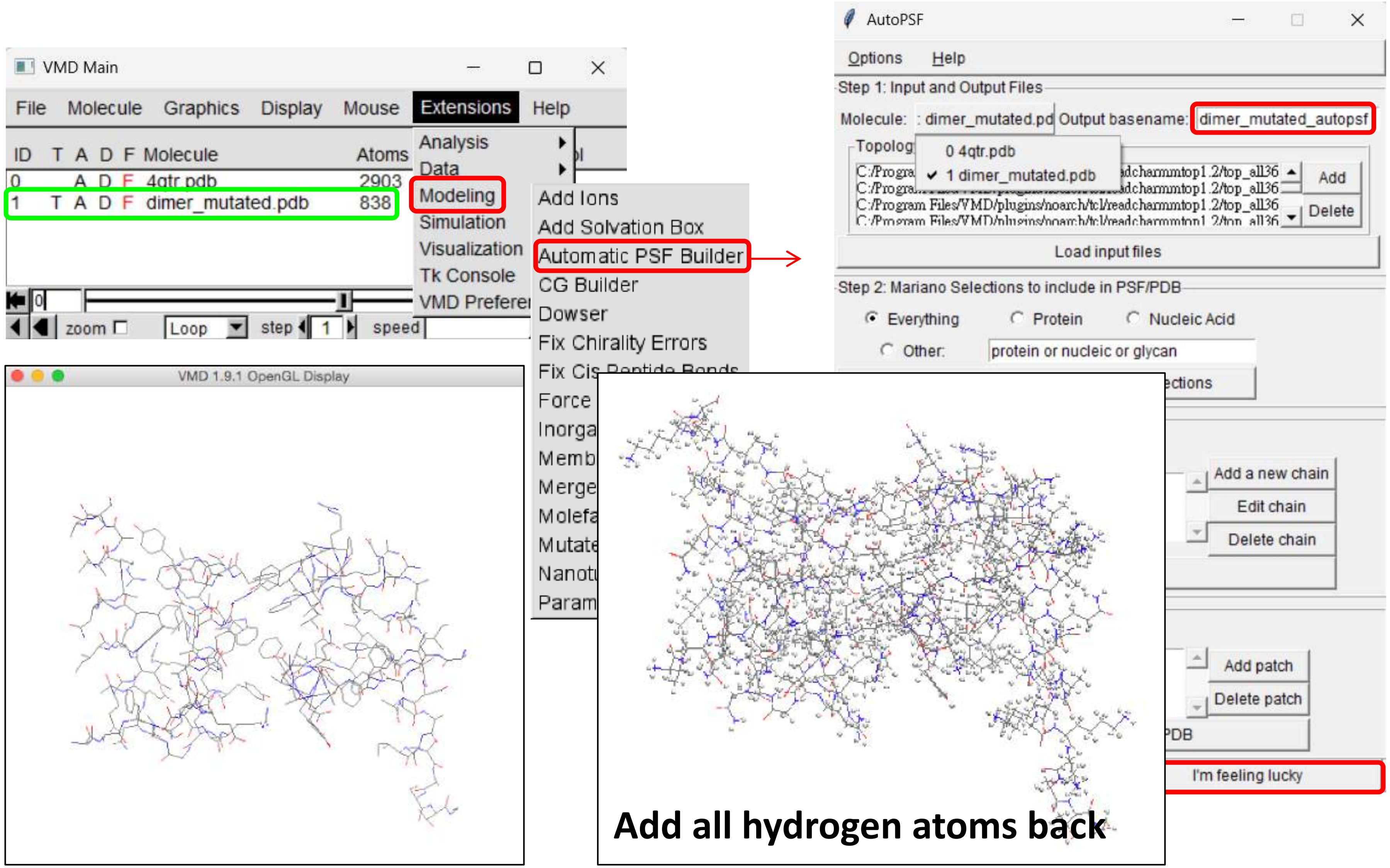
… Chain A B

… ColorID → 10

… NewCartoon

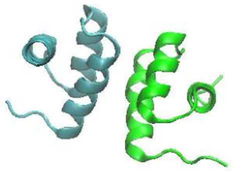


# Use PSF Builder to Generate **All-Atom** Model



# NAMD Energy Plug-in

The ***NAMD Energy*** plugin provides an interface for evaluating energies using *NAMD*. *NAMD Energy* will apply the desired energy calculations to each frame of the selected molecule.



input

coordinates →

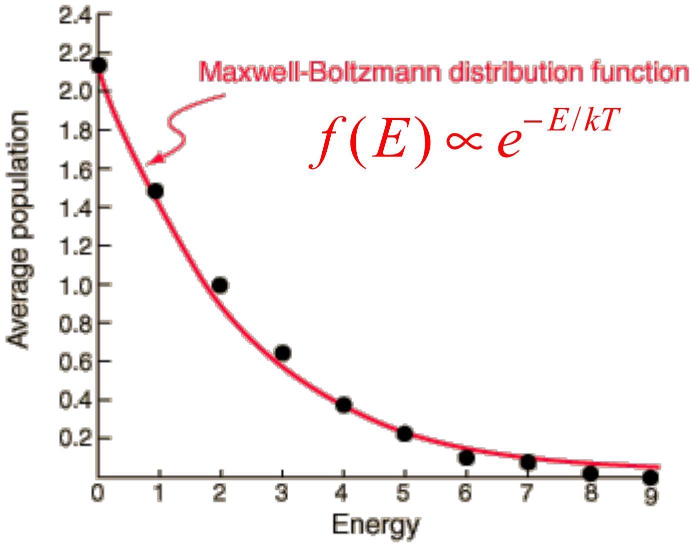
Simulator
(NAMD)

CHARMM, GROMOS, AMBER…

compute →

*E* (Energy)

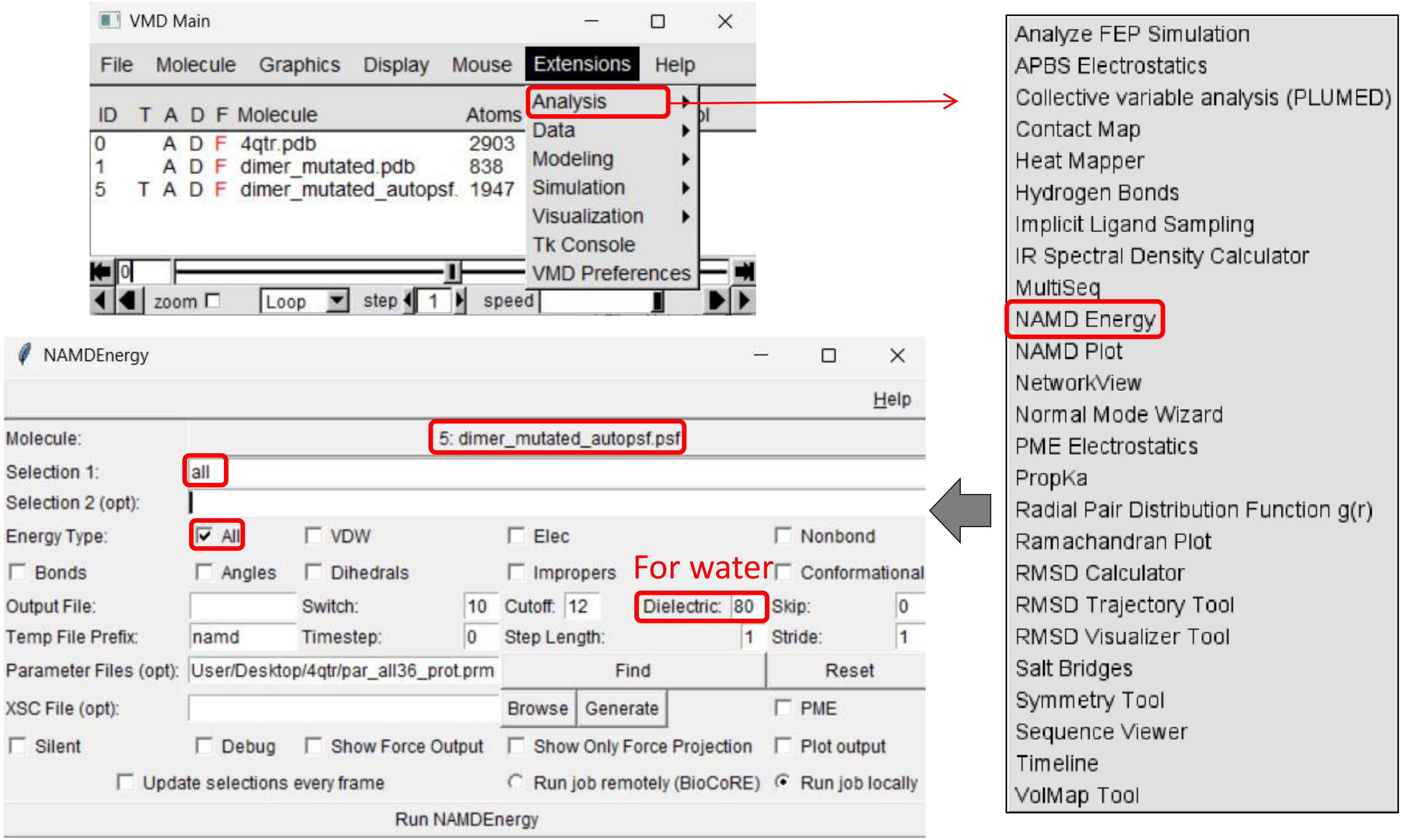
output



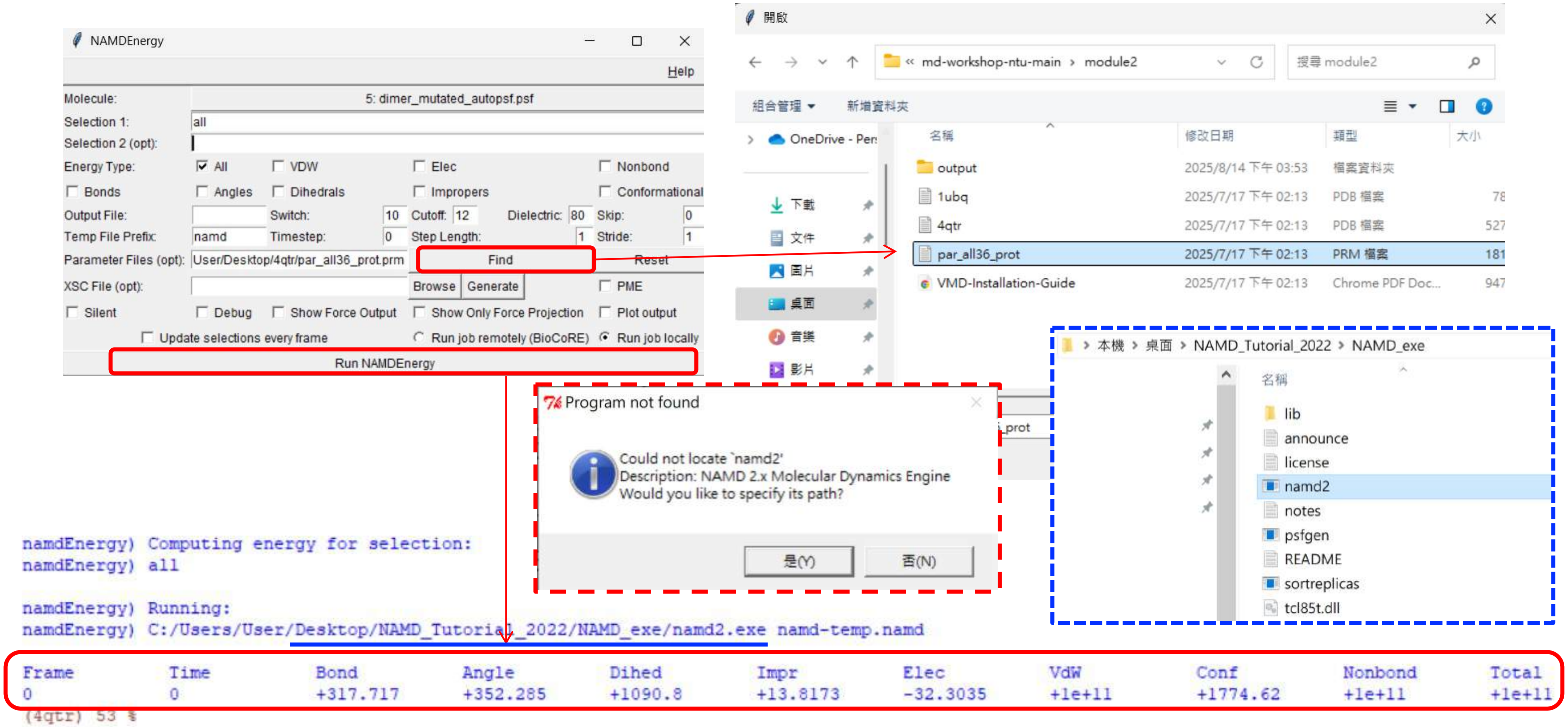

The energy of the system correlates with the system's population through *Boltzmann distribution*.

http://hyperphysics.phy-astr.gsu.edu

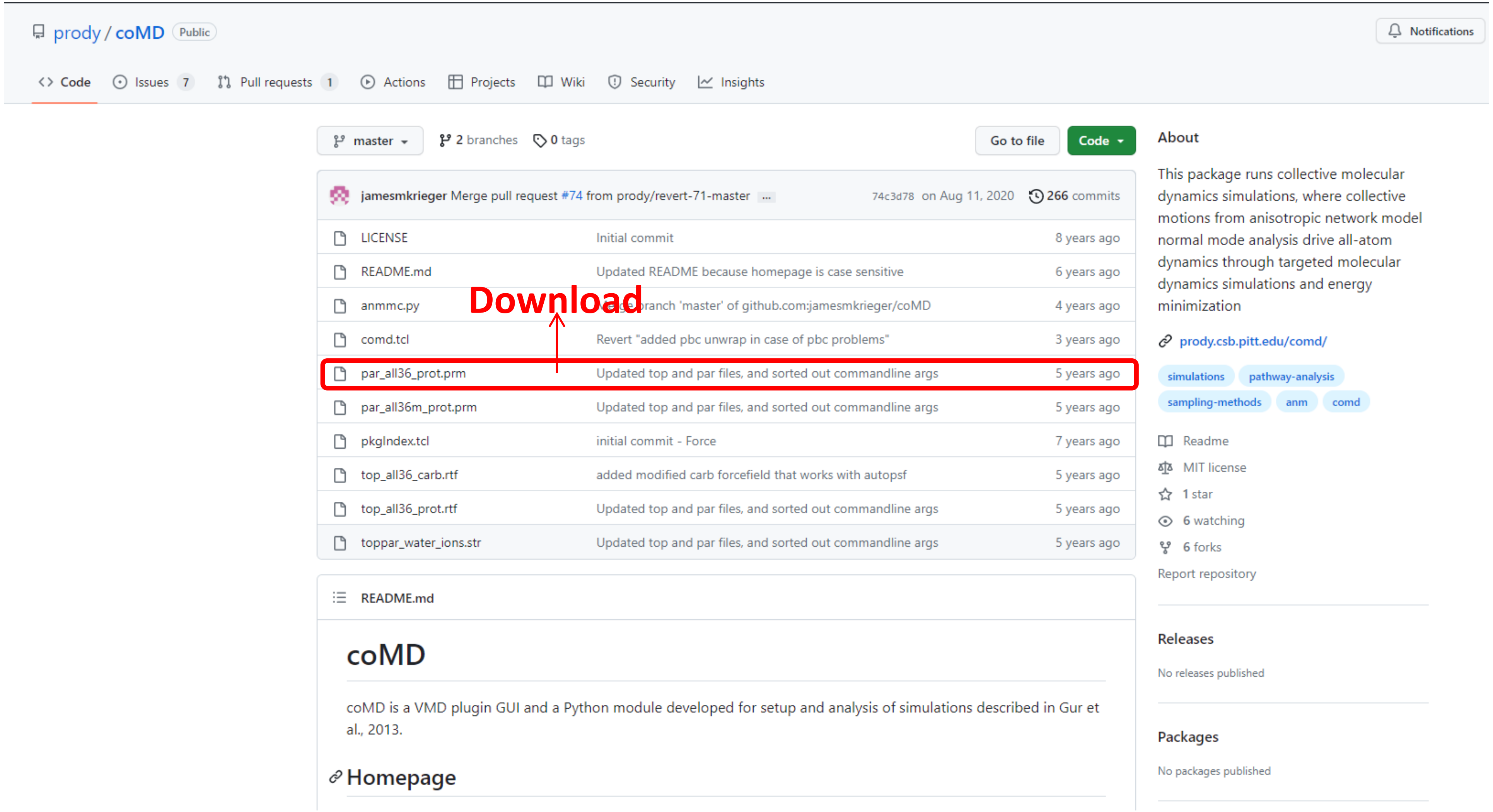
# NAMD Energy Plug-in



# NAMD Energy Plug-in

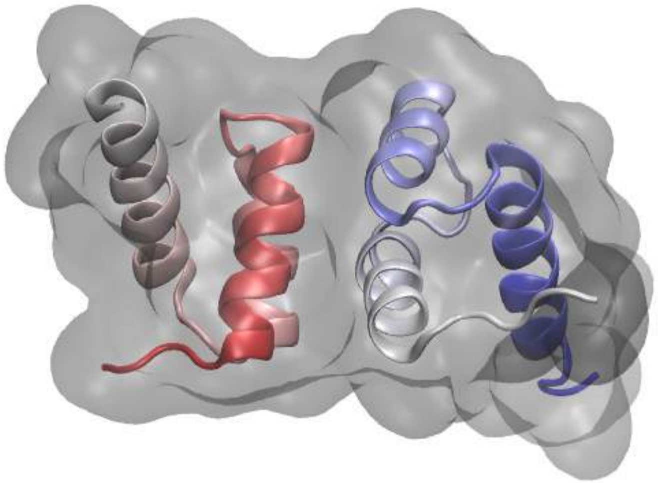


# CHARMM Force Field Files (if necessary)

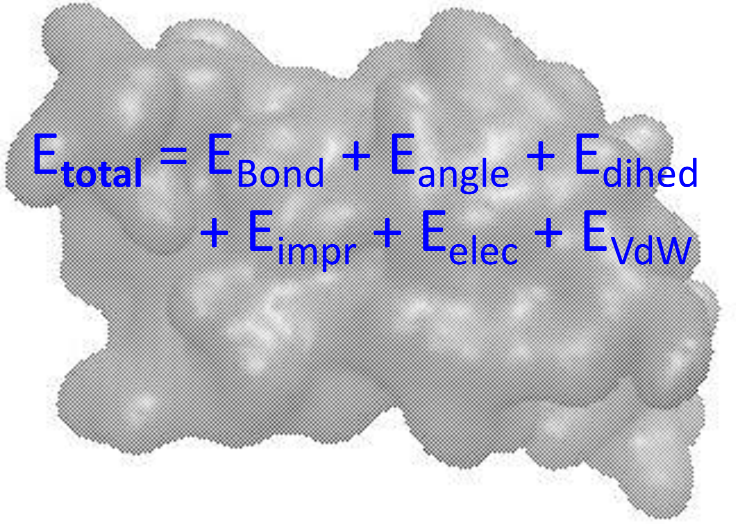



https://github.com/prody/coMD

# NAMD Energy Plug-in



**Total Energy**



$$E_{\mathbf{total}} = E_{Bond} + E_{angle} + E_{dihed} + E_{impr} + E_{elec} + E_{VdW}$$

**Electronic energy as the effective potential experienced by the nuclei**
**➔ Molecular Mechanics**

# NAMD Energy Plug-in

Check the result with
***Extensions → Tk console***

```
namdEnergy) Computing energy for selection:
namdEnergy) all

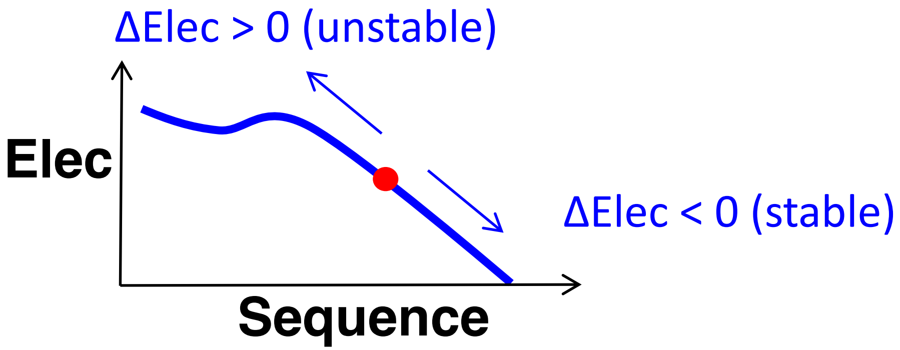
namdEnergy) Running:
namdEnergy) C:/Users/User/Desktop/NAMD_Tutorial_2022/NAMD_exe/namd2.exe namd-temp.namd
```

| Frame | Time | Bond | Angle | Dihed | Impr | Elec | VdW | Conf | Nonbond | Total |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | +317.717 | +352.285 | +1090.8 | +13.8173 | -32.3035 | +1e+11 | +1774.62 | +1e+11 | +1e+11 |

```
(4qtr) 53 %
```

**In kcal/mol**

**Evaluate the contribution of electrostatic energy from all the charged residues in the molecule.**

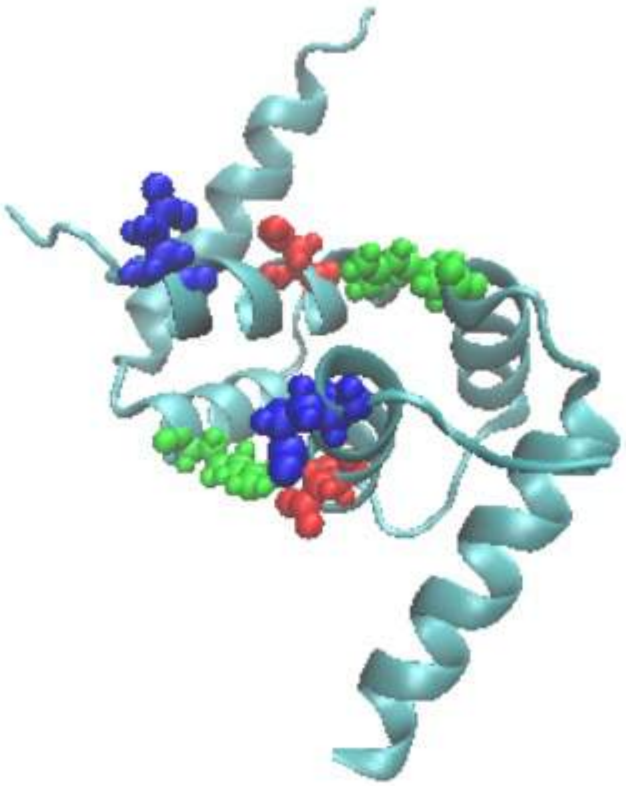


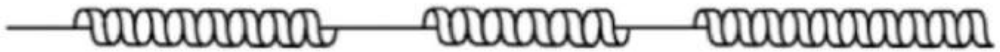
# Homework

For the sequence of the engineered dimer structure, use the **NAMD Energy** plug-in to calculate its energy and compare the results with those of the individual mutation versions **(E10S, D17K, and R37E; see the table below)**. Which of the sequences is electrostatically more favorable?
(Use ***dimer_mutated_autopsf.pdb*** and ***dimer_mutated_autopsf.psf***)
Then, use VMD to highlight these residues and render an image.

[Hint] Use the NAMD Energy plug-in to compute the electrostatic energy. In VMD, add a new representation to highlight the selected residues, choose a coloring method of interest, and then render the resulting image.

What do you think is the disadvantage of using this approach in designing a new protein dimer? Please comment on it.

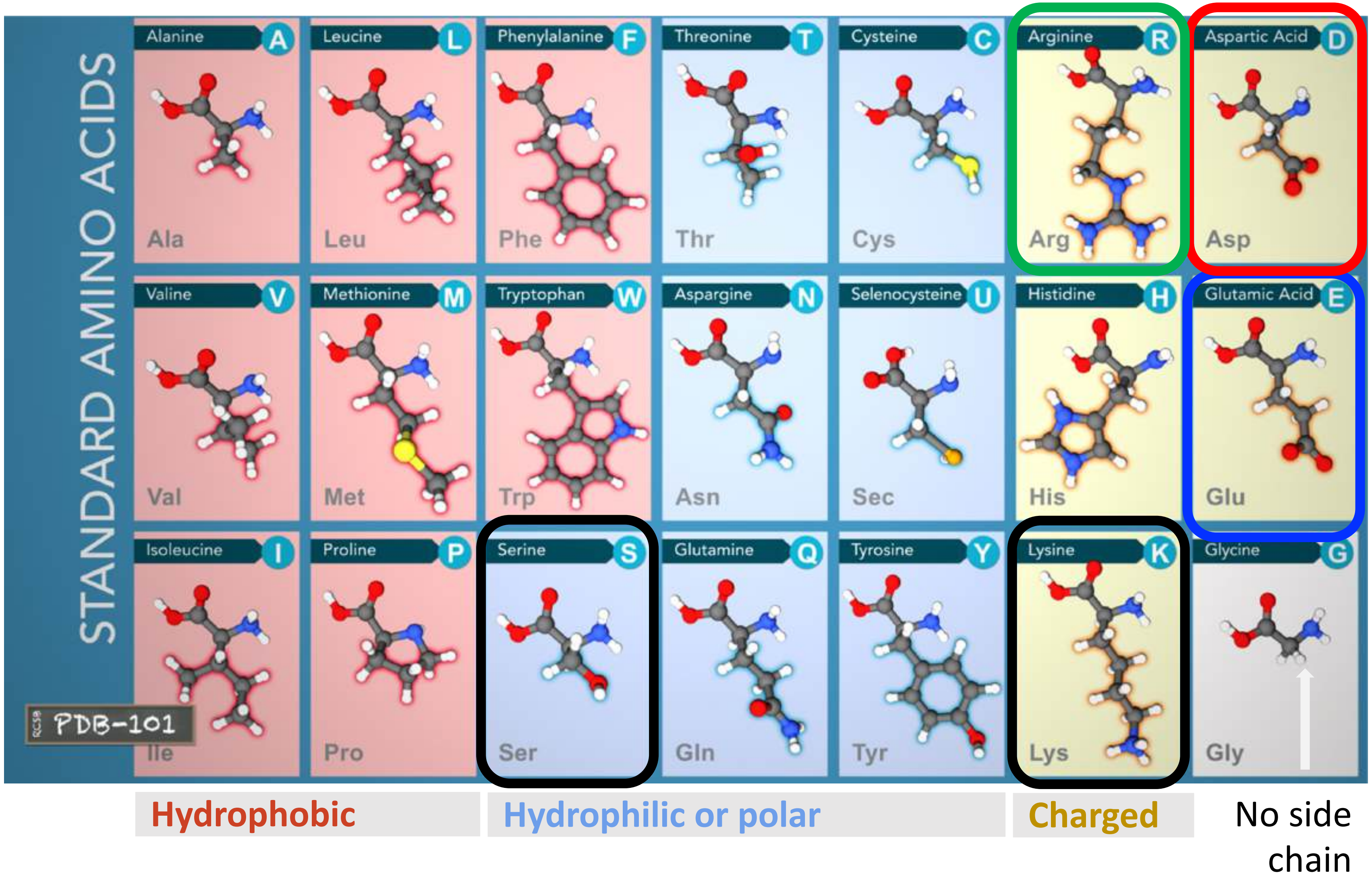




| | |
|---|---|
| ENH (WT) | EKRPRTAFSSEQLARLKREFNENRYLTERRRQQLSSELGLNEAQIKIWFQNKRAKIKKST |
| dualENH | ---------E--KKA-DLA-YFD----PEW-RY--QR----------------------- |
| | 1 10 20 30 40 50 60 |

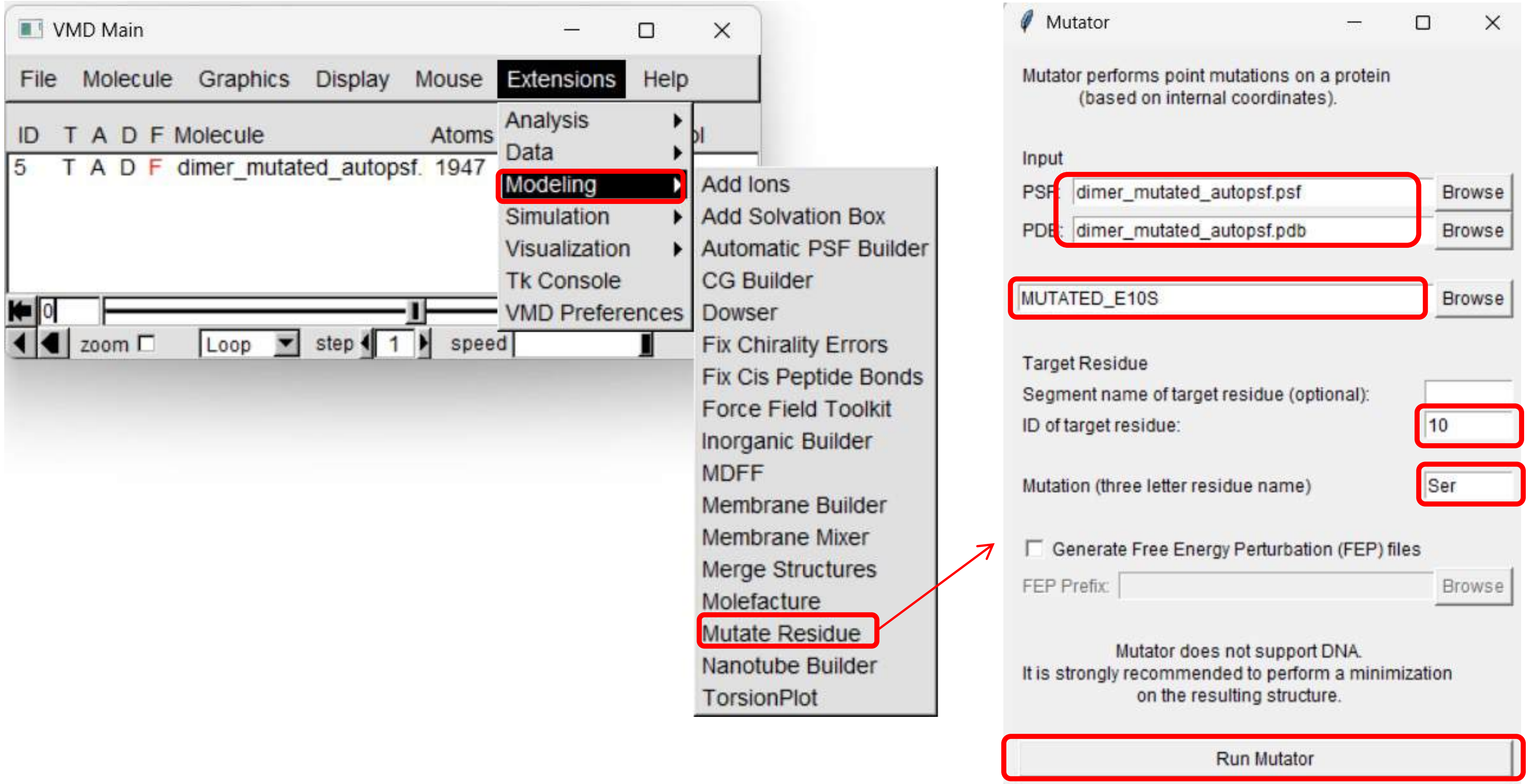
**Blue: GLU10**
**Red: ASP17**
**Green: ARG37**

Mou et al., *Nature* **2015**, 525, 230


STANDARD AMINO ACIDS
Alanine A
Ala
Leucine L
Leu
Phenylalanine F
Phe
Threonine T
Thr
Cysteine C
Cys
Arginine R
Arg
Aspartic Acid D
Asp
Valine V
Val
Methionine M
Met
Tryptophan W
Trp
Aspargine N
Asn
Selenocysteine U
Sec
Histidine H
His
Glutamic Acid E
Glu
Isoleucine I
Ile
Proline P
Pro
Serine S
Ser
Glutamine Q
Gln
Tyrosine Y
Tyr
Lysine K
Lys
Glycine G
Gly
PDB-101
Hydrophobic
Hydrophilic or polar
Charged
No side chain

# Mutator Plug-in

The ***mutator*** plugin provides a very simple method for mutating a target residue selected by its ID, and the three character residue code for the mutant amino acid.



# Resources

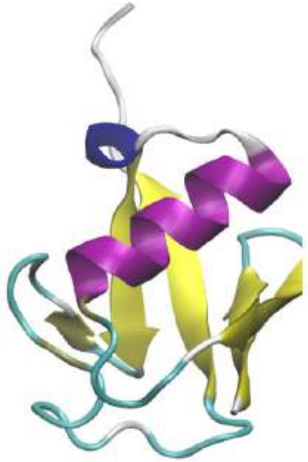
VMD Tutorials
http://www.ks.uiuc.edu/Research/vmd/current/docs.html#tutorials

VMD plugins
http://www.ks.uiuc.edu/Research/vmd/plugins/

Tutorials and Scientific Case Studies
http://www.ks.uiuc.edu/Training/Tutorials/

# Part III

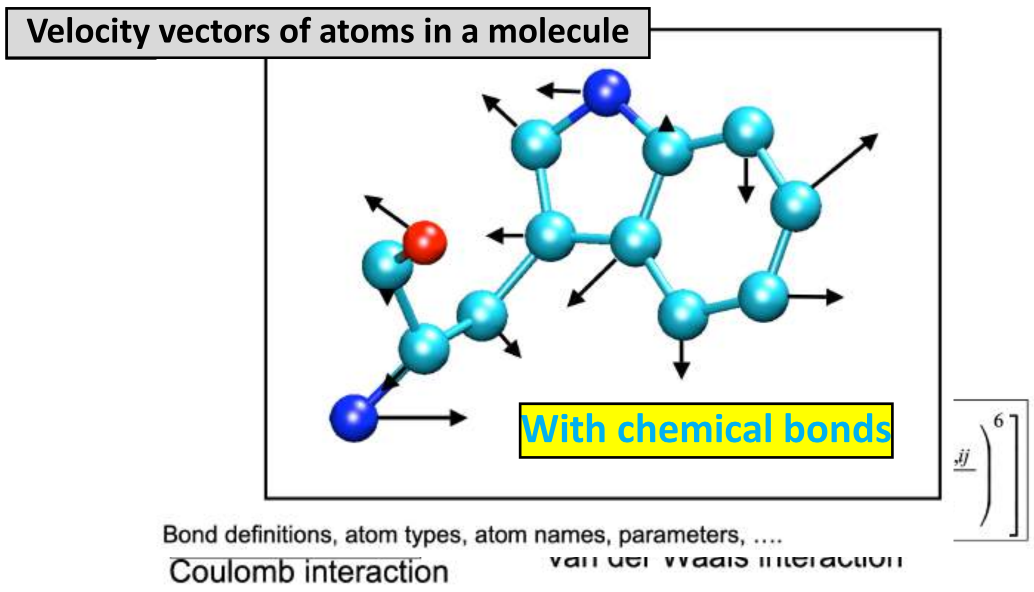
(MD in Bio)

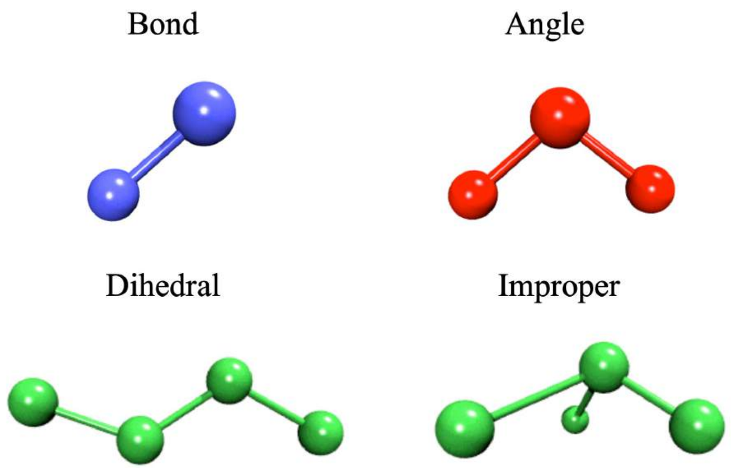


**III. MD (Bio)**

- NAMD/VMD
  - Protein
  - Solvation
  - Add ions

# 分子力場/Molecular Force Fields




**Electronic energy as the effective potential experienced by the nuclei**

$$U(\vec{R}) = \underbrace{\sum_{bonds} k_i^{bond}(r_i - r_0)^2}_{U_{bond}} + \underbrace{\sum_{angles} k_i^{angle}(\theta_i - \theta_0)^2}_{U_{angle}} + \underbrace{\sum_{dihedrals} k_i^{dihe}[1 + \cos(n_i\phi_i + \delta_i)]}_{U_{dihedral}} + \underbrace{\sum_i \sum_{j \neq i} 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \sum_i \sum_{j \neq i} \frac{q_i q_j}{\epsilon r_{ij}}}_{U_{nonbond}}$$

$U_{bond}$ = oscillations about the equilibrium bond length

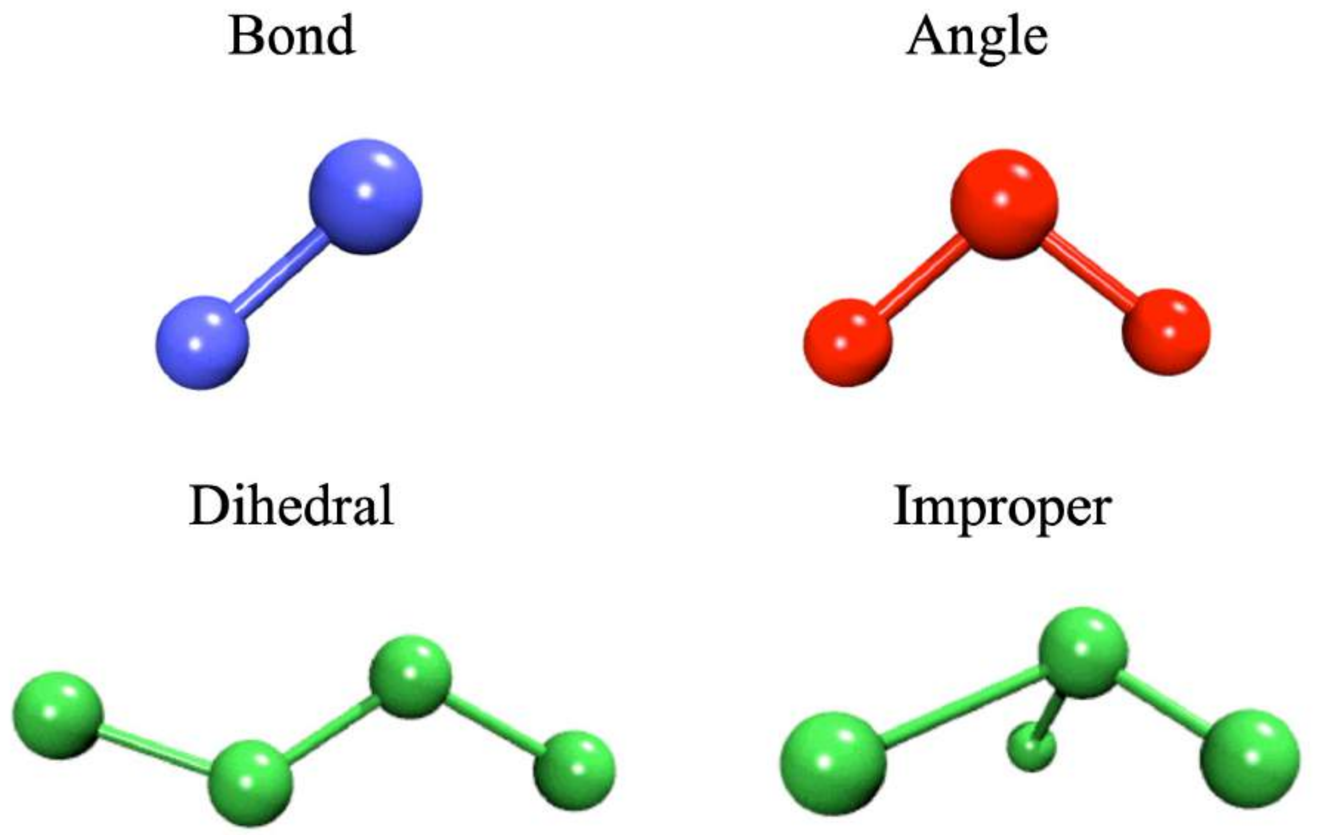
$U_{angle}$ = oscillations of 3 atoms about an equilibrium bond angle

$U_{dihedral}$ = torsional rotation of 4 atoms about a central bond

$U_{nonbond}$ = non-bonded energy terms (electrostatics and Lenard-Jones)

Material from Computational Biophysics Workshop at UIUC 2009

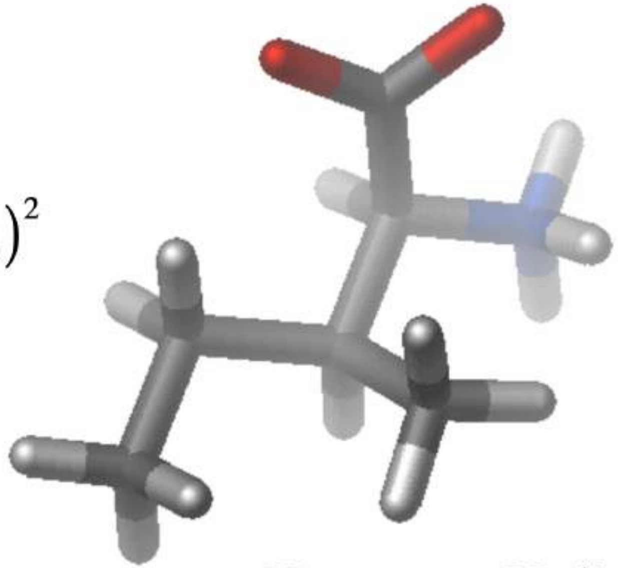
# 能量項/CHARMm Force Field



# 透過化學鍵的作用力/Bonded Interactions

$$V_{angle} = K_\theta(\theta - \theta_o)^2$$

$$V_{bond} = K_b(b - b_o)^2$$

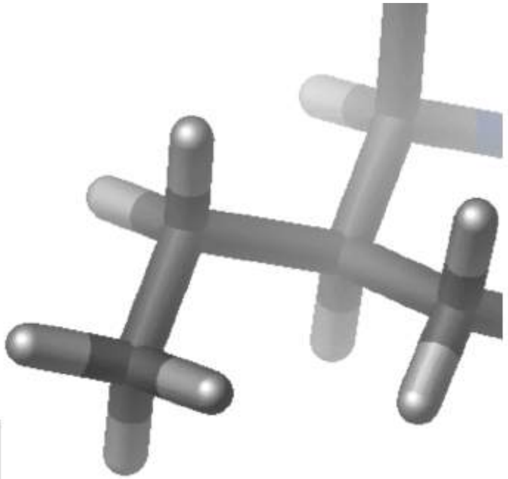


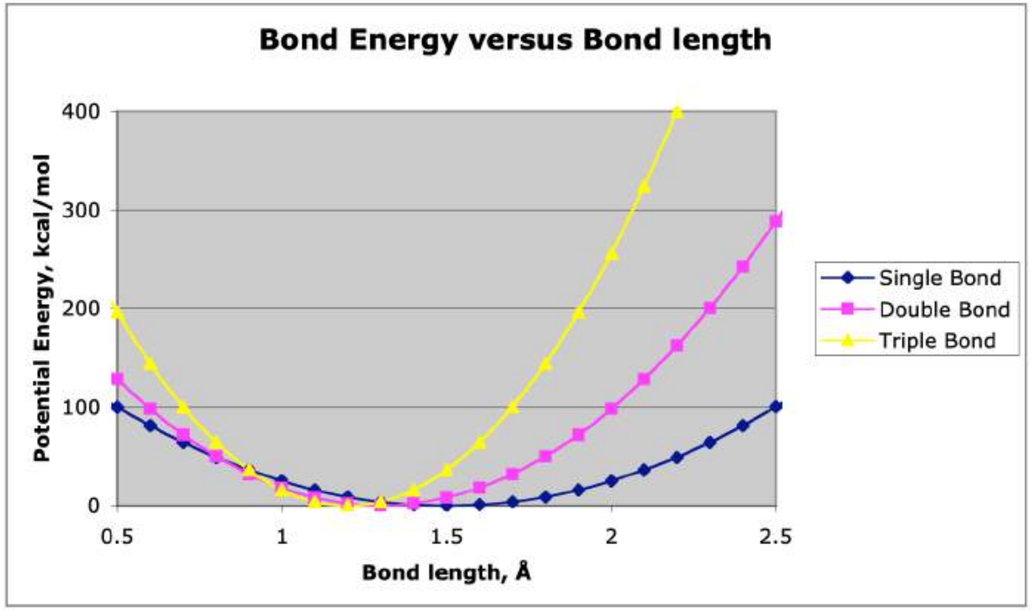
$$V_{dihedral} = K_\phi(1 + \cos(n\phi - \delta))$$

# 鍵長/ Bond Length

$$V_{bond} = K_b (b - b_o)^2$$

| Chemical type | $K_{bond}$ | $b_o$ |
|---|---|---|
| C-C | 100 kcal/mole/Å $^2$ | 1.5 Å |
| C=C | 200 kcal/mole/Å $^2$ | 1.3 Å |
| C≡C | 400 kcal/mole/Å $^2$ | 1.2 Å |





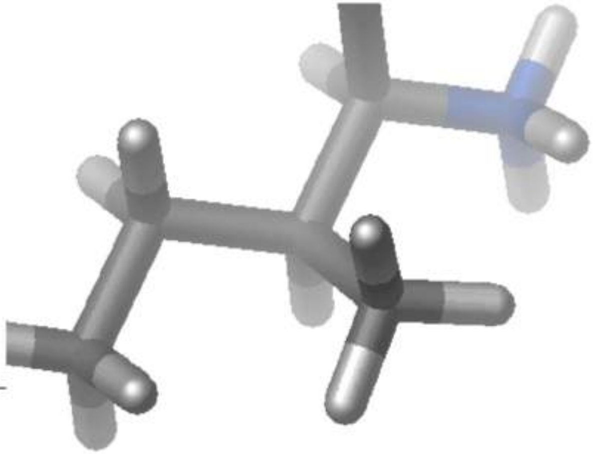

*Bond angles* and *improper* terms have similar quadratic forms, but with softer spring constants. The force constants can be obtained from vibrational analysis of the molecule (experimentally or theoretically).

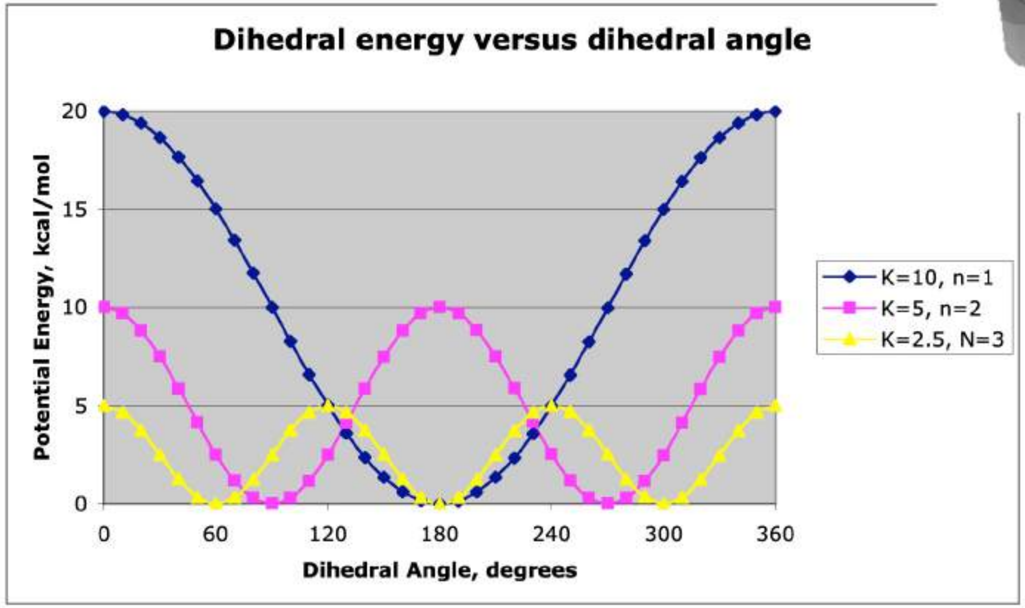
Material from Computational Biophysics Workshop at UIUC 2009

# 雙面角 / Dihedral Angles

## Dihedral Potential

$$V_{dihedral} = K_{\phi}(1+\cos(n\phi-\delta))$$






$\delta = 0^{\circ}$

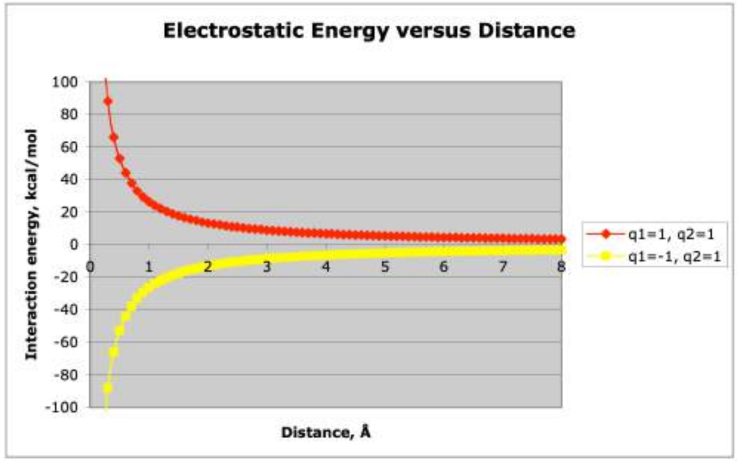
Material from Computational Biophysics Workshop at UIUC 2009

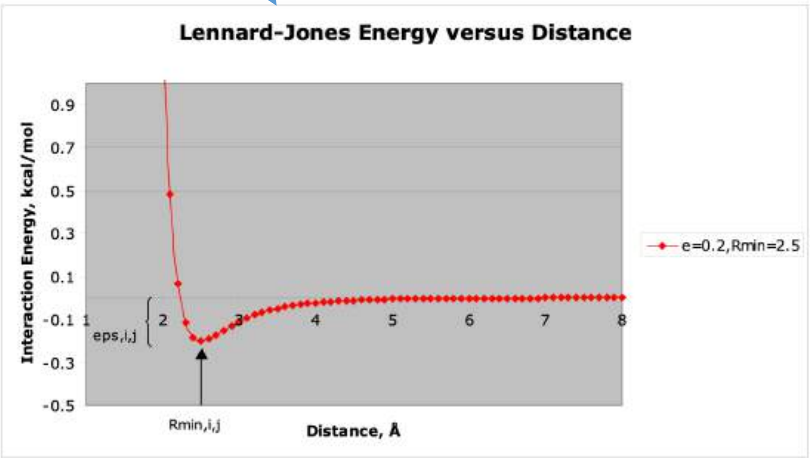
# 非鍵結作用力/Non-bonded Interactions

靜電作用力

$$\sum_{nonbonded} \frac{q_i q_j}{4\pi D r_{ij}} + \varepsilon_{ij}\left[\left(\frac{R_{\min,ij}}{r_{ij}}\right)^{12} - 2\left(\frac{R_{\min,ij}}{r_{ij}}\right)^{6}\right]$$

凡德瓦力




Note that the effect is long range.

$D$: *dielectric constant*

$q_i$: *partial atomic charge*

$\epsilon_{i,j}$: Lennard–Jones (LJ), van der Waals well–depth

$$\epsilon_{i,j} = \sqrt{\epsilon_i \cdot \epsilon_j}$$

$R_{min}$: LJ radius ($R_{\min,ij}/2$ in CHARMM)

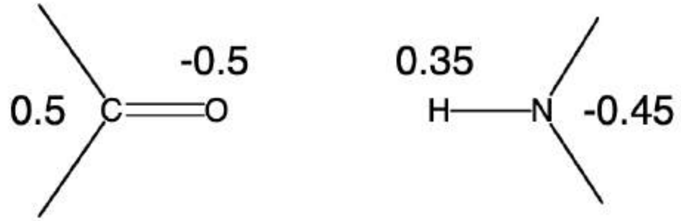
$$R_{min,ij} = R_{min,i} + R_{min,j}$$

Material from Computational Biophysics Workshop at UIUC 2009

# 電荷擬合方法/Charge Fitting Strategy

CHARMM- Mulliken*　　　AMBER(ESP/RESP)

Partial atomic charges




*Modifications based on interactions with TIP3 water

# 主流生物分子力場/Popular Force Fields

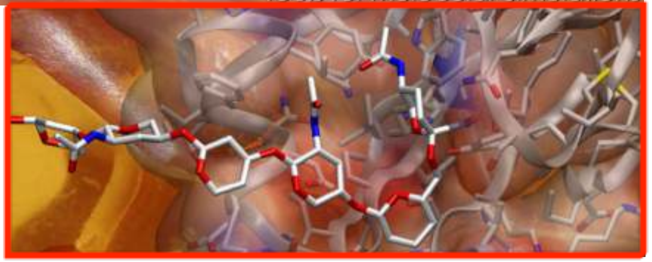
CHARMM
Chemistry at HARvard Macromolecular Mechanics

https://www.charmm.org/

CHARMM是一種用於分子動力學的分子力場，同時，採用這種力場的分子動力學軟體包也採用了這個名稱。CHARMM開發項目包括一個與**馬丁·卡普拉斯**及其在哈佛大學的研究小組合作的開發者網絡，這個網絡包含了美國等地的開發者，協同開發及維護CHARMM軟體包。這個軟體允許進行學術研究的個人及小組在支付一定費用之後得到使用許可。

The Amber Home Page
Tools for Molecular Simulations

**AMBER力場**是在生物大分子的模擬計算領域有著廣泛應用的一個分子力場。開發這個力場的是**Peter Kollman**課題組(UCSF)，最初AMBER力場是專門為了計算蛋白質和核酸體系而開發的，計算其力場參數的數據均來自實驗值，後來隨著AMBER力場的廣泛應用，包括Kollman在內的很多課題組對AMBER力場的內容不斷進行豐富，逐漸開發出了一個可以用於生物大分子、有機小分子和高分子模擬計算的力場體系。但是總體來講，AMBER力場的優勢在於對生物大分子的計算，其對小分子體系的計算結果常常不能令人滿意。



https://ambermd.org/

GROMACS
FAST. FLEXIBLE. FREE.

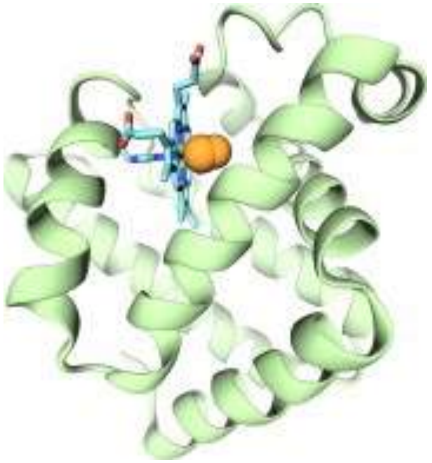
http://www.gromacs.org/

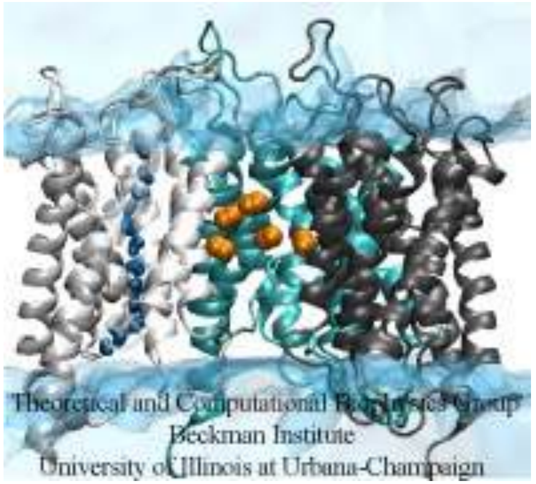
GROMACS是一套分子動力學模擬程序包，主要用來模擬研究蛋白質、脂質、核酸等生物分子的性質。它起初由荷蘭格羅寧根大學生物化學系開發，目前由來自世界各地的大學和研究機構維護。GROMACS是**開源免費**的軟體，4.6版之前的版本使用GNU通用公共許可證發行，而4.6版之後使用GNU寬通用公共許可證發行。

# Visual Molecular Dynamics

http://www.ks.uiuc.edu/Research/vmd/

VMD is designed for **visualization**, **analysis**, and modeling of biological systems such as proteins, nucleic acids, lipid bilayer assemblies, etc.
VMD can read standard Protein Data Bank (**PDB**) files and display the contained structure.






- Support all major computer platforms (ex. MacOS X, Windows, Linux)
- Many molecular rendering and coloring methods

# CHARMM Potential Function

$$U(\vec{R}) = \underbrace{\sum_{bonds} k_i^{bond}(r_i - r_0)^2}_{U_{bond}} + \underbrace{\sum_{angles} k_i^{angle}(\theta_i - \theta_0)^2}_{U_{angle}} +$$

$$\underbrace{\sum_{dihedrals} k_i^{dihe}[1 + \cos(n_i\phi_i + \delta_i)]}_{U_{dihedral}} +$$

$$\underbrace{\sum_i \sum_{j \neq i} 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \sum_i \sum_{j \neq i} \frac{q_i q_j}{\epsilon r_{ij}}}_{U_{nonbond}}$$

PDB file

**geometry**

幾何構型

Topology

PSF file

Parameter file

**parameters**

參數

# File Format/Structure

- The structure of a pdb file
- The structure of a psf file
- The topology file
- The parameter file
- Connection to potential energy terms

# Structure of a PDB File

index name resname chain resid x y z segname

```
ATOM     22  N   ALA B   3      -4.073  -7.587  -2.708  1.00  0.00      BH
ATOM     23  HN  ALA B   3      -3.813  -6.675  -3.125  1.00  0.00      BH
ATOM     24  CA  ALA B   3      -4.615  -7.557  -1.309  1.00  0.00      BH
ATOM     25  HA  ALA B   3      -4.323  -8.453  -0.704  1.00  0.00      BH
ATOM     26  CB  ALA B   3      -4.137  -6.277  -0.676  1.00  0.00      BH
ATOM     27  HB1 ALA B   3      -3.128  -5.950  -0.907  1.00  0.00      BH
ATOM     28  HB2 ALA B   3      -4.724  -5.439  -1.015  1.00  0.00      BH
ATOM     29  HB3 ALA B   3      -4.360  -6.338   0.393  1.00  0.00      BH
ATOM     30  C   ALA B   3      -6.187  -7.538  -1.357  1.00  0.00      BH
ATOM     31  O   ALA B   3      -6.854  -6.553  -1.264  1.00  0.00      BH
ATOM     32  N   ALA B   4      -6.697  -8.715  -1.643  1.00  0.00      BH
ATOM     33  HN  ALA B   4      -6.023  -9.463  -1.751  1.00  0.00      BH
ATOM     34  CA  ALA B   4      -8.105  -9.096  -1.934  1.00  0.00      BH
ATOM     35  HA  ALA B   4      -8.287  -8.878  -3.003  1.00  0.00      BH
ATOM     36  CB  ALA B   4      -8.214 -10.604  -1.704  1.00  0.00      BH
ATOM     37  HB1 ALA B   4      -7.493 -11.205  -2.379  1.00  0.00      BH
ATOM     38  HB2 ALA B   4      -8.016 -10.861  -0.665  1.00  0.00      BH
ATOM     39  HB3 ALA B   4      -9.245 -10.914  -1.986  1.00  0.00      BH
ATOM     40  C   ALA B   4      -9.226  -8.438  -1.091  1.00  0.00      BH
ATOM     41  O   ALA B   4     -10.207  -7.958  -1.667  1.00  0.00      BH
0000000000000000000000000000000000000000000000000000000000000000000000000
        10        20        30        40        50        60        70
```

>>> It is an ascii, fixed-format file <<<

"No connectivity information"

# 拓樸/Topology File

**From top_all22_model.inp**

```
RESI PHEN          0.00   ! phenol, adm jr.
GROUP
ATOM CG    CA     -0.115  !
ATOM HG    HP      0.115  !          HD1   HE1
GROUP                     !           |     |
ATOM CD1   CA     -0.115  !          CD1--CE1
ATOM HD1   HP      0.115  !          //     \\
GROUP                     !     HG--CG       CZ--OH
ATOM CD2   CA     -0.115  !          \      /    \
ATOM HD2   HP      0.115  !          CD2==CE2     HH
GROUP                     !           |     |
ATOM CE1   CA     -0.115  !          HD2   HE2
ATOM HE1   HP      0.115
GROUP
ATOM CE2   CA     -0.115
ATOM HE2   HP      0.115
GROUP
ATOM CZ    CA      0.110
ATOM OH    OH1    -0.540
ATOM HH    H       0.430
BOND CD2 CG CE1 CD1 CZ CE2 CG HG CD1 HD1
BOND CD2 HD2 CE1 HE1 CE2 HE2 CZ OH OH HH
DOUBLE CD1 CG CE2 CD2  CZ CE1
```

From MacKerell

Top_all22_model.inp contains all protein model compounds. Lipid, nucleic acid and carbohydate model compounds are in the full topology files.

HG will ultimately be deleted. Therefore, move HG (hydrogen) charge into CG, such that the CG charge becomes 0.00 in the final compound.

Use remaining charges/atom types without any changes.

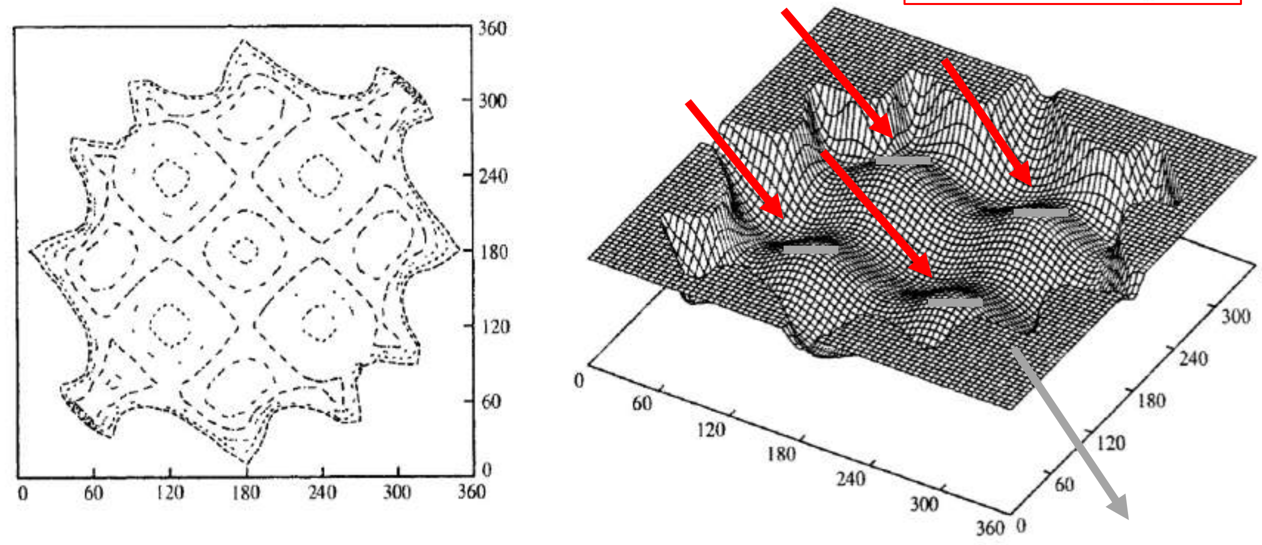
Do the same with indole

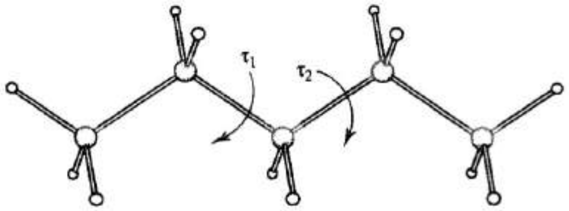
# 能量最佳化/Energy Minimization

- Energy landscape of pentane with two torsion angles

Energy minima (能量極小值)

$$\frac{\partial f}{\partial x_i} = 0; \qquad \frac{\partial^2 f}{\partial x_i^2} > 0$$

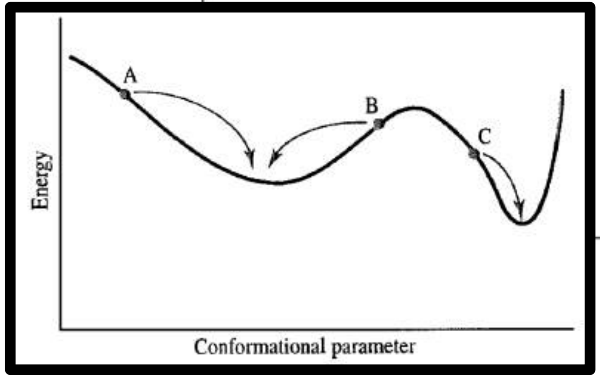



Energy maxima (能量極大值)

$$\frac{\partial f}{\partial x_i} = 0; \qquad \frac{\partial^2 f}{\partial x_i^2} < 0$$

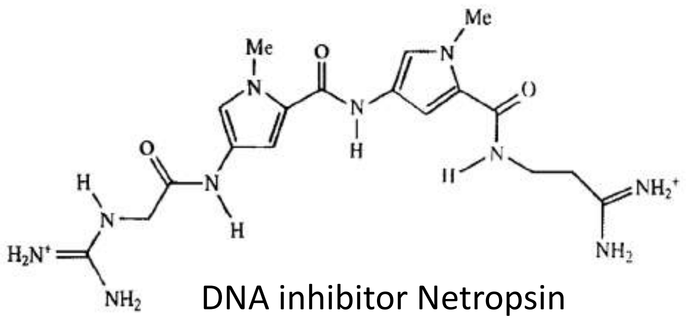
1D 能量示意圖




Leach, Molecular Modelling: Principles and Applications 2nd 2001 Chapter 5

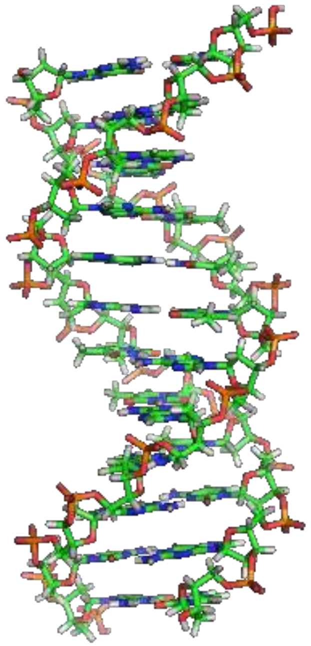
# 能量最佳化/Energy Minimization

- Compare two different methods



DNA inhibitor Netropsin
(紡錘黴素)

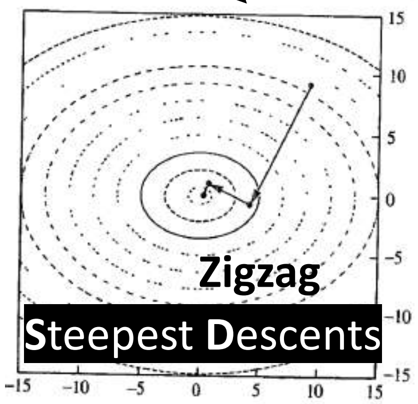
Docking
(對接/結合)

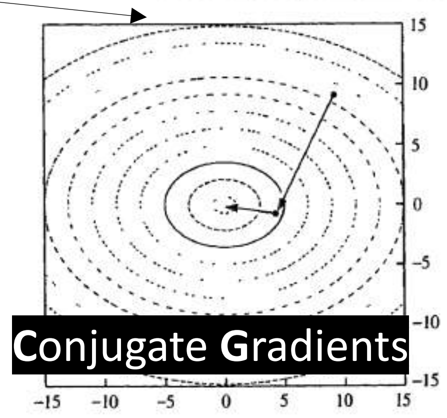


WikiWand:DNA

| | 初步結構 | | 細部結構 | |
|---|---|---|---|---|
| | Initial refinement (Av. gradient <1 kcal Å$^{-2}$) | | Stringent minimisation (Av. gradient <0.1 kcal Å$^{-2}$) | |
| Method | CPU time (s) | Number of iterations | CPU time (s) | Number of iterations |
| Steepest descents | 67 | 98 | 1405 | 1893 |
| Conjugate gradients | 149 | 213 | 257 | 367 |

最速下降法(SD) → Steepest descents
共軛梯度法(CG) → Conjugate gradients




- Steepest Descents → 穩定、慢，但適合「救急」去掉高能接觸。
- Conjugate Gradients → 高效、快，但需要已經「不太差」的初始結構。
- 實務上常常兩者**搭配使用**，形成「粗略 → 精細」的最佳化流程。

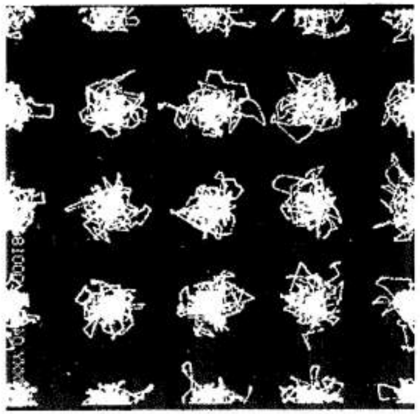
Leach, Molecular Modelling: Principles and Applications 2nd 2001 Chapter 5

# Molecular Dynamics Simulation Method

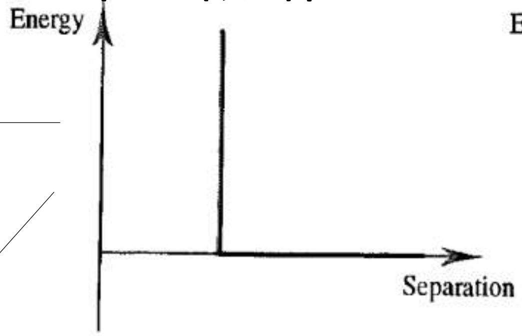
- Integrating Newton's law of motion

$$\frac{d^2x_i}{dt^2} = \frac{F_{x_i}}{m_i}$$

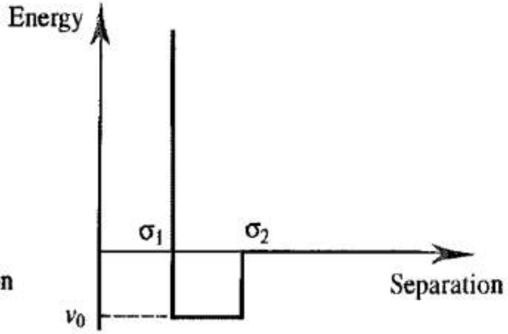
固態 Hard-sphere 模擬



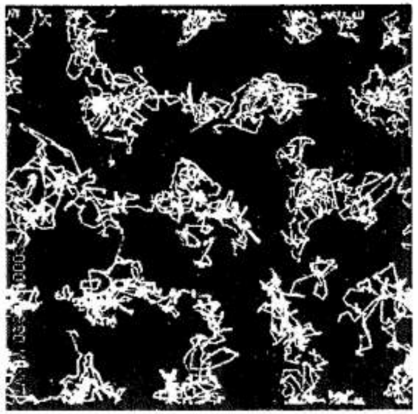
Hard-sphere (硬球) potential




Square-well (方井) potential




液態 Hard-sphere 模擬



Alder and Wainwright, JCP 1959

Leach, Molecular Modelling: Principles and Applications 2nd 2001 Chapter 7

# Molecular Dynamics Simulation Method

- Position and dynamic properties can be approximated as Taylor series expansions

(位置以及動態性質可用泰勒展開式近似)

Position(位置)→ $\mathbf{r}(t+\delta t) = \mathbf{r}(t) + \delta t\mathbf{v}(t) + \frac{1}{2}\delta t^2\mathbf{a}(t) + \frac{1}{6}\delta t^3\mathbf{b}(t) + \frac{1}{24}\delta t^4\mathbf{c}(t) + \cdots$

Velocity(速度)→ $\mathbf{v}(t+\delta t) = \mathbf{v}(t) + \delta t\mathbf{a}(t) + \frac{1}{2}\delta t^2\mathbf{b}(t) + \frac{1}{6}\delta t^3\mathbf{c}(t) + \cdots$

Acceleration(加速度)→ $\mathbf{a}(t+\delta t) = \mathbf{a}(t) + \delta t\mathbf{b}(t) + \frac{1}{2}\delta t^2\mathbf{c}(t) \cdots$

$\mathbf{b}(t+\delta t) = \mathbf{b}(t) + \delta t\mathbf{c}(t) + \cdots$

Truncated
(截斷)
越高次項越準喔~

## Verlet algorithm

$$\mathbf{r}(t+\delta t) = \mathbf{r}(t) + \delta t\mathbf{v}(t) + \frac{1}{2}\delta t^2\mathbf{a}(t) + \quad \cdot$$

$$\mathbf{r}(t-\delta t) = \mathbf{r}(t) - \delta t\mathbf{v}(t) + \frac{1}{2}\delta t^2\mathbf{a}(t) - \cdots$$

相加

Truncated at O($\delta t^4$)
→Fourth-order method, Why?
(仍準確至三次項)

$$\mathbf{r}(t+\delta t) = 2\mathbf{r}(t) - \mathbf{r}(t-\delta t) + \delta t^2\mathbf{a}(t)$$

Leach, Molecular Modelling: Principles and Applications 2nd 2001 Chapter 7

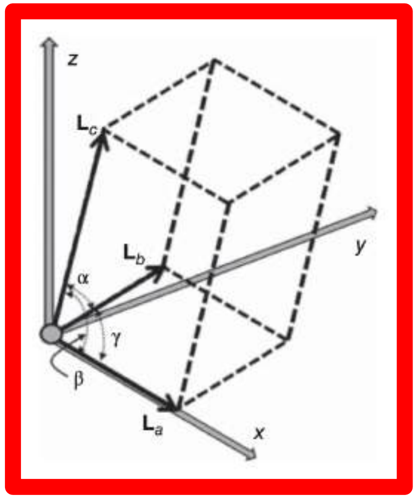
# [補充] 分子模擬的技術細節

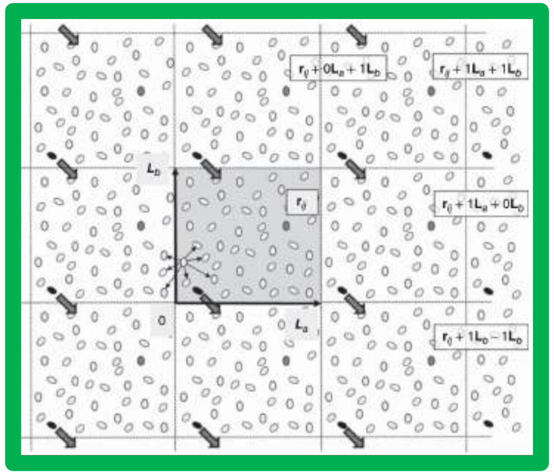
## 1. Initial Setup of the System

→ Determining the shape and size of the container to confine the molecules of interest

→ Defining the initial positions and velocities of the atoms

→ Creating the ***simulation cell***




## 2. Boundary Conditions

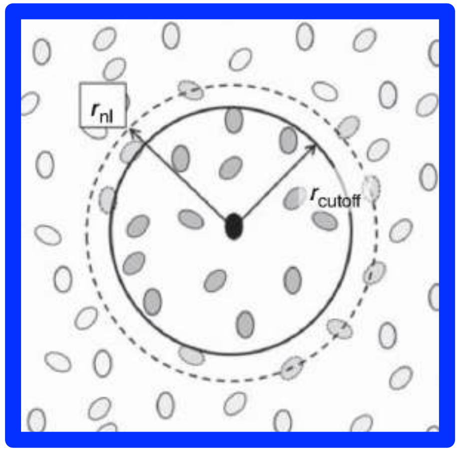
→ Modeling a macroscopic (bulk) system with a single *simulation cell* ($10^6$ instead of $10^{23}$)

→ Applying ***Periodic boundary conditions (PBC)*** to simulate an infinite system

## 3. Treatment of Short-Range and Long-Range Forces

→ Handling short-range forces between nearby atoms using cutoff distances or ***neighbor lists***

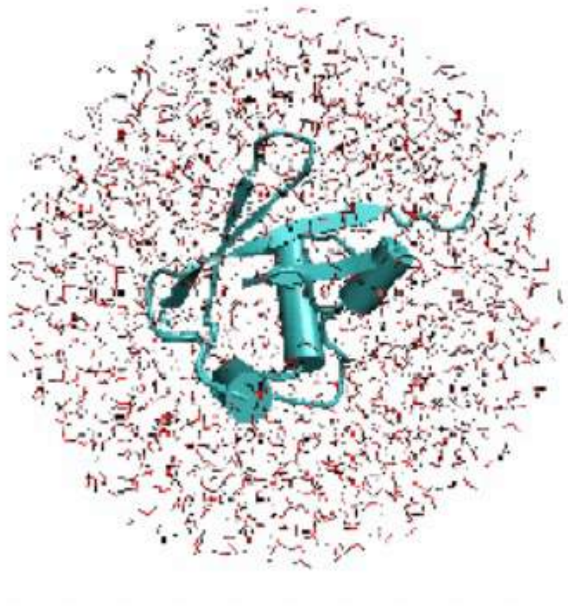
→ Accounting for long-range forces through methods like Ewald summation or *particle mesh Ewald (PME)*



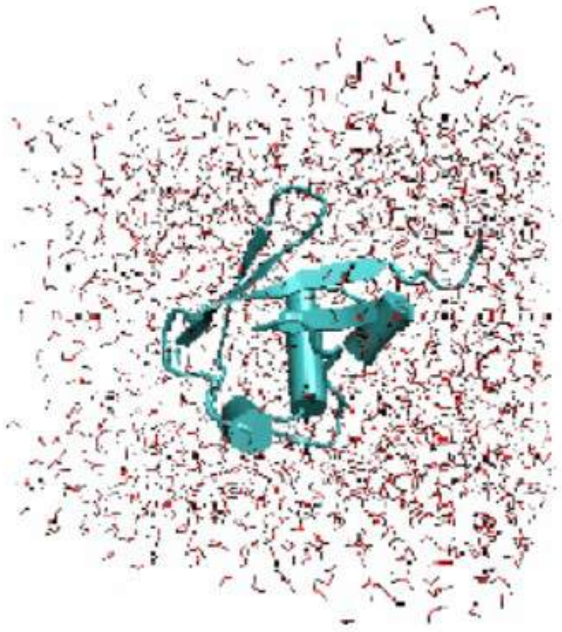



S. Alavi, Molecular Simulations 2020 Chapter 4

# [補充] 表面張力 & 週期邊界條件



vs

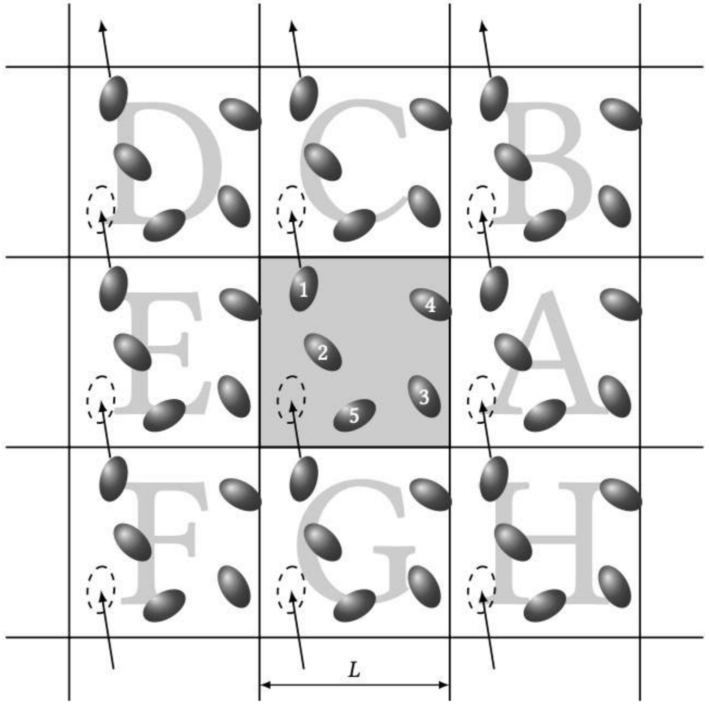


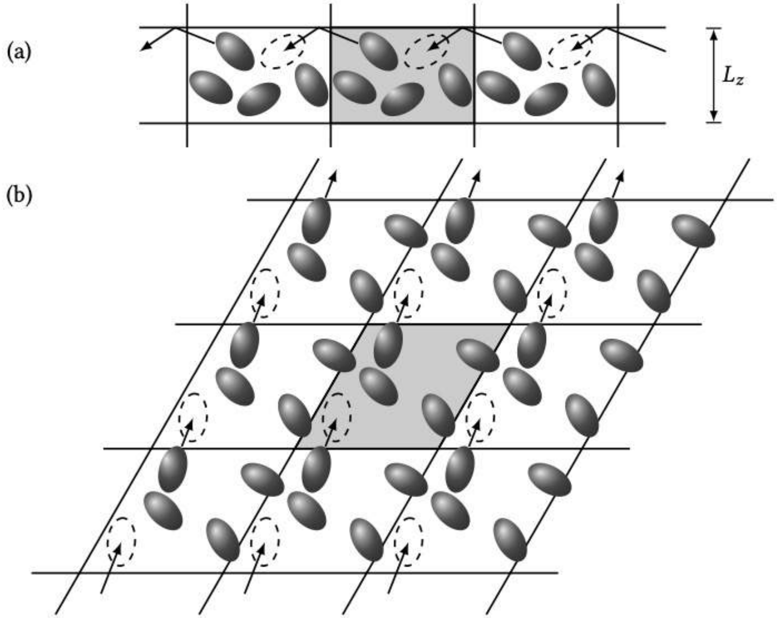
Water sphere

Water box

Section 1.6 in Allen and Tildesley, Computer Simulation of Liquids 2nd, Oxford

# [補充] 表面張力 & 週期邊界條件




Section 1.6 in Allen and Tildesley, Computer Simulation of Liquids 2nd, Oxford

# Tutorial: 蛋白質模擬 (Protein)

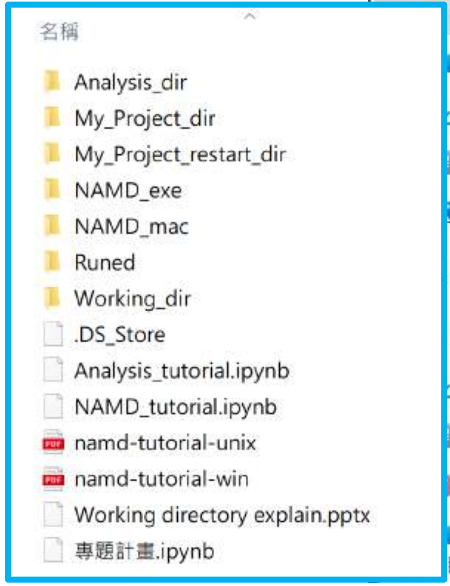
方法一：

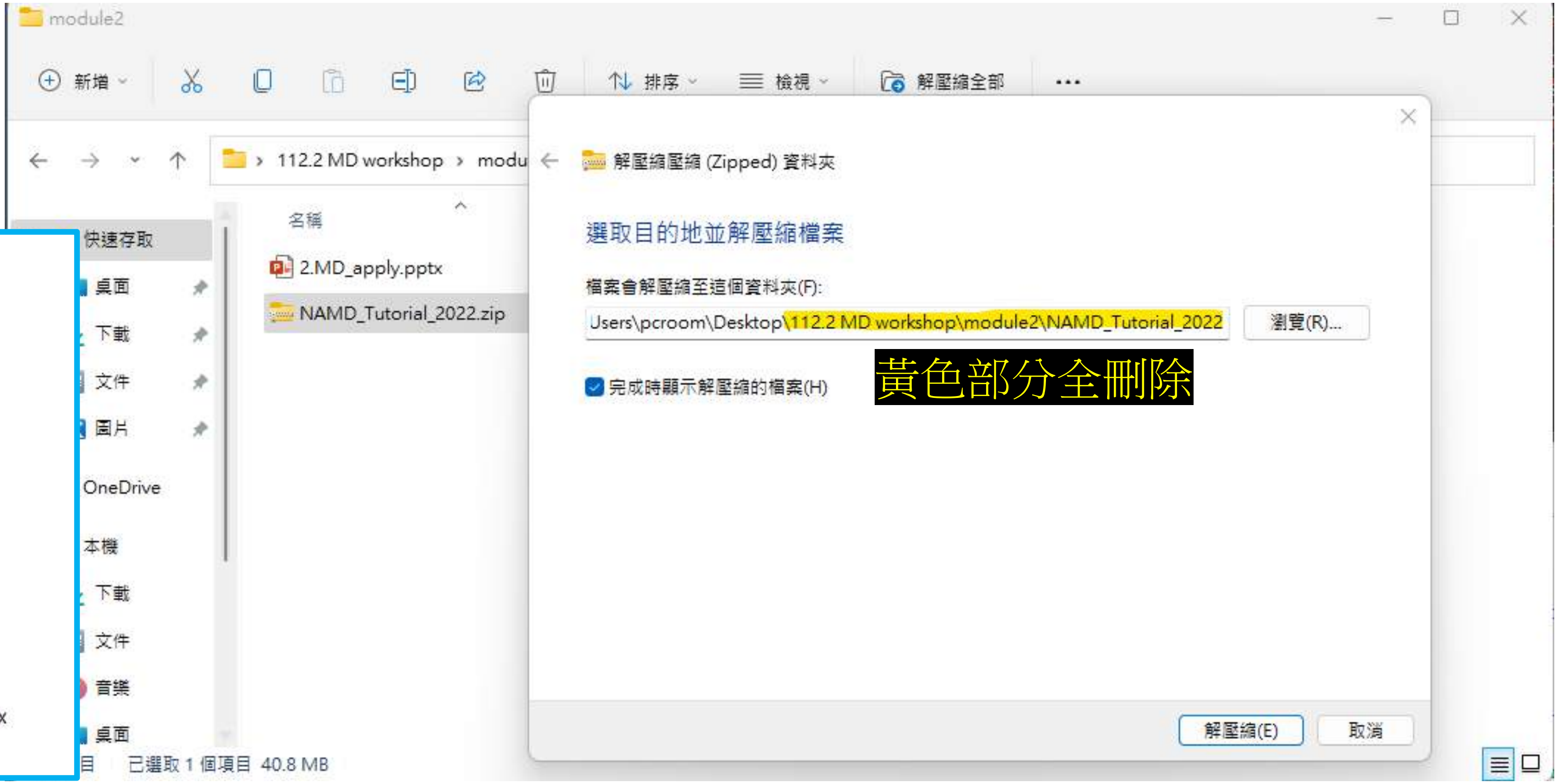
1. 從 113.2 MD workshop/module3 中，將 **NAMD_Tutorial_2025.zip** 資料夾拖曳至桌面
2. 按滑鼠右鍵 → 解壓縮檔案… → 目的地路徑 \\.....\\.....\\Desktop\\(此處全刪除)
3. 解壓縮

方法二：

或是直接從 **NAMD_Tutorial_2025.zip** 上按右鍵 → 解壓縮檔案…

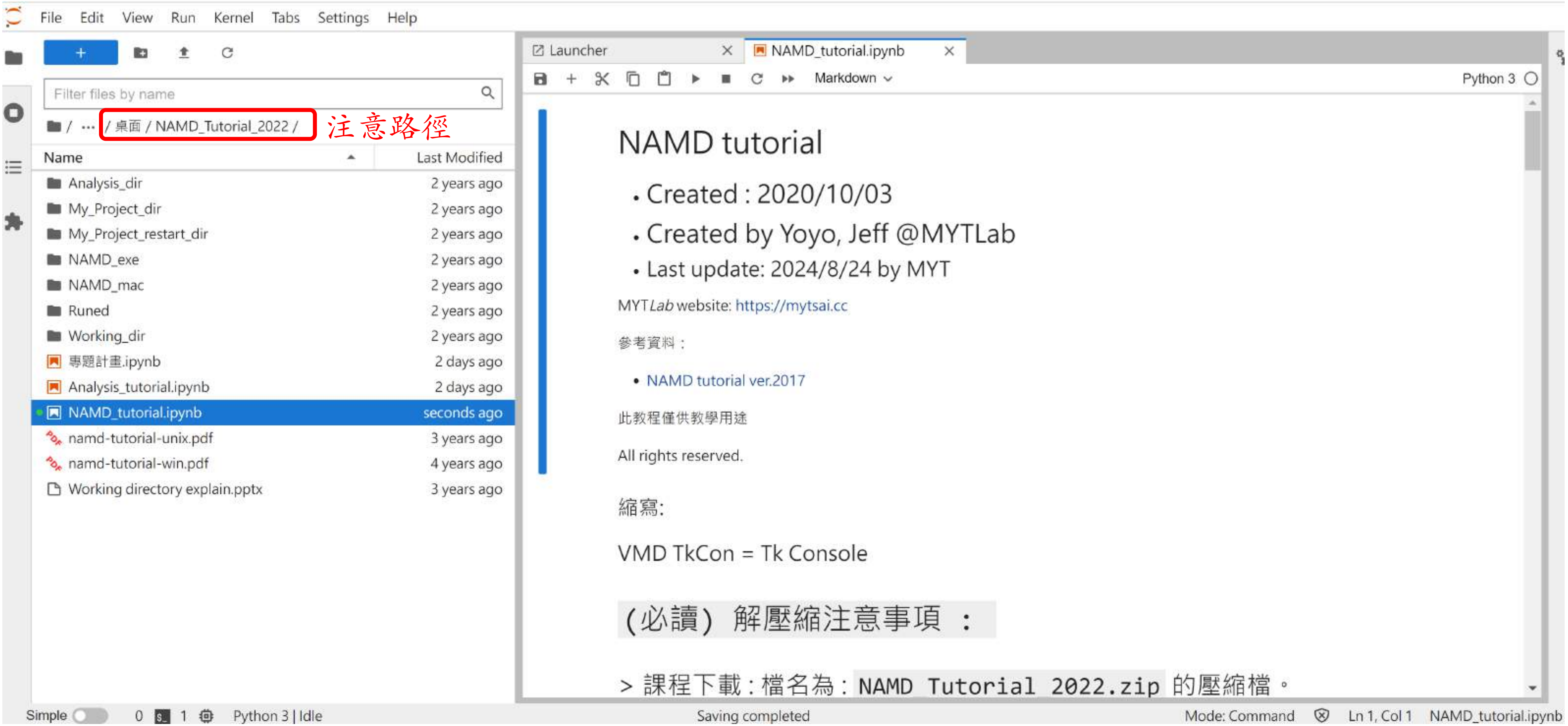
桌面

# Tutorial: 蛋白質模擬 (Protein)

## 開啟 JupyterLab



# 補充 configuration file 參數說明
(以 NAMD_Tutorial_2022/Working_dir 內檔案為例說明)

# Configuration File (ubq_wb_eq.conf)

ubq_wb_0.15

```
structure              ubq_wb.psf
coordinates            ubq_wb.pdb
```

**structure:** 指定系統的 PSF 檔案（Protein Structure File），定義了原子的拓撲結構 (Topology)，如鍵結、角度、質量、原子類型等。

**coordinates:** 指定系統的 PDB 檔案 (Protein Data Bank )，含有每個原子的初始三維座標（但不含鍵結資訊）。

```
set temperature     310
set outputname      ubq_wb_0.15_eq
firsttimestep       0
```

**temperature:** 設定模擬的目標溫度（單位 K）。

**outputname:** 定義輸出的檔名前綴。

**firsttimestep:** 模擬開始的步數。

```
paraTypeCharmm      on
parameters          par_all27_prot_lipid.inp
temperature         $temperature
```

**paraTypeCharmm:** 是否使用 CHARMM 格式的力場 (此處為 "on")。

**parameters:** 指定參數檔（含力場常數、非鍵結參數等）。

**temperature:** 設定模擬的初始溫度 (溫度與上方相同)。

```
# Force-Field Parameters
exclude             scaled1-4
1-4scaling          1.0
cutoff              12.0
switching           on
switchdist          10.0
pairlistdist        14.0
```

**exclude scaled1-4:** 排除共價鍵連接的 1-2, 1-3 原子間交互作用，1-4 原子進行縮放。

**1-4scaling:** 縮放 1-4 靜電作用力大小 (此處為 1.0，表示不縮放)。

**cutoff:** 非鍵結力截斷半徑，計算距離內凡德瓦力 (此處為 12 Å)。

**switching :** 是否啟用 switching function 平滑處理在 cutoff 區間內的力變化 (此處為 **on**)。

**switchdist:** switching function 開始的距離（此處為 10 Å）。

**pairlistdist:** 原子對列表更新的距離（此處為 14 Å ）。

```
# Integrator Parameters
timestep            2.0   ;# 2fs/step
rigidBonds          all   ;# needed for 2fs steps
nonbondedFreq       1
fullElectFrequency  2
stepspercycle       10
```

**timestep:** 積分步長，單位為 fs (此處為 2 fs)。

**rigidBonds:** 使用 SHAKE 讓氫鍵長固定 (需使用 2 fs 步長)。

**nonbondedFreq:** 非鍵結力計算頻率 (此處為 1 step)。

**fullElectFrequency:** 完全電場計算頻率 (此處為 2 steps) 。

**stepspercycle:** 定義內部時間整合器如何平衡力的更新頻率與效率 (此處為 10 steps) 。

```
# Constant Temperature Control
langevin            on     ;# do langevin dynamics
langevinDamping     1      ;# damping coefficient (gamma) of 1/ps
langevinTemp        $temperature
langevinHydrogen    off     ;# don't couple langevin bath to hydrogens
```

**langevin :** 是否啟用 Langevin 熱浴模擬恆溫條件 (此處為 on) 。

**langevinDamping:** 阻尼係數，控制與熱浴耦合強度 (此處為1 $ps^{-1}$) 。

**langevinTemp:** 設定目標溫度 (溫度與上方相同)。

**langevinHydrogen :** 是否對氫原子施加 Langevin 力 (此處為 off)。

```
# Periodic Boundary Conditions
cellBasisVector1     40.6     0.    0.0
cellBasisVector2      0.0  42.2    0.0
cellBasisVector3      0.0     0    46.2
cellOrigin           30.8   29.0   17.7
wrapAll              on
```

**cellBasisVector[1-3]:** 定義模擬盒的邊界向量，這裡是正交盒。

**cellOrigin:** 模擬盒的原點位置 。

**wrapAll:** 是否將穿越邊界原子包裹回盒內 (此處為 on)。

```
# PME (for full-system periodic electrostatics)
PME                 yes
PMEGridSpacing      1.0
```

```
# Constant Pressure Control (variable volume)
useGroupPressure      yes ;# needed for rigidBonds
useFlexibleCell       no
useConstantArea       no

langevinPiston        on
langevinPistonTarget  1.01325 ;#  in bar -> 1 atm
langevinPistonPeriod  100.0
langevinPistonDecay   50.0
langevinPistonTemp    $temperature
```

**useGroupPressure:** 是否使用群體壓力方法 (此處為 yes) 。

**useFlexibleCell:** 是否讓模擬盒在所有方向均等調整 (此處為 no) 。

**useConstantArea:** 模擬盒是否固定 XY 平面面積、僅允許 Z 軸改變體積 (此處為 no)。

**langevinPiston:** 啟用 piston 模型模擬等壓系統 (此處為 on) 。

**langevinPistonTarget:** 設定目標壓力 (此處為 1 atm) 。

**langevinPistonPeriod:** 活塞震盪的時間週期 (此處為 100fs)。

**langevinPistonDecay:** 阻尼時間 (此處為 50fs)。

**langevinPistonTemp:** 活塞也需與溫度耦合，設為與主模擬相同。

**PME:** 是否使用 Particle Mesh Ewald 計算長距離靜電力 (此處為 yes)。

**PMEGridSpacing:** 網格間距設定，1Å 網格點間距通常能提供足夠精度。

```
# Output
outputName          $outputname

restartfreq         500     ;# 500steps = every 1ps
dcdfreq             250
xstFreq             250
outputEnergies      100
outputPressure      100
```

**outputName:** 所有輸出檔案（如 DCD, XST, restart, log）之檔名前綴。

**restartfreq:** 幾步輸出一次重啟檔 (此處為 500 steps) 。

**dcdfreq:** 幾步輸出一次 DCD 座標(此處為 250 steps) 。

**xstFreq:** 幾步輸出一次盒子大小 (此處為 250 steps)。

**outputEnergies:** 幾步輸出一次能量資訊至 log 檔 (此處為 100 steps) 。

**outputPressure:** 幾步輸出一次壓力資訊至 log 檔 (此處為 100 steps) 。

```
# Minimization
minimize            100
reinitvels          $temperature

run 2500 ;# 5ps
```

**minimize:** 進行幾步的能量最小化 (此處為 100 steps) 。

**reinitvels:** 以目標溫度初始化系統原子速度分布 (溫度與上方相同) 。

**run:** 模擬步數設定 (此處為 2500 steps，5 ps) 。

MYTLab Computation Bio

# Restart (ubq_wb_eq_restart.conf)

如果要接續先前的模擬工作（restart）需要對 Configuration file (.conf) 進行修改:

Step 1 ~ Step 4

Step 1. 加上下方參數和對應檔案路徑，以讀取前次模擬之結果與系統參數

**透過設定參數 “1” 開啟讀取前此資料；”0” 則關閉**

**輸入前一次模擬的輸出檔路徑與前綴**

```
# Continuing a job from the restart files
if {1} {
set inputname      ../ubq_wb_0.15_eq
binCoordinates     $inputname.restart.coor
binVelocities      $inputname.restart.vel  ;# remove the "temperature" entry if you use this!
extendedSystem     $inputname.restart.xsc
}
```

**binCoordinates: 讀取前一次模擬結束時儲存的原子位置 (*.restart.coor)。**

**binVelocities: 讀取前一次模擬的原子速度 (*.restart.vel)。**

**extendedSystem: 讀取模擬盒子大小與形狀（PBC）的資訊，保持空間一致性 (*.restart.xsc) 。**

Step 2. 將力場設定參數下方的 temperature 前方加上“#”使其變成註解

```
paraTypeCharmm      on
parameters          par_all27_prot_lipid.inp
temperature         $temperature
```

```
paraTypeCharmm      on
parameters          par_all27_prot_lipid.inp
#temperature         $temperature
```

Step 3. PBC設定參數則添加 “if” 指令用於控制此參數組

```
# Periodic Boundary Conditions
cellBasisVector1    40.6    0.    0.0
cellBasisVector2     0.0   42.2   0.0
cellBasisVector3     0.0    0    46.2
cellOrigin          30.8   29.0  17.7
wrapAll             on
```

```
# Periodic Boundary Conditions
# if run restart change 1 to 0 #
if {0} {
cellBasisVector1    40.6    0.    0.0
cellBasisVector2     0.0   42.2   0.0
cellBasisVector3     0.0    0    46.2
cellOrigin          30.8   29.0  17.7
}
wrapAll             on
```

透過設定參數 0 關閉讀取此參數；1 則開啟

Step 4. 將minimize 、reinitvels 前方加上 “#” 使其成為註解

```
# Minimization
minimize            100
reinitvels          $temperature

run 2500 ;# 5ps
```

```
# Minimization
# minimize          100
# reinitvels        $temperature
```

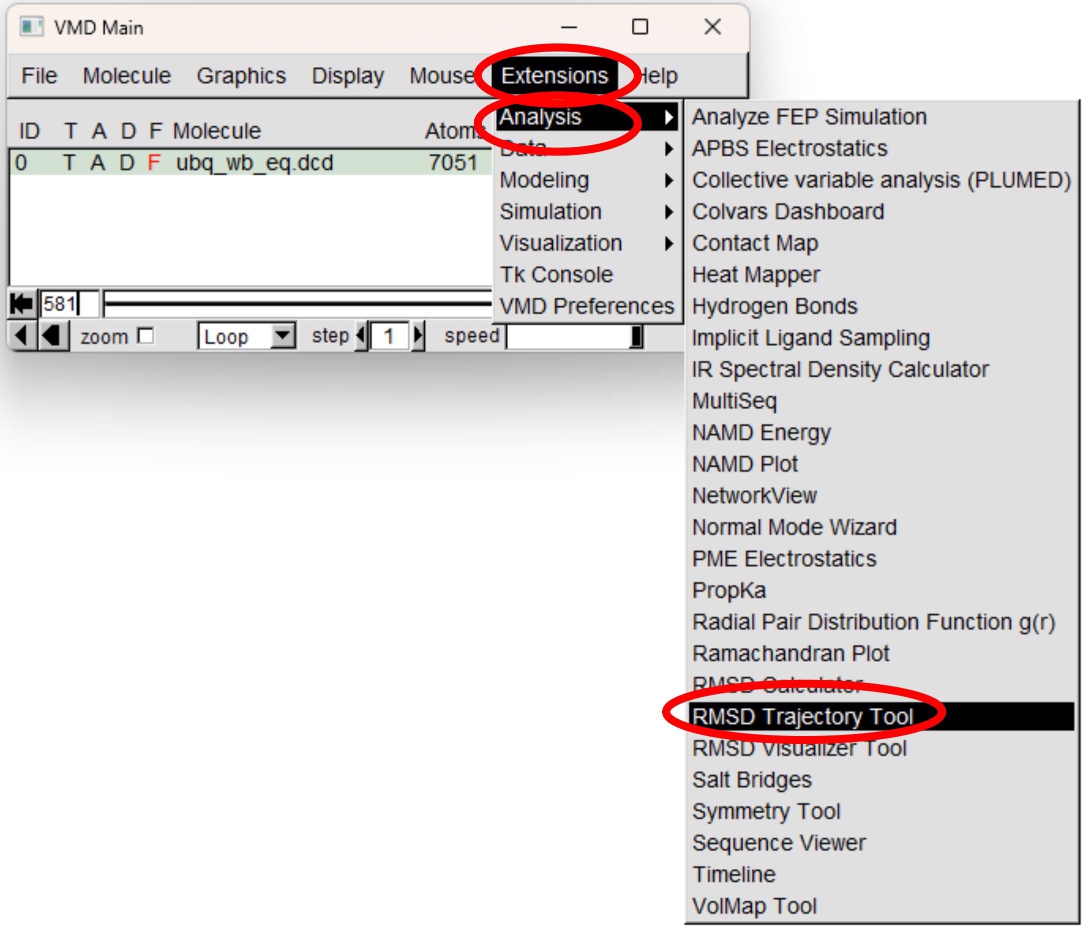
~~run 50000 ;# 100ps~~

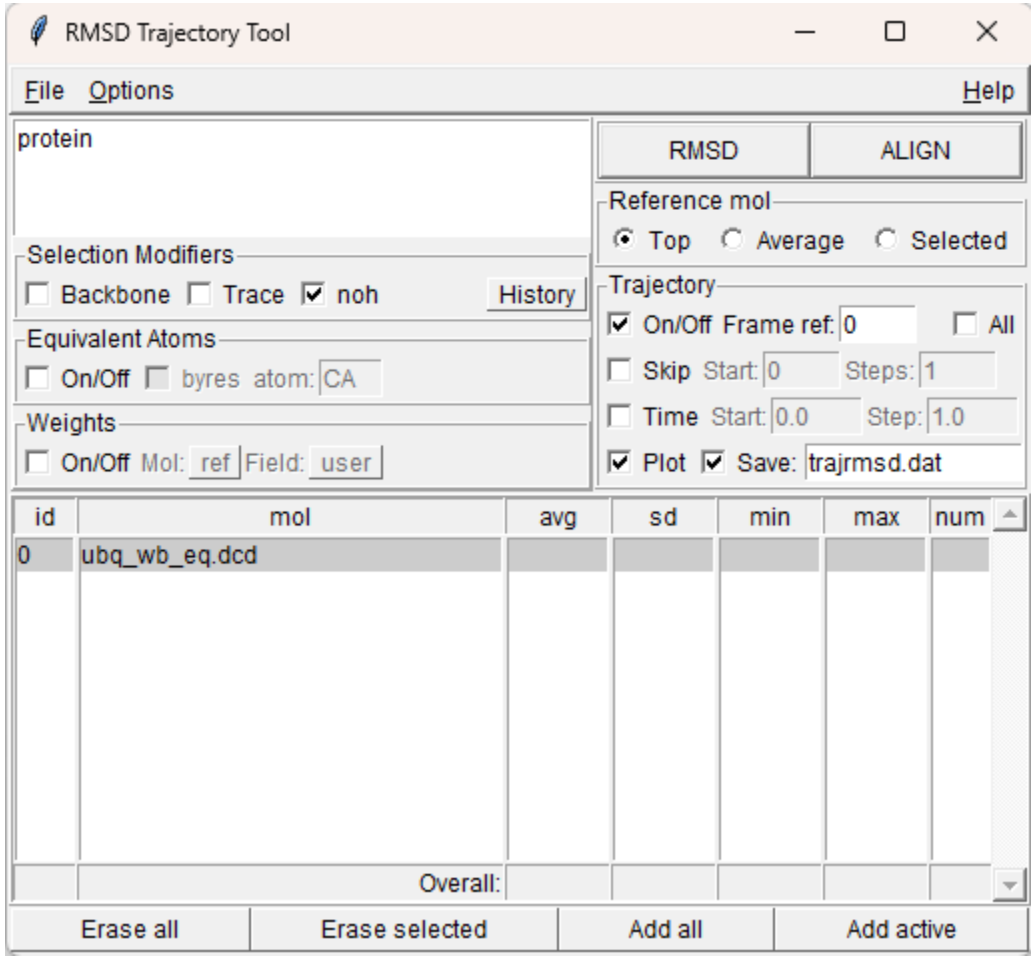
run 2500 ; # 5ps

# RMSD Trajectory Tool Plug-in

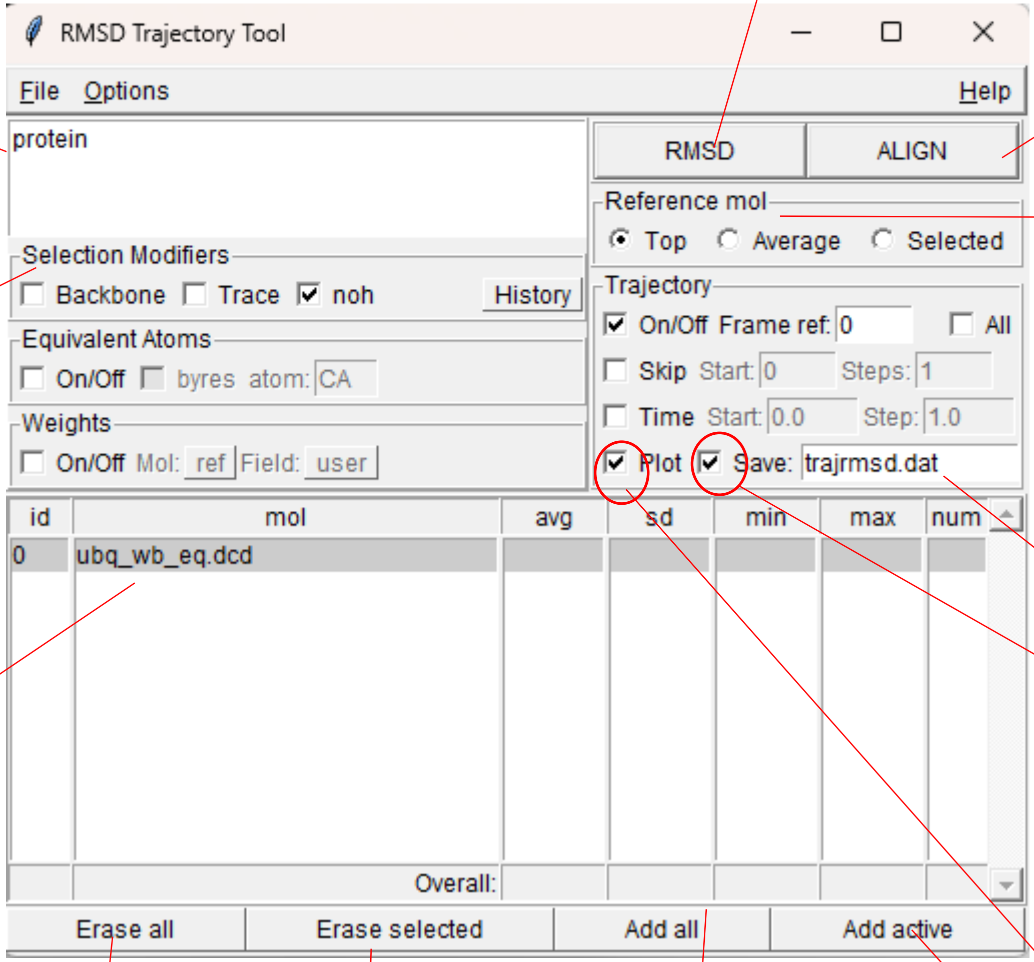
GUI 視窗化介面

# 開啟 RMSD Trajectory Tool



# 介紹 RMSD Trajectory Tool




RMSD：計算RMSD

ALIGN：將結構對齊至參考分子

定義選取分子結構的關鍵字

軟體預先定義的選取分子結構選項

Backbone: 選擇分子骨架的部分

Trace: 選擇所有Cα原子

noh: 排除氫原子

選取參考結構:

Top: VMD Main 正在操作的分子ID (前面有個 ”T”)

Average: 以所有結構的平均結構為參考（不適用於 ALIGN)

Selected：使用表格中選取的第一個分子作為參考。

文字檔的檔名

讓插件輸出:文字檔

顯示出RMSD變化的趨勢圖

加入的結構會顯示在這裡

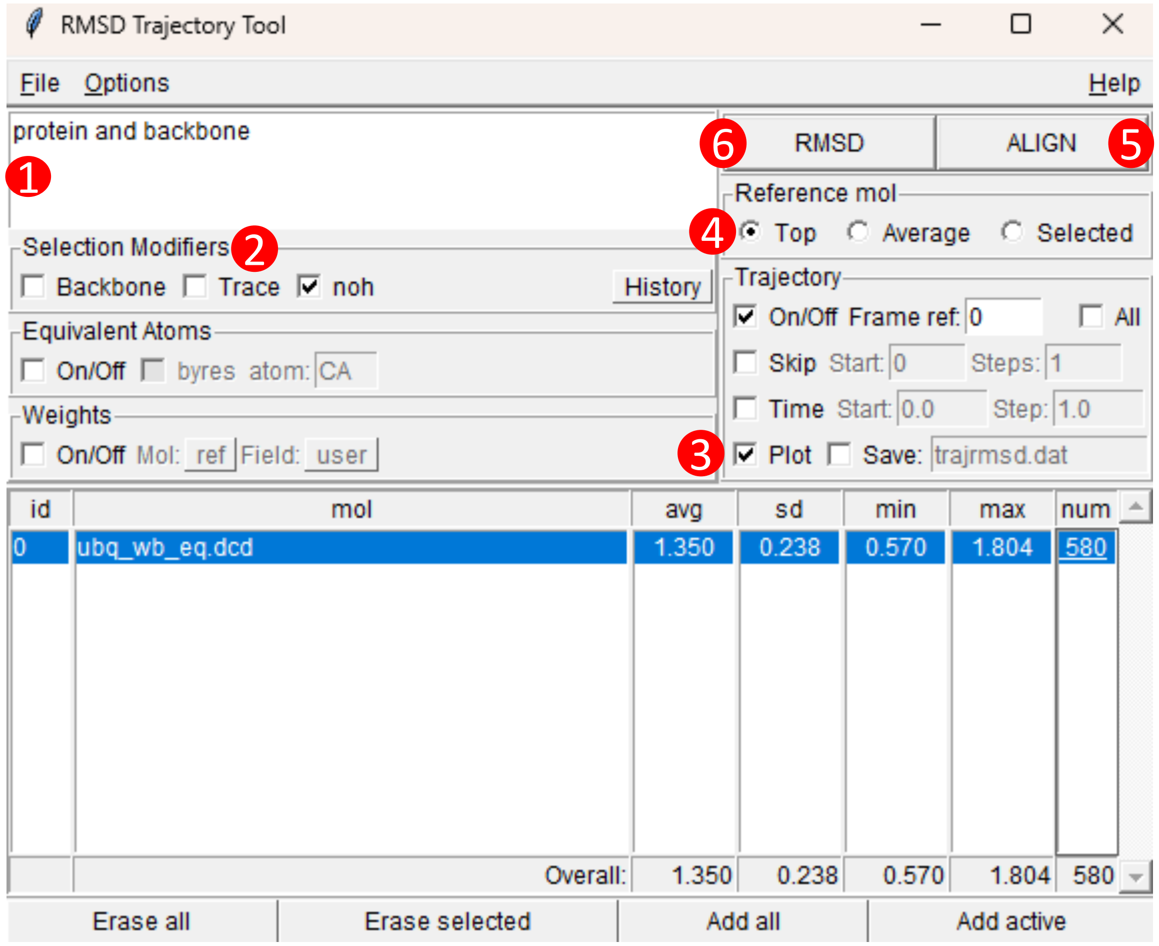
去除所有結構

去除選擇結構

加入所有結構

加入新結構

# 使用範例



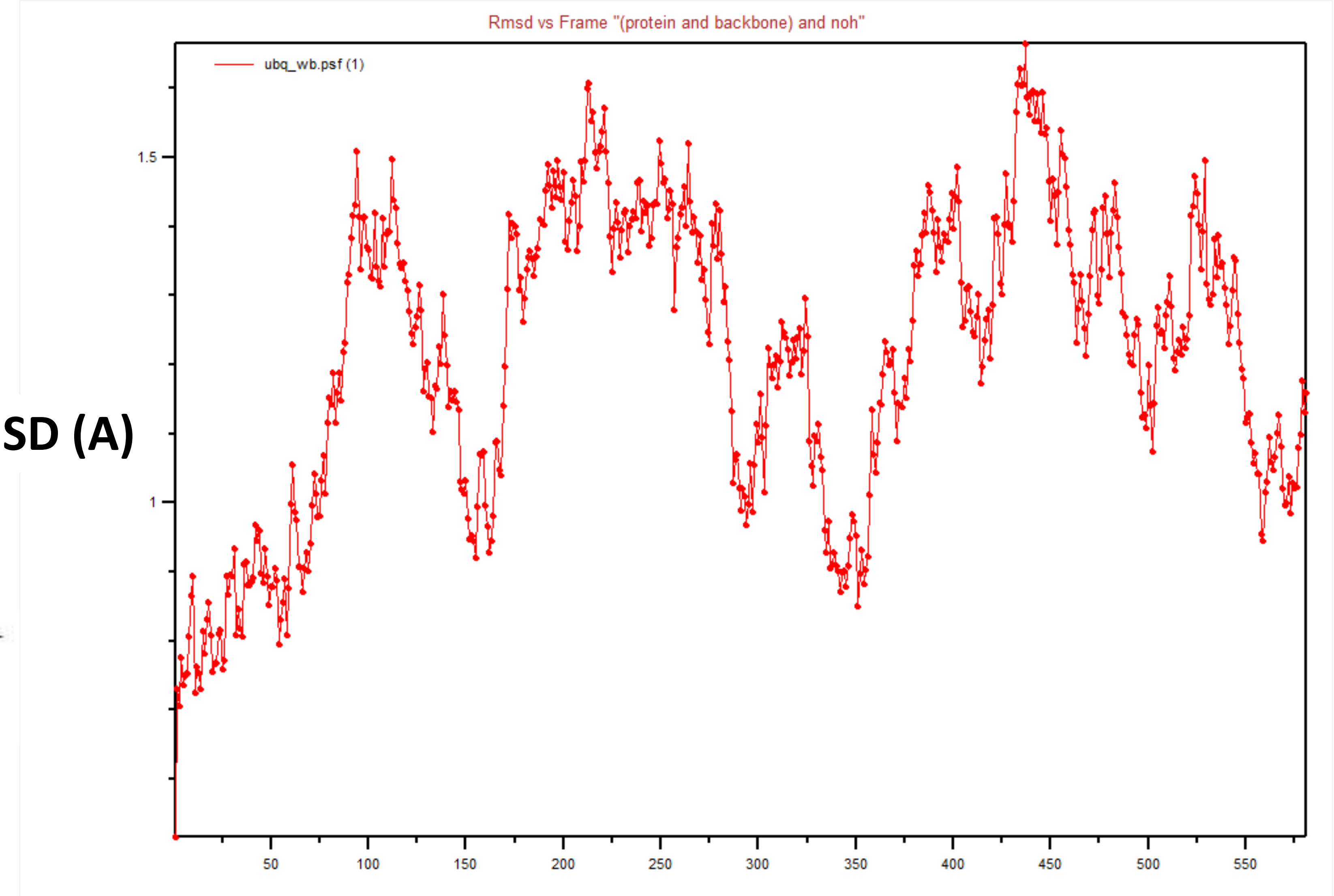

1. 選擇蛋白質且只選擇骨架的部分
2. 勾選 noh是去除氫(它一次只能選一樣)
3. 勾選 plot 軟體才會畫出圖來
4. 點選Top的意思是，VMD Main 正在操作的分子ID (前面有個 ”T”) ，而下方Trajectory的部分，


則表示我們是選取第 0 個 frame 作為參考結構。

5. 結構對齊(以第 4 步的 Frame 0 為參考結構)
6. 進行RMSD計算

# 結果 →

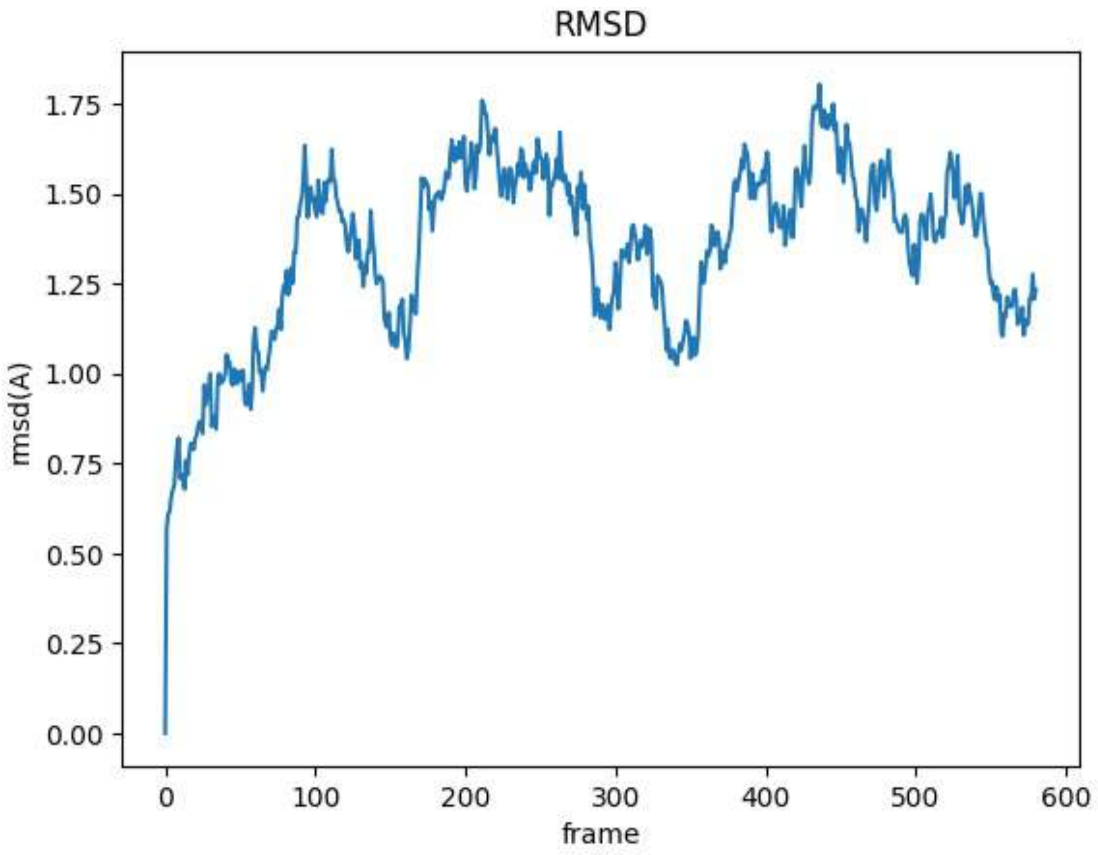
RMSD (A)



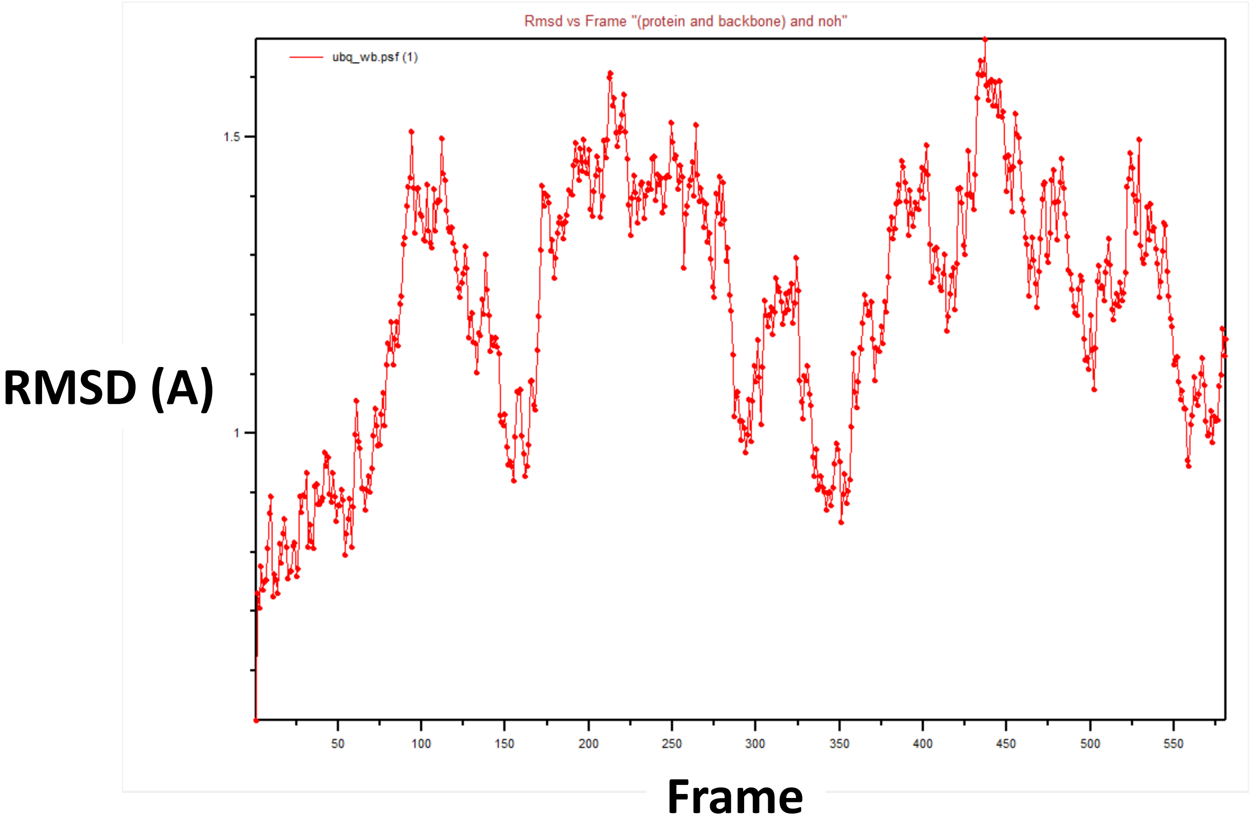

Frame

$$RMSD_{\alpha}(t_j) = \sqrt{\frac{\sum_{\alpha=1}^{N_\alpha} (\vec{r}_\alpha(t_j) - \langle \vec{r}_\alpha \rangle)^2}{N_\alpha}}$$

# Script vs GUI 結果比較




Python




GUI

# [補充] 程式碼解讀

```
# 開啟一個寫入檔案 sample.dat，並將檔案控制權賦值給變數 outfile
set outfile [open sample.dat w]

# 取得當前(top)分子模型的總幀數，並賦值給變數 nf
set nf [molinfo top get numframes]

# 選取第 0 幀中的 backbone 非氫原子，作為對齊與計算的參考結構，存為 frame0
set frame0 [atomselect top "protein and backbone and noh" frame 0]

# 選取整個軌跡中的 backbone 非氫原子，準備對齊與 RMSD 計算
set sel [atomselect top "protein and backbone and noh"]

                         0
set i 0    # 使用 for 迴圈，從第 1 幀開始，逐一與 frame0 對齊並計算 RMSD
for {set i 1} {$i < $nf} {incr i} {
   # 將 sel 切換到第 i 幀
   $sel frame $i

   # 對齊第 i 幀與第 0 幀的 backbone 結構，並將 sel 移動到最佳重疊位置
   $sel move [measure fit $sel $frame0]

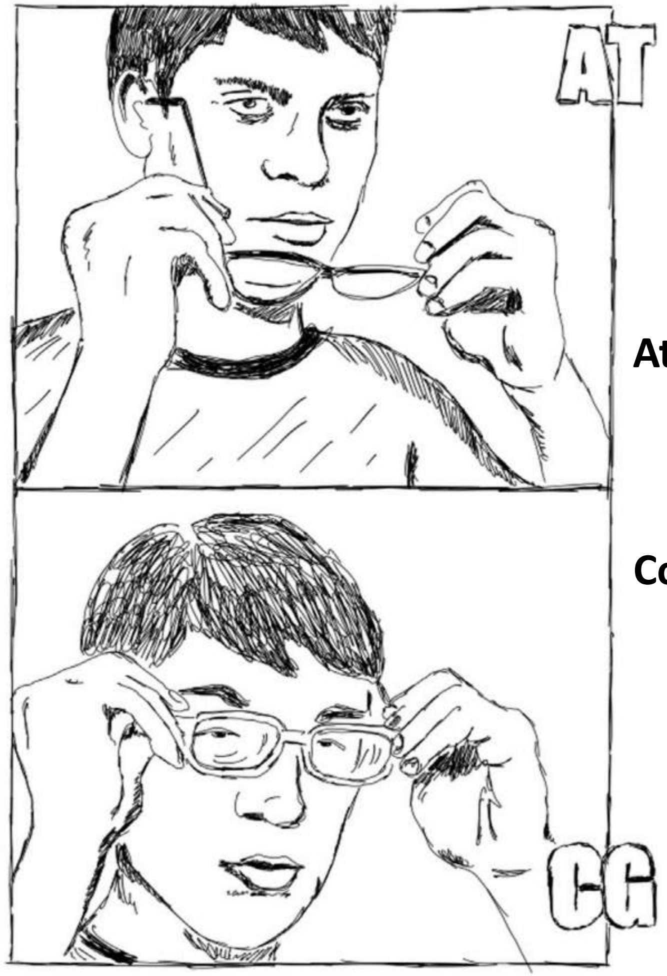
   # 計算對齊後的 RMSD，並寫入 outfile
   puts $outfile "[measure rmsd $sel $frame0]"
}

# 關閉檔案，完成寫入
close $outfile
```

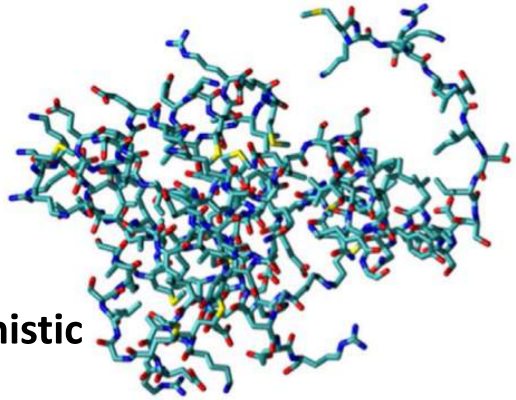
# [補充] 粗粒化模型

- The loss of detail is a defect, however, it does have the advantage of simplicity and parsimony

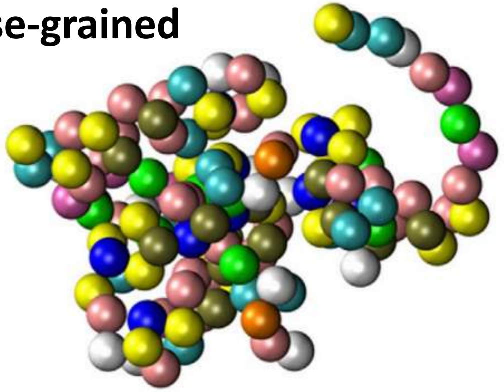
(粗粒化有缺點，但也具有簡化後的優勢)




Atomistic



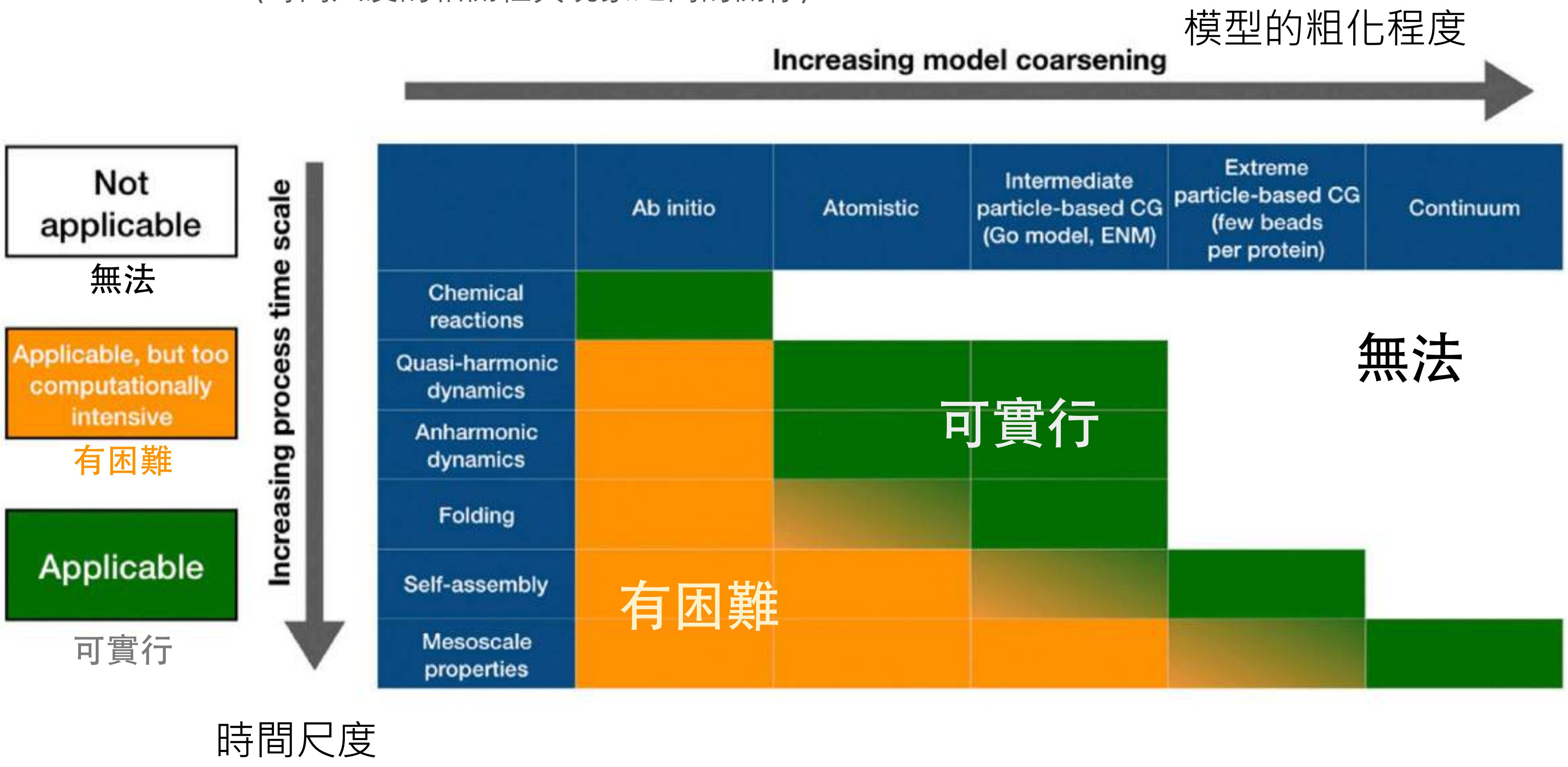
Coarse-grained



Giulini, M.; Rigoli, M.; Mattiotti, G.; Menichetti, R.; Tarenzi, T.; Fiorentini, R.; Potestio, R. *Frontiers in Molecular Biosciences* **2021**, *8*. https://doi.org/10.3389/fmolb.2021.676976.

# [補充] 時間尺度與解析程度

- (時間尺度的相關性與現象之間的關係)




Giulini, M.; Rigoli, M.; Mattiotti, G.; Menichetti, R.; Tarenzi, T.; Fiorentini, R.; Potestio, R. *Frontiers in Molecular Biosciences* **2021**, *8*. https://doi.org/10.3389/fmolb.2021.676976.

# [補充] 實驗、理論與模擬之間的連結

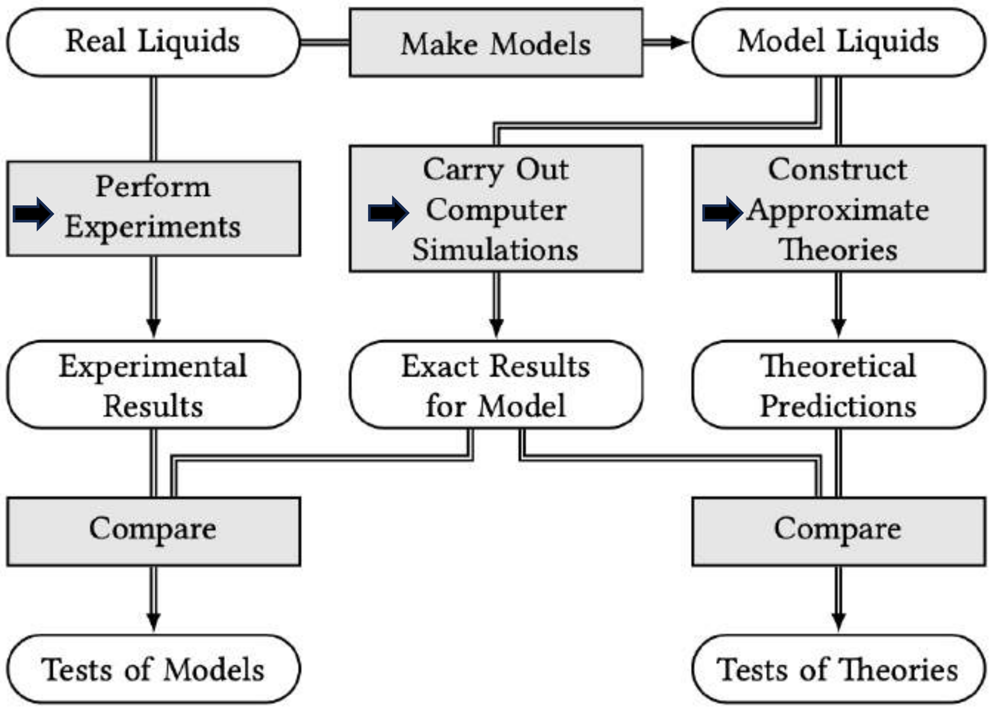
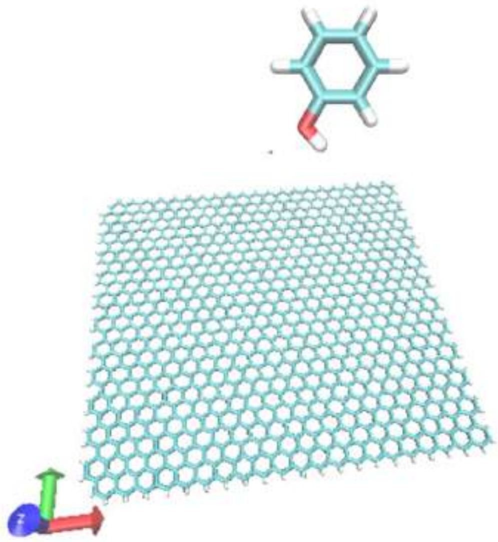



**Fig. 1.2** The connection between experiment, theory, and computer simulation.

Section 1.2 in Allen and Tildesley, Computer Simulation of Liquids 2nd, Oxford

# Part IV

(MD in Nano)



**IV. MD (Nano)**

- NAMD/VMD
  - Graphene
  - Small molecules

NAMD Scalable Molecular Dynamics

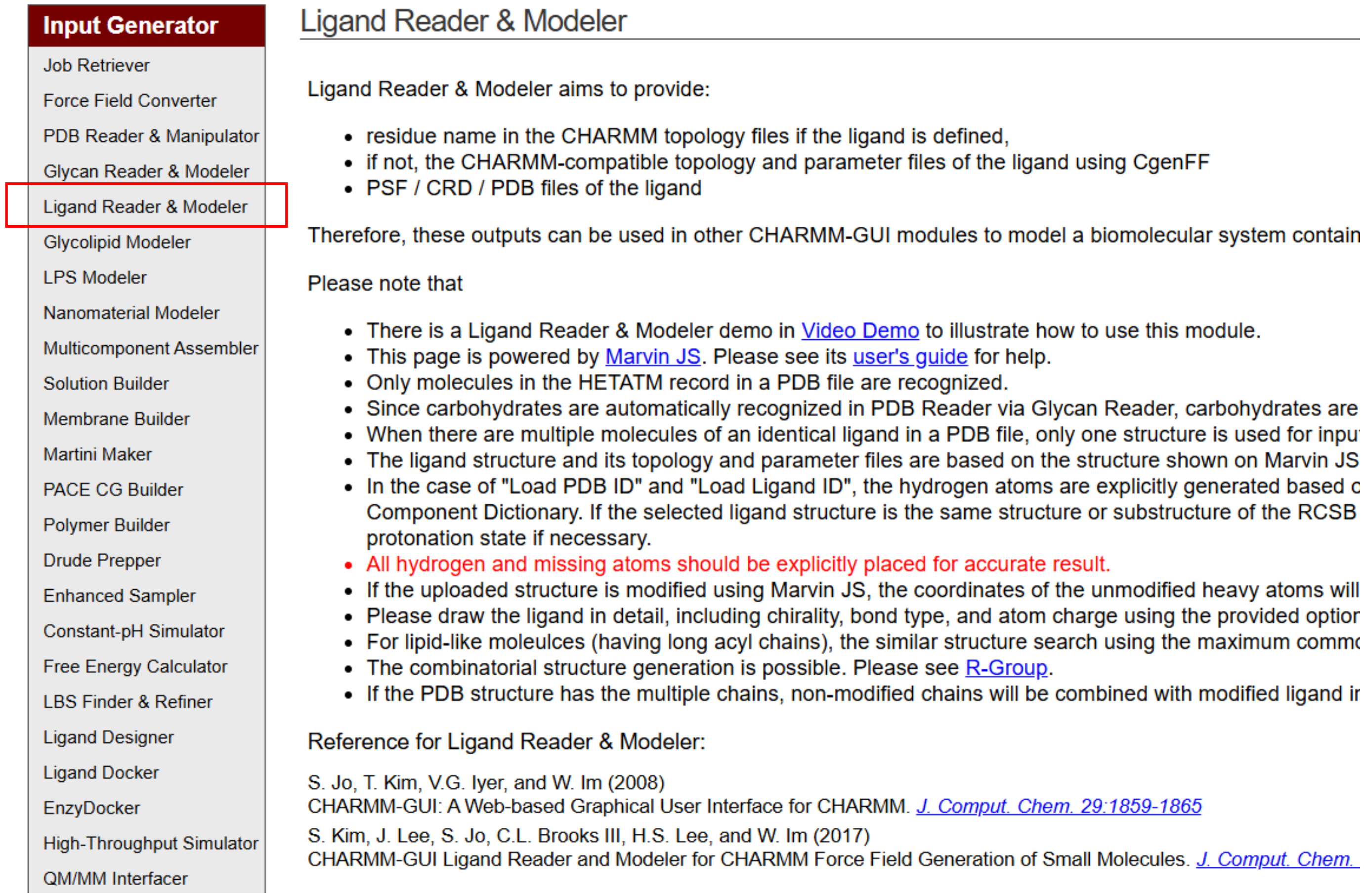
# CHARMM-GUI 建模教學

CHARMM-GUI
Effective Simulation Input Generator and More

- Small molecules EX: phenol
- Graphene
- Carbon Nanotube(CNT)

→ 產出 pdb、psf 等檔案

# Small molecules 建置



Input Generator

- Job Retriever
- Force Field Converter
- PDB Reader & Manipulator
- Glycan Reader & Modeler
- Ligand Reader & Modeler
- Glycolipid Modeler
- LPS Modeler
- Nanomaterial Modeler
- Multicomponent Assembler
- Solution Builder
- Membrane Builder
- Martini Maker
- PACE CG Builder
- Polymer Builder
- Drude Prepper
- Enhanced Sampler
- Constant-pH Simulator
- Free Energy Calculator
- LBS Finder & Refiner
- Ligand Designer
- Ligand Docker
- EnzyDocker
- High-Throughput Simulator
- QM/MM Interfacer

## Ligand Reader & Modeler

Ligand Reader & Modeler aims to provide:

- residue name in the CHARMM topology files if the ligand is defined,
- if not, the CHARMM-compatible topology and parameter files of the ligand using CgenFF
- PSF / CRD / PDB files of the ligand

Therefore, these outputs can be used in other CHARMM-GUI modules to model a biomolecular system contain

Please note that

- There is a Ligand Reader & Modeler demo in Video Demo to illustrate how to use this module.
- This page is powered by Marvin JS. Please see its user's guide for help.
- Only molecules in the HETATM record in a PDB file are recognized.
- Since carbohydrates are automatically recognized in PDB Reader via Glycan Reader, carbohydrates are
- When there are multiple molecules of an identical ligand in a PDB file, only one structure is used for inpu
- The ligand structure and its topology and parameter files are based on the structure shown on Marvin JS
- In the case of "Load PDB ID" and "Load Ligand ID", the hydrogen atoms are explicitly generated based c Component Dictionary. If the selected ligand structure is the same structure or substructure of the RCSB protonation state if necessary.
- All hydrogen and missing atoms should be explicitly placed for accurate result.
- If the uploaded structure is modified using Marvin JS, the coordinates of the unmodified heavy atoms will
- Please draw the ligand in detail, including chirality, bond type, and atom charge using the provided optior
- For lipid-like moleulces (having long acyl chains), the similar structure search using the maximum commo
- The combinatorial structure generation is possible. Please see R-Group.
- If the PDB structure has the multiple chains, non-modified chains will be combined with modified ligand ir

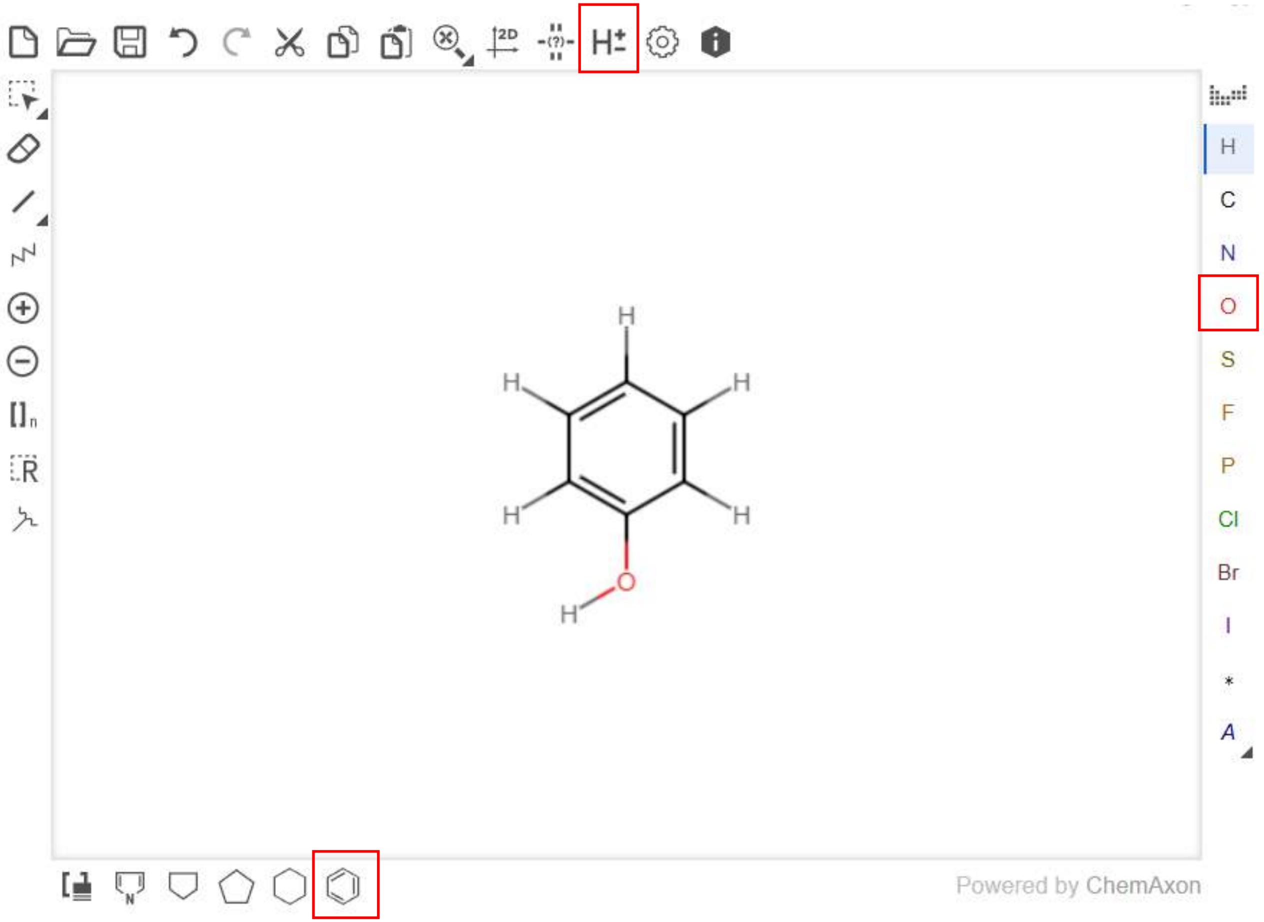
Reference for Ligand Reader & Modeler:


S. Jo, T. Kim, V.G. Iyer, and W. Im (2008)
CHARMM-GUI: A Web-based Graphical User Interface for CHARMM. J. Comput. Chem. 29:1859-1865

S. Kim, J. Lee, S. Jo, C.L. Brooks III, H.S. Lee, and W. Im (2017)
CHARMM-GUI Ligand Reader and Modeler for CHARMM Force Field Generation of Small Molecules. J. Comput. Chem.

# Ex: 建置 Phenol

3. 加氫



2. 從碳上拉出氧

1.

4. 按右下角下一步

Next Step:
Search ligand

5. 確認為phenol的parameter

**Exact**

Click the residue name to visualize the structure.

| residue name | Charge | Residue in | Comment |
|---|---|---|---|
| PHEN | 0.00 | toppar_all36_prot_model.str | PHENOL, ADM JR. |

**Isomer**

No isomer in CHARMM forcefield.

6. 按右下角產出pdb、psf檔

Next Step:
Generate PDB

Ligand Reader & Modeler

User Pro

Bookmark this link, if you want to comeback to this page

PDB Info

JOB ID: 5446577346

download.tgz

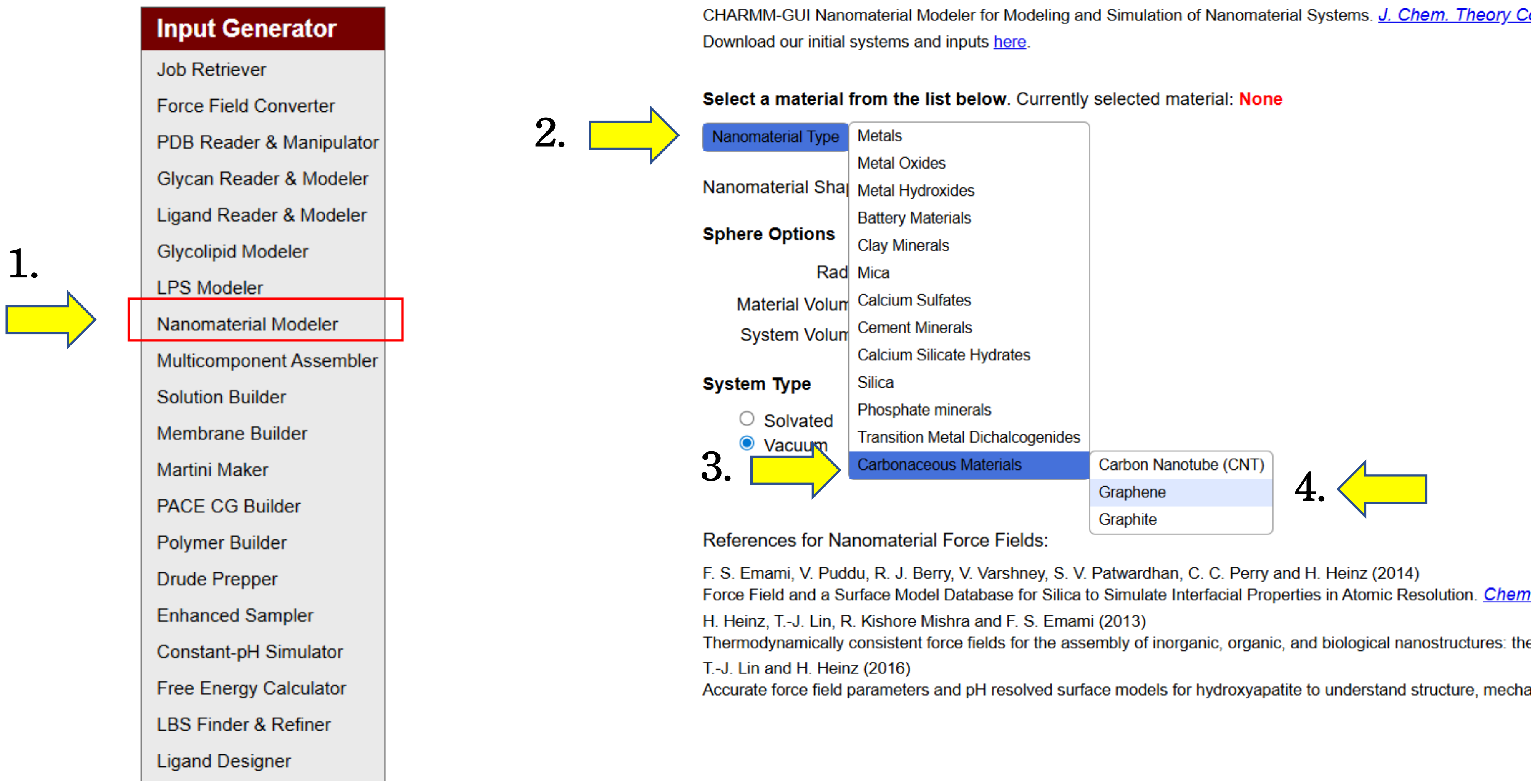
CHARMM Input: ligandrm.inp
CHARMM Output: ligandrm.out
CHARMM PDB: ligandrm.pdb (view structure)
CHARMM CRD: ligandrm.crd
CHARMM PSF: ligandrm.psf

7.點擊下載

8.將ligandrm.pdb & psf檔名改為phenol.pdb & psf

| | | |
|---|---|---|
| phenol | 2025/3/3 下午 03:40 | PDB 檔案 |
| phenol.psf | 2025/3/3 下午 03:40 | PSF 檔案 |

# Graphene 建置


Input Generator
Job Retriever
Force Field Converter
PDB Reader & Manipulator
Glycan Reader & Modeler
Ligand Reader & Modeler
Glycolipid Modeler
LPS Modeler
Nanomaterial Modeler
Multicomponent Assembler
Solution Builder
Membrane Builder
Martini Maker
PACE CG Builder
Polymer Builder
Drude Prepper
Enhanced Sampler
Constant-pH Simulator
Free Energy Calculator
LBS Finder & Refiner
Ligand Designer
1.
2.
3.
4.
CHARMM-GUI Nanomaterial Modeler for Modeling and Simulation of Nanomaterial Systems. J. Chem. Theory C
Download our initial systems and inputs here.
Select a material from the list below. Currently selected material: None
Nanomaterial Type
Metals
Metal Oxides
Metal Hydroxides
Battery Materials
Clay Minerals
Mica
Calcium Sulfates
Cement Minerals
Calcium Silicate Hydrates
Silica
Phosphate minerals
Transition Metal Dichalcogenides
Carbonaceous Materials
Carbon Nanotube (CNT)
Graphene
Graphite
Nanomaterial Sha
Sphere Options
Rad
Material Volum
System Volum
System Type
Solvated
Vacuum
References for Nanomaterial Force Fields:
F. S. Emami, V. Puddu, R. J. Berry, V. Varshney, S. V. Patwardhan, C. C. Perry and H. Heinz (2014)
Force Field and a Surface Model Database for Silica to Simulate Interfacial Properties in Atomic Resolution. Chem
H. Heinz, T.-J. Lin, R. Kishore Mishra and F. S. Emami (2013)
Thermodynamically consistent force fields for the assembly of inorganic, organic, and biological nanostructures: the
T.-J. Lin and H. Heinz (2016)
Accurate force field parameters and pH resolved surface models for hydroxyapatite to understand structure, mecha

5. 設定好Graphene性質

**Select a material from the list below.** Currently selected material: **Graphene**

Nanomaterial Shape: Box

Percent Defect 0

Use virtual electron model

**Box Options**

| | | |
|---|---|---|
| Miller Index | 001 | |
| X length (Å) | 50 | (54.3) |
| Y length (Å) | 50 | (51.3) |
| Z length (Å) | 1 | (3.3) |
| Material Volume ($Å^3$) | 9,317.3 | |
| System Volume ($Å^3$) | 9,317.3 | |

**Unit Cell Info**

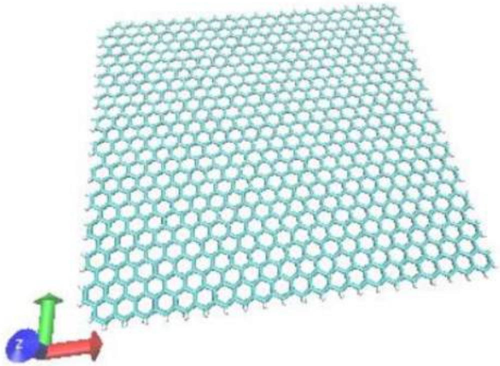
| | |
|---|---|
| Unit Cell X | 4.934 Å |
| Unit Cell Y | 4.273 Å |
| Unit Cell Z | 3.348 Å |
| Layer Dimension | Z |
| Number of Layers | 1 |

**Periodic Options**

☑ X ☐ Y ☐ Z

**System Type**

○ Solvated
◉ Vacuum



6. 按右下角下一步

Next Step:
Build Nanomaterial


7. 點擊下載



8. 將step1_nanomaterial.pdb & psf 檔名改為graphene.pdb & psf

# Carbon nanotube 建置

4. 設定好CNT性質

Nanomaterial Type

Chirality (n, m) 6 , 5

Cell Copies 2

Use virtual electron model ☐

**Calculated Tube Info**

| | |
|---|---|
| Tube Length | 81.3 Å |
| Tube Diameter | 7.5 Å |
| Tube Conductivity | **Semiconductive** |
| System Volume | 114,212.9 $Å^3$ |

**Periodic Options**

☑ X ☐ Y ☐ Z

**System Type**

○ Solvated
◉ Vacuum

5. 按右下角下一步

Next Step:
Build Nanomaterial


6. 點擊下載



7. 將step1_nanomaterial.pdb & psf
檔名改為nanotube.pdb & psf

# 補充

## 改變 cell copy 數可產生不同長度的碳管 (n倍長)

Nanomaterial Type

Chirality (n, m) 6 , 5
Cell Copies 1
Use virtual electron model ☐

**Calculated Tube Info**

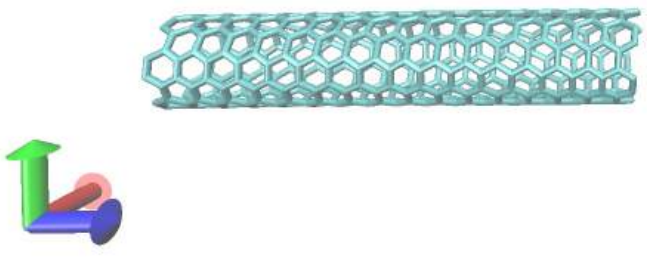
| | |
|---|---|
| Tube Length | 40.7 Å |
| Tube Diameter | 7.5 Å |
| Tube Conductivity | **Semiconductive** |
| System Volume | 57,106.4 $Å^3$ |

**Periodic Options**

☑ X ☐ Y ☐ Z

**System Type**

○ Solvated
◉ Vacuum



Nanomaterial Type

Chirality (n, m) 6 , 5
Cell Copies 2
Use virtual electron model ☐

**Calculated Tube Info**

| | |
|---|---|
| Tube Length | 81.3 Å |
| Tube Diameter | 7.5 Å |
| Tube Conductivity | **Semiconductive** |
| System Volume | 114,212.9 $Å^3$ |

**Periodic Options**

☑ X ☐ Y ☐ Z

**System Type**

○ Solvated
◉ Vacuum

Nanomaterial Type

Chirality (n, m) 6 , 5
Cell Copies 3
Use virtual electron model ☐

**Calculated Tube Info**

| | |
|---|---|
| Tube Length | 122.0 Å |
| Tube Diameter | 7.5 Å |
| Tube Conductivity | **Semiconductive** |
| System Volume | 171,319.3 $Å^3$ |

**Periodic Options**

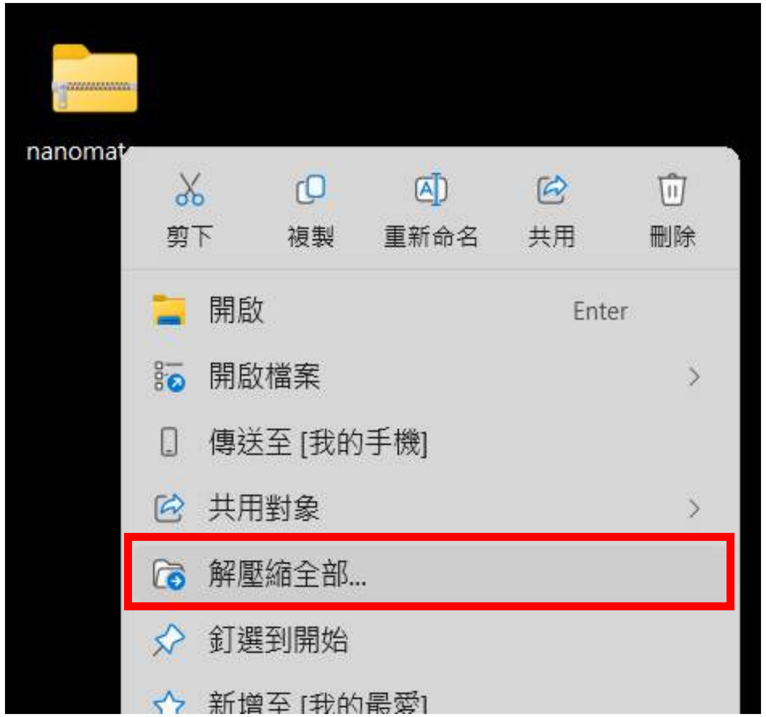
☑ X ☐ Y ☐ Z

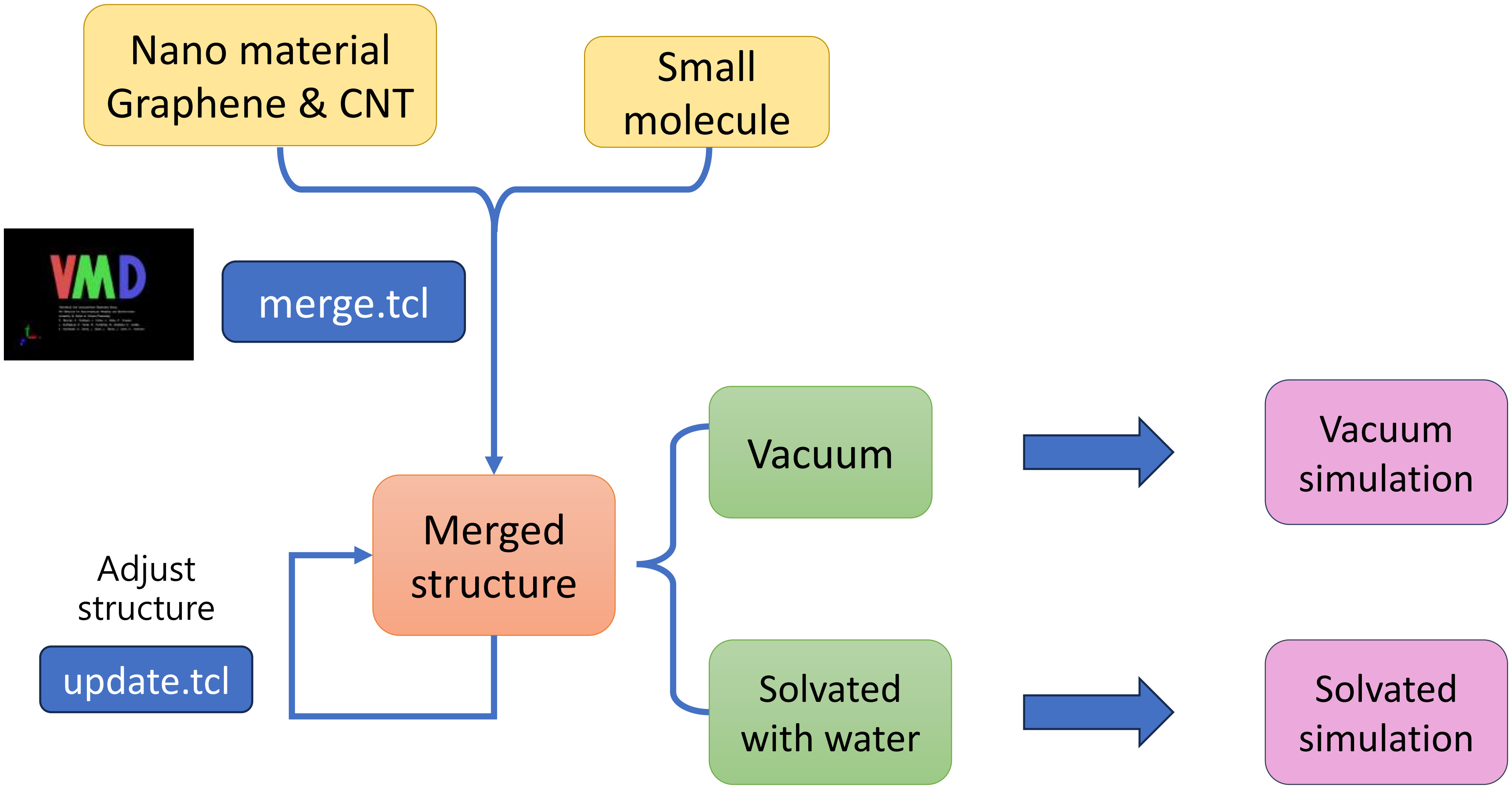
**System Type**

○ Solvated
◉ Vacuum

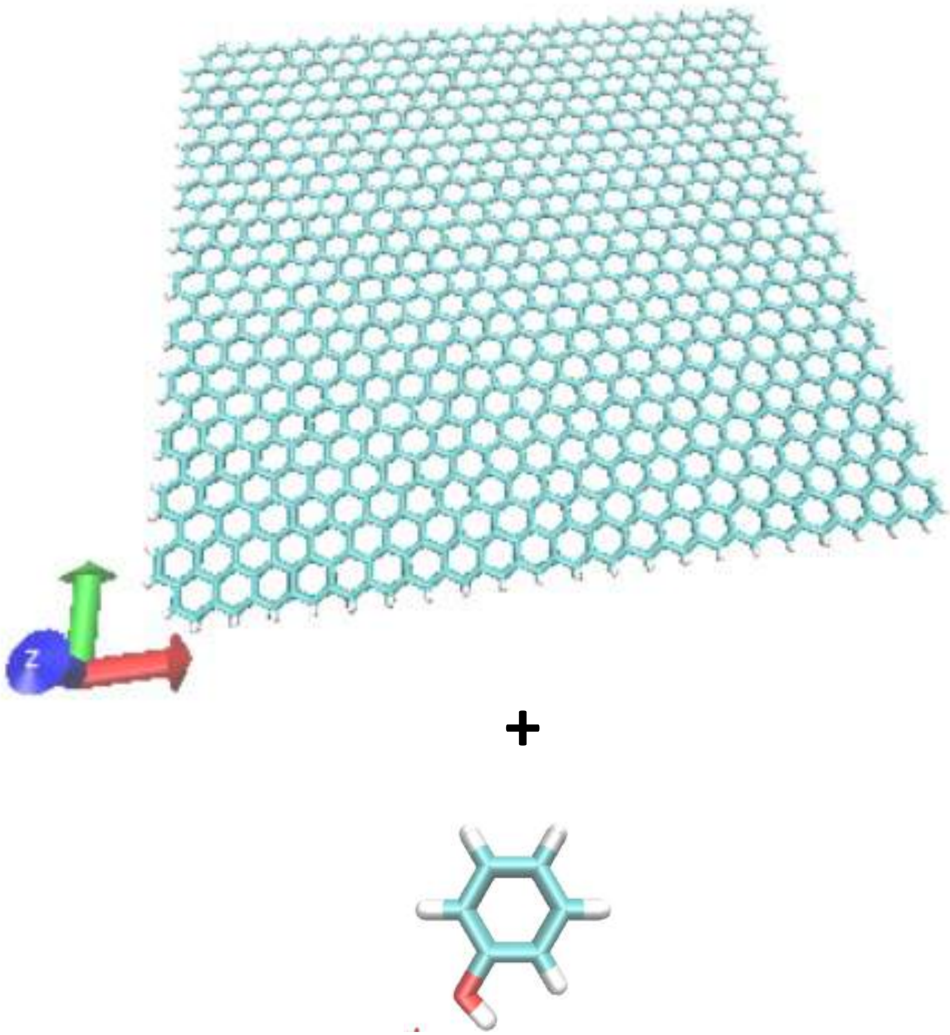
# Nanomaterial 解壓縮

1. 從 113.2 MD workshop/module4 中，將 **nanomaterial.zip** 資料夾拖曳至桌面
2. 按滑鼠右鍵 → 解壓縮全部... → 目的地路徑 \.....\.....\Desktop\(此處全刪除)
3. 解壓縮

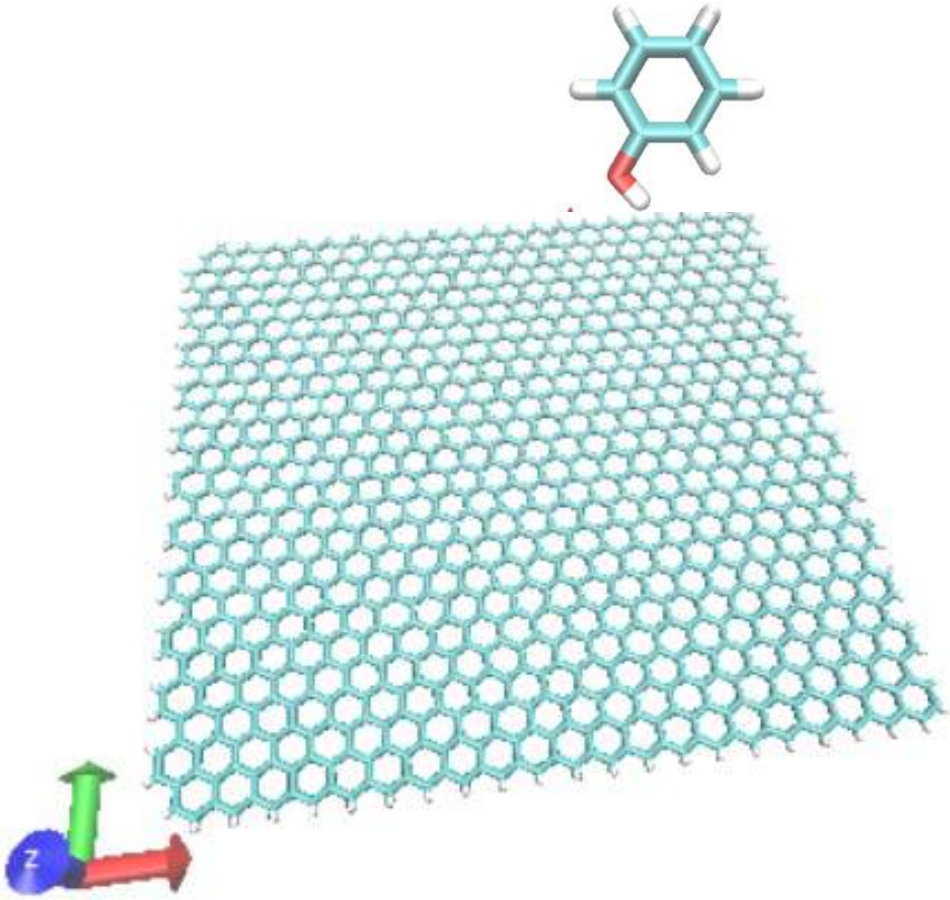




Nano material
Graphene & CNT
Small
molecule
VMD
merge.tcl
Merged
structure
Adjust
structure
update.tcl
Vacuum
Solvated
with water
Vacuum
simulation
Solvated
simulation

# 奈米材料模型建立 (1. Merge)



Merge



# Merge.tcl介紹

```
#load nanomaterial
package require topotools 1.7
set midlist {}
set mol [mol new graphene.psf waitfor all]
mol addfile graphene.pdb

#load molecule in system
lappend midlist $mol
set mol [mol new phenol.psf waitfor all]
mol addfile phenol.pdb $mol

#moving molecule
set sel [atomselect top all]
$sel moveby {0.0 20.0 0.0}
    lappend midlist $mol
#do the magic
set mol [::TopoTools::mergemols $midlist]

animate write psf graphene_sim_temp.psf $mol
animate write pdb graphene_sim_temp.pdb $mol
```

搭配VMD 1.9.3
(VMD 1.9.4則使用1.8)

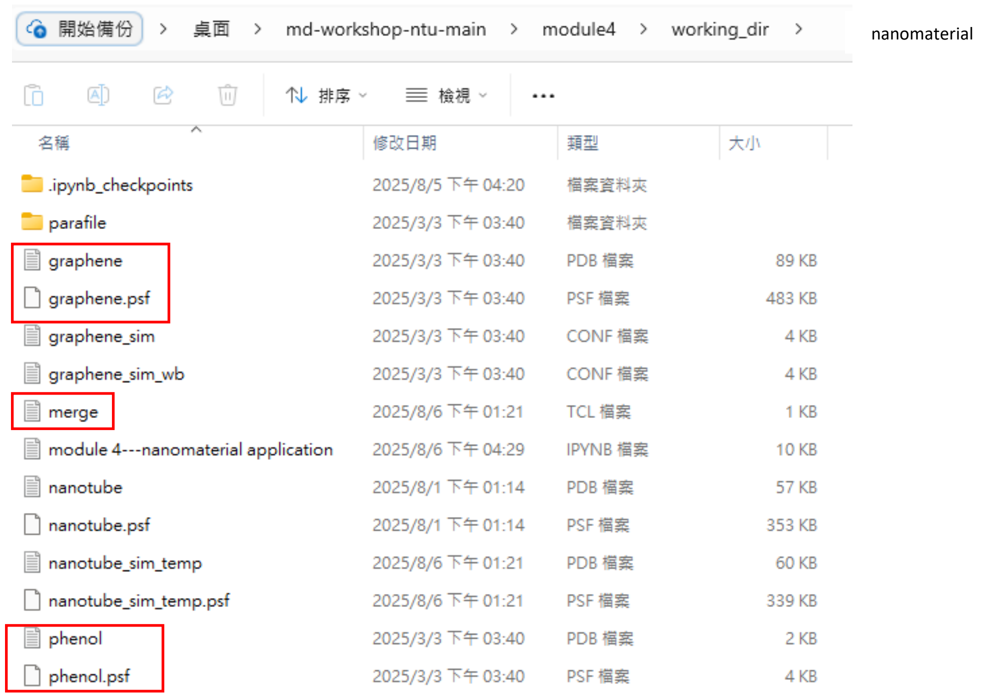
匯入Graphene or CNT structure

匯入phenol structure

選擇merge位置

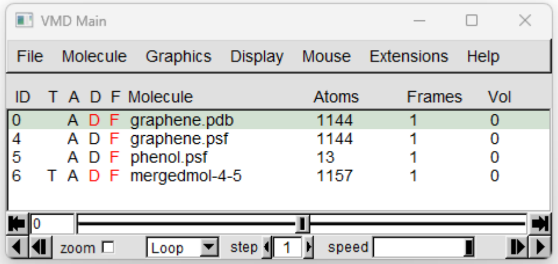
輸出merge檔案名稱

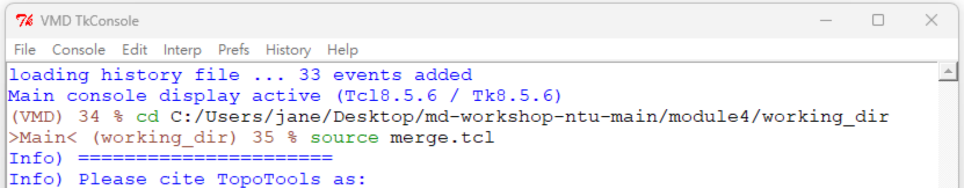
# 使用方式(以Graphene為例)



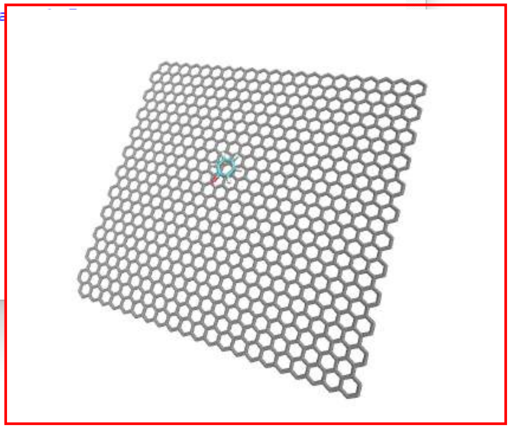

1. 將右側框起檔案放置同個路徑中
( Ex: ~ ./nanomaterial/Working_dir/ )

2.
於vmd中匯入graphene.pdb後
打開tk console
cd至放置檔案的路徑後按enter
再鍵入source merge.tcl後按enter




3. 獲得merged structure



# 補充：Merge 不同位置示範

```
#load nanomaterial
package require topotools 1.7
set midlist {}
set mol [mol new graphene.psf waitfor all]
mol addfile graphene.pdb

#load molecule in system
lappend midlist $mol
set mol [mol new phenol.psf waitfor all]
mol addfile phenol.pdb $mol

#moving molecule
set sel [atomselect top all]
$sel moveby {0.0 0.0 10.0}
    lappend midlist $mol
#do the magic
set mol [::TopoTools::mergemols $midlist]

animate write psf graphene_sim_temp.psf $mol
animate write pdb graphene_sim_temp.pdb $mol
```

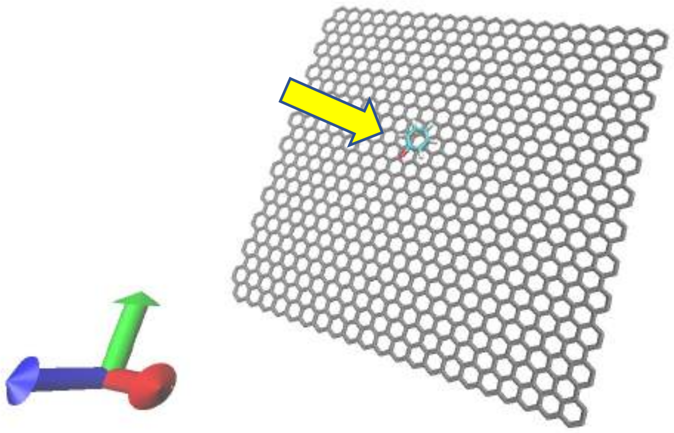
將 x 值改成15

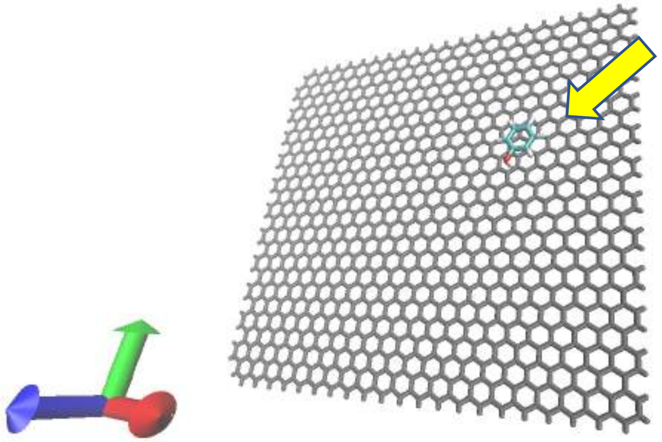
```
#load nanomaterial
package require topotools 1.7
set midlist {}
set mol [mol new graphene.psf waitfor all]
mol addfile graphene.pdb

#load molecule in system
lappend midlist $mol
set mol [mol new phenol.psf waitfor all]
mol addfile phenol.pdb $mol

#moving molecule
set sel [atomselect top all]
$sel moveby {15.0 0.0 10.0}
    lappend midlist $mol
#do the magic
set mol [::TopoTools::mergemols $midlist]

animate write psf graphene_sim_temp_1.psf $mol
animate write pdb graphene_sim_temp_1.pdb $mol
```

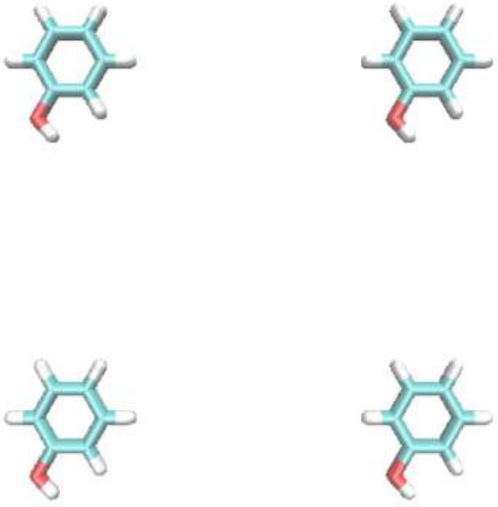


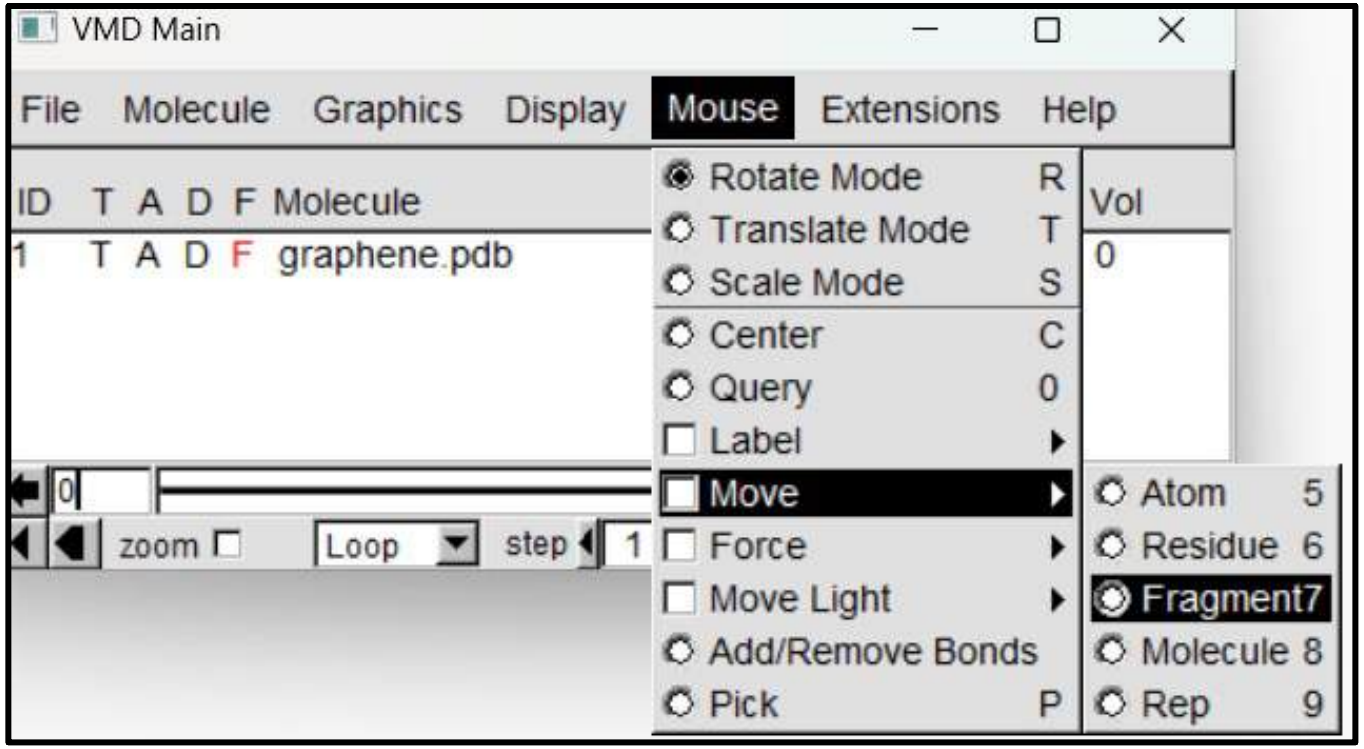


# 奈米材料模型建立 (2. Update)

## Adjustment

按住**shift**與左鍵可以旋轉分子

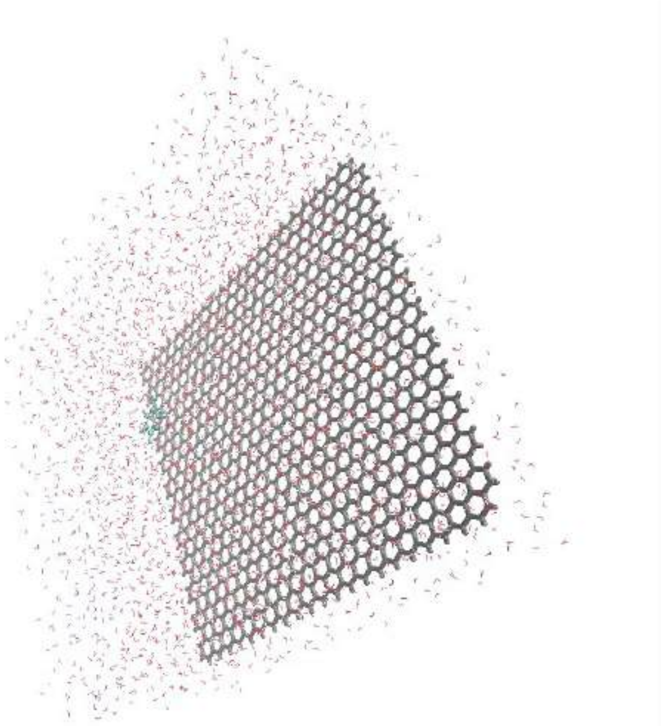





完成旋轉與調整後，在 tk console 輸入 *source update.tcl*

# Solvated simulation (Graphene)

將: graphene_sim_wb.conf graphene_sim_wb.pdb graphene_sim_wb.psf 放入/graphene/solvated/資料夾內



graphene_sim_wb.conf ↓

```
# if run restart change 1 to 0 #
temperature          $temperature


# Force-Field Parameters
exclude              scaled1-4
1-4scaling           1.0
cutoff               12.0
switching            on
switchdist           10.0
pairlistdist         14.0
margin               5

# Integrator Parameters
timestep             2.0  ;# 2fs/step
rigidBonds           all  ;# needed for 2fs steps
nonbondedFreq        1
fullElectFrequency   2
stepspercycle        10


# Constant Temperature Control
langevin             on    ;# do langevin dynamics
langevinDamping      1     ;# damping coefficient (gamma) of 1/ps
langevinTemp         $temperature
langevinHydrogen     off    ;# don't couple langevin bath to hydrogens


# Periodic Boundary Conditions
cellBasisVector1     18.0    0.0     0.0
cellBasisVector2      0.0   37.0     0.0
cellBasisVector3      0.0    0.0    93.0
cellOrigin            0.0   10.4     0.0

wrapAll              on


# PME (for full-system periodic electrostatics)
PME                  yes
PMEGridSpacing       1.0

#manual grid definition
#PMEGridSizeX         45
#PMEGridSizeY         45
#PMEGridSizeZ         48


# Constant Pressure Control (variable volume)
useGroupPressure       yes ;# needed for rigidBonds
useFlexibleCell        no
useConstantArea        no

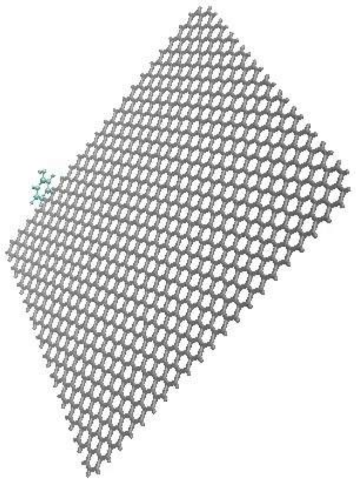
langevinPiston         on
langevinPistonTarget   1.01325 ;#  in bar -> 1 atm
langevinPistonPeriod   100.0
langevinPistonDecay    50.0
langevinPistonTemp     $temperature


# Output
outputName           $outputname

restartfreq          250     ;# 500steps = every 1ps
dcdfreq              250
xstFreq              250
outputEnergies       100
outputPressure       100
```

# Vacuum simulation (Graphene)

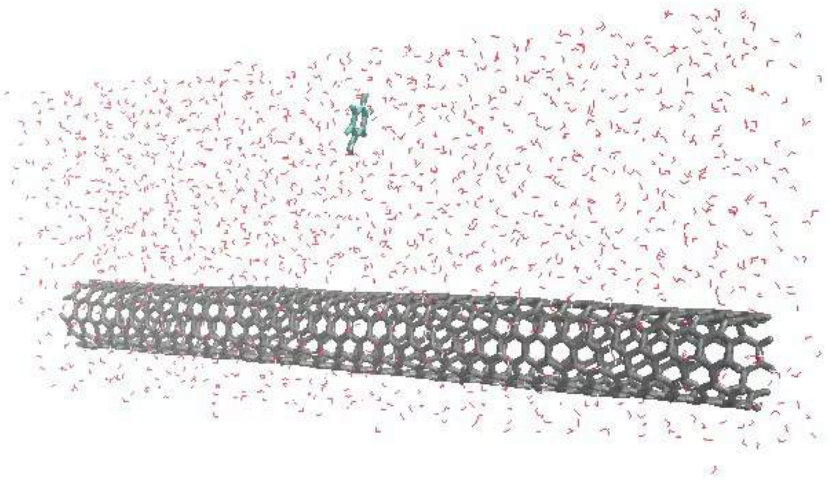
將: graphene_sim.conf graphene_sim_temp.pdb
graphene_sim_temp.psf 放入/graphene/vacuum/資料夾內



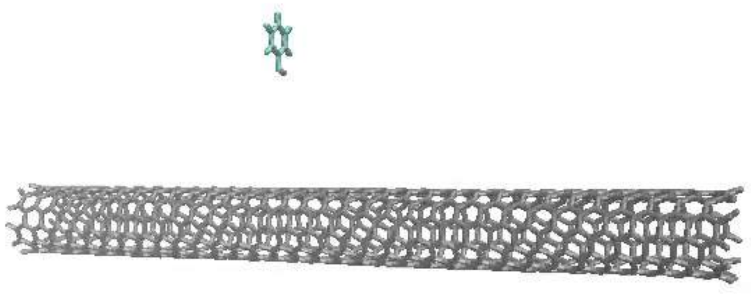
和solvated 系統指令差在↓

```
# Constant Pressure Control (variable volume)
useGroupPressure       no    ;#no need in vaccum
useFlexibleCell        no
useConstantArea        no
```

# Solvated simulation (CNT)

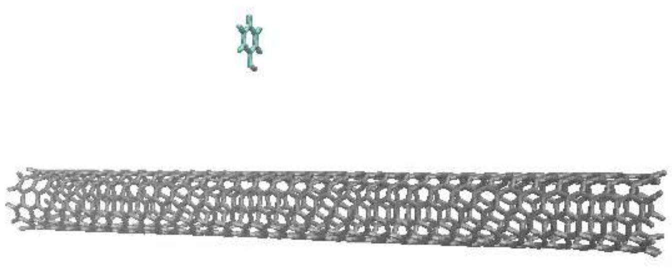


# Vacuum simulation (CNT)



# 計算小分子與CNT的距離

- 在Tk Console輸入 (注意放置路徑)：
  source distance.tcl



Frame數　phenol相對CNT端點距離

```
0     20.78642318055843
1     20.786781819098746
2     21.43881348889818
3     21.925169780145072
4     22.49873621900503
5     23.05626825633667
6     23.734683241321928
7     23.568969645829565
8     22.944020405255017
9     22.72318032861033
10     22.5102304295546
11     22.045578774801303
12     22.225801333196134
13     22.467258795174175
14     23.599905198295826
15     24.175286507557995
16     24.650381070596282
17     25.717776770317602
18     27.31156748112046
19     28.05961128886521
20     28.292052402519854
```

產出distance.dat↑

phenol相對CNT端點距離　統計次數

```
7.088063155391156 5
7.449060043707225 3
7.8100569320232935 4
8.171053820339361 1
8.532050708655431 5
8.893047596971499 2
9.254044485287569 1
9.615041373603637 4
9.976038261919706 2
10.337035150235774 0
10.698032038551844 4
11.059028926867912 2
11.42002581518398 7
11.78102270350005 9
12.14201959181612 7
12.503016480132187 9
12.864013368448255 9
13.225010256764325 4
13.586007145080394 2
13.947004033396462 3
14.30800092171253 1
14.6689978100286 0
15.02999469834467 0
15.390991586660737 0
15.751988474976805 1
```

產出distribution.dat↑